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Towards Fine-Grained Surgical Scene Understanding through Grounded Instrument–Tissue Interaction Modelling

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A dissertation submitted in partial fulfilment
of the requirements for the degree of
Doctor of Philosophy

August 13, 2026

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I, Oluwatosin Alabi, confirm that the work presented in this thesis is my own. Where information has been derived from other sources, I confirm that this has been indicated in the thesis.

Abstract

Minimally invasive surgery offers important benefits to patients, including reduced blood loss, less pain, faster recovery, and fewer post-operative complications. However, it also replaces direct observation of the operative field with a restricted view through an endoscope or laparoscope, making complex surgical events difficult to interpret. Computer-aided intervention seeks to augment this limited visual information and support a richer understanding of the operative scene. Although surgical artificial intelligence has advanced tasks such as instrument segmentation, workflow recognition, and action analysis, modelling surgical interactions in a clinically meaningful and interpretable way remains difficult. Surgical action triplets, represented as ⟨`instrument, verb, target`⟩ tuples, provide a compact description of surgical activity by specifying which instrument performs which action on which anatomical structure. Existing triplet understanding methods largely operate at the frame level, identifying interactions without explicitly grounding them to the responsible instrument instance.

This thesis advances fine-grained surgical scene understanding by developing datasets, benchmarks, and models for grounded instrument–tissue interaction modelling in laparoscopic surgery. First, it reviews the broader landscape of multitask learning and surgical scene understanding in minimally invasive surgical vision, identifying the progression from perceptual and workflow tasks towards fine-grained action understanding, as well as the lack of explicit spatial grounding for surgical interactions.

Second, this thesis introduces CholecInstanceSeg, a large-scale tool instance segmentation dataset for laparoscopic cholecystectomy. CholecInstanceSeg contains 41,933 annotated frames from 85 procedures and 64,483 instrument instances, enabling individual tools to be localised and distinguished in complex multi-instrument scenes. This dataset establishes the spatial foundation required for grounded surgical action understanding.

Third, this thesis introduces triplet segmentation, a task that extends surgical action triplet recognition from frame-level prediction to instance-level spatial grounding. To support this task, CholecTriplet-Seg links instrument masks with verb and anatomical target labels across 30,955 frames and 49,866 grounded triplets. TargetFusionNet is then proposed as a target-aware architecture that integrates weak anatomical priors into an instance segmentation framework. Strong instance supervision substantially improves triplet grounding, while TargetFusionNet increases mask-based complete-triplet AP from 12.23 to 13.47 relative to the strongly supervised Mask2Former-Triplet baseline.

Finally, this thesis formulates surgical triplet segmentation as specialist instrument grounding followed by interaction prediction with a multimodal large language model. Direct supervised fine-tuning establishes a viable second-stage baseline, while sequential dialogue prediction and three-frame temporal context improve complete-triplet prediction when instrument instances are provided. The complete system demonstrates that this two-stage formulation remains effective with predicted instances, achieving substantially stronger component scores and comparable complete-triplet AP to TargetFusionNet (13.52 versus 13.47).

Collectively, this work moves surgical action understanding beyond frame-level recognition towards grounded, interpretable modelling of instrument–tissue interactions. These contributions provide foundations for future context-aware surgical assistance, workflow analysis, and intelligent computer-assisted intervention systems.

Acknowledgments

I would first like to express my sincere gratitude to my supervisors, Professor Tom Vercauteren and Professor Miaojing Shi, for their guidance, encouragement, and support throughout my doctoral studies. Their insight and advice have shaped both this thesis and my development as a researcher, and their belief in me sustained me even at times when I did not believe in myself.

I am deeply grateful to Charlie Budd and Meng Wei for being such dependable pillars throughout this journey. Their friendship, practical help, and willingness to listen and offer support made the difficult periods easier and the successful moments more meaningful.

Finally, I would like to thank my parents and my siblings, Tolu and Tomi, for their unwavering love, patience, and support. Their encouragement has sustained me throughout my education and made the completion of this thesis possible.

List of Publications

The following publications and activities arose from work undertaken during the doctoral research period.

Publications forming this thesis

Peer-reviewed journal articles

1. **Oluwatosin Alabi**, Tom Vercauteren, and Miaojing Shi. *Multitask Learning in Minimally Invasive Surgical Vision: A Review.* Medical Image Analysis, 101:103480, 2025.

2. **Oluwatosin Alabi**, Ko Ko Zayar Toe, Zijian Zhou, Charlie Budd, Nicholas Raison, Miaojing Shi, and Tom Vercauteren. *CholecInstanceSeg: A Tool Instance Segmentation Dataset for Laparoscopic Surgery.* Scientific Data, 12:825, 2025.

3. **Oluwatosin Alabi**, Meng Wei, Charlie Budd, Tom Vercauteren, and Miaojing Shi. *Grounding Surgical Action Triplets with Instrument Instance Segmentation: A Dataset and Target-Aware Fusion Approach.* International Journal of Computer Assisted Radiology and Surgery, 21(5):901–908, 2026.

Manuscript in preparation

1. **Oluwatosin Alabi**. *Multimodal Large Language Models for Interaction Prediction in Two-Stage Surgical Triplet Segmentation.* Manuscript in preparation.

Other publications during doctoral research

For challenge reports with exceptionally large consortium author lists, authorship is abbreviated below; the full author lists are available in the corresponding publisher records.

Peer-reviewed journal and conference articles

1. Tobias Rueckert et al. (including **Oluwatosin Alabi**). *Comparative Validation of Surgical Phase Recognition, Instrument Keypoint Estimation, and Instrument Instance Segmentation in Endoscopy: Results of the PhaKIR 2024 Challenge*. Medical Image Analysis, 109:103945, 2026.

2. Adriana Namour, **Oluwatosin Alabi**, Minghan Zhao, Charalampos Komninos, Tom Vercauteren, Lyndon da Cruz, Sebastien Ourselin, and Christos Bergeles. *Semi-Automated Retinal Microsurgery Video Annotation with SAM2: Comparative Analysis of Prompt Strategies*. International Workshop on Ophthalmic Medical Image Analysis, Lecture Notes in Computer Science, 16209:95–104, 2025.

3. Zijian Zhou, **Oluwatosin Alabi**, Meng Wei, Tom Vercauteren, and Miaojing Shi. *Text Promptable Surgical Instrument Segmentation with Vision–Language Models*. Advances in Neural Information Processing Systems, 36:28611–28623, 2023.

Preprints

1. Meng Wei, Charlie Budd, **Oluwatosin Alabi**, Miaojing Shi, and Tom Vercauteren. *SurgPIS: Surgical-Instrument-Level Instances and Part-Level Semantics for Weakly-Supervised Part-Aware Instance Segmentation*. arXiv preprint arXiv:2507.19592, 2025.

2. Miaojing Shi, Tianyu Cen, Zijie Yue, Meng Wei, **Oluwatosin Alabi**, and Tom Vercauteren. *Multimodal Large Language Model Driven Radiology Report Generation with Clinical Knowledge Enhancement*. arXiv preprint arXiv:2403.06728, 2024.

3. Aneeq Zia et al. (including **Oluwatosin Alabi**). *Intuitive Surgical SurgToolLoc and SurgVU Challenges Results: 2022–2025*. arXiv preprint arXiv:2305.07152, first released 2023 and revised 2026.

Relationship to thesis chapters

Chapter 2 draws on the review article, Chapter 3 draws on the CholecInstanceSeg article, and Chapter 4 draws on the accepted article introducing CholecTriplet-Seg and TargetFusionNet. Each has been integrated and contextualised as a thesis chapter while retaining the corresponding publication's scientific content. The manuscript in preparation forms the scientific basis of Chapter 5.

Academic service and research engagement

- Co-organiser of the *Large Models Meet Surgical Data Science* (LM-SURG) workshop at the 2026 IEEE International Symposium on Biomedical Imaging (ISBI), London, UK.
- Co-initiated a surgical artificial intelligence collaboration between King's College London and Obafemi Awolowo University Teaching Hospitals Complex (OAUTHC), Nigeria, with Professor Adewale Adisa and Tom Vercauteren, and received support from the King's Global Engagement Partnership Fund to establish a pilot open dataset of Nigerian laparoscopic cholecystectomy videos for surgical safety and scene-understanding research.
- Helped establish the collaboration's research infrastructure at OAUTHC by deploying secure local video storage and de-identification workflows, training the surgical team in video-data management, organising a growing multi-procedure laparoscopic video collection, and developing clinical annotation protocols with local collaborators.
- Contributed to the organisation of the 2025 conference of the Association of Gynaecological Endoscopic Surgeons of Nigeria (AGES), Lagos, Nigeria.

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Chapter 1

Introduction

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1.1 Motivation

Minimally invasive surgery (MIS) has transformed surgical care by reducing tissue trauma, blood loss, post-operative pain, recovery time, and complication rates compared with open surgery [Jaffray 2005]. These benefits arise partly because the operative field is accessed through small incisions and viewed through an endoscope or laparoscope. However, the same visual mediation that makes MIS less invasive also makes the operative scene difficult to interpret computationally. The camera view is narrow, indirect, and continuously changing, while smoke, blood, specular reflections, motion blur, occlusion, tissue deformation, and variable illumination can obscure important visual evidence.

Computer-assisted intervention seeks to extract useful information from these complex visual streams. Potential applications include surgical workflow analysis, skill assessment, context-aware assistance, robotic support, and automatic documentation. All of these applications depend on a fundamental capability: a computational system must represent not only which objects are visible, but also what is happening between them.

Substantial progress has been made on individual surgical vision tasks. Models can recognise instruments, segment their visible regions, identify operative phases, and classify surgical activities. Nevertheless, surgical scene understanding requires more than a collection of independent labels. This thesis addresses the relational understanding of surgical activity by progressing from instrument perception to spatially grounded instrument–tissue interaction prediction.

1.2 From Recognition to Grounded Interaction Understanding

Early work in surgical video analysis commonly separated perceptual and workflow tasks. Instrument detection and segmentation described the visible tools, while phase and action recognition described the progression or content of an operation. These tasks provide complementary information, but none alone gives a complete account of an instrument–tissue interaction. Tool recognition does not explain how a tool is being used; action recognition without spatial evidence does not identify the responsible object; and anatomical recognition without an associated action does not describe the event taking place.

Surgical action triplets provide a compact representation that connects these elements. An interaction is expressed as the $\langle \texttt{instrument, verb, target} \rangle$ tuple, such as <Grasper, Retract, Gallbladder> or <Hook, Dissect, Gallbladder>. This formulation describes both an action and its anatomical context, and therefore captures finer-grained surgical meaning than instrument presence or workflow phase alone.

The CholecT series of datasets, particularly CholecT50, has been central to the development of surgical action triplet recognition [Nwoye 2022b]. TripNet and Rendezvous established influential modelling approaches for learning relationships between instruments, verbs, and targets [Nwoye 2020; Nwoye 2022b]. Their associated benchmarks enabled systematic comparison on fine-grained activity recognition in laparoscopic cholecystectomy. However, progress in recognising triplet labels did not by itself resolve where each interaction occurred.

Most surgical action triplet methods have historically treated the task as frame-level recognition. A model predicts that a triplet is present somewhere in an image, but does not necessarily associate it with a particular visible instrument. This distinction becomes important when a frame contains several tools, especially multiple instances of the same instrument class. If two graspers are visible, a frame-level annotation may indicate that a grasper retracts the liver and a grasper grasps the gallbladder, yet it does not explicitly identify which physical instance performs either interaction.

Spatial awareness therefore requires more than attaching a coarse location to a frame-level label. It requires an instance representation that distinguishes individual instruments and preserves the association between each instance and its predicted verb and target. Instance segmentation is well suited to this role because it separates overlapping or repeated tools while retaining their visible spatial extent. At the outset of this research, however, large-scale instance-level annotations for instruments in routine human laparoscopic surgery were scarce. This limited both the development of specialist instance segmentation models and the construction of strongly grounded interaction benchmarks.

This thesis connects these lines of research by progressing from instrument instance segmentation to grounded triplet prediction and, finally, interaction prediction for individually grounded instruments.

1.3 Research Gap

The literature review in Chapter 2 examines multitask learning as one important framework for surgical scene understanding. Its broader finding is that surgical understanding depends on multiple related perceptual, temporal, and semantic outputs. Multitask learning can exploit relationships between such outputs, but it is a means rather than the central subject of this thesis. The more fundamental question is whether the outputs needed for fine-grained interaction understanding are defined, annotated, and spatially connected in the first place.

The review identifies a progression from coarse workflow recognition towards increasingly fine-grained activity descriptions. However, surgical action triplets remained predominantly frame-level, while instrument instance segmentation had received substantially less attention than presence recognition, detection, or semantic segmentation. The field therefore lacked the combined supervision needed to connect semantically rich interactions to their responsible instrument instances.

These limitations define two connected gaps. First, a sufficiently large and diverse dataset of instrument instances is needed to support reliable instance segmentation in laparoscopic video. Second, action-triplet annotations must be aligned with those instances to establish a benchmark for grounded interaction understanding. Triplet labels without instance masks describe the interaction but leave its actor spatially ambiguous. Instance masks without verb and target labels localise the actor but do not describe its surgical role. Bringing these forms of supervision together is consequently a central contribution of this thesis.

Grounding also makes the prediction problem more demanding. A complete grounded triplet is correct only when the instrument instance is localised and its instrument, verb, and target components are all correctly identified. An error in any component can invalidate the complete prediction. Component-level results are therefore necessary for interpreting where performance is gained or lost, while complete-triplet performance measures whether the full relationship has been recovered. This conjunctive structure is particularly challenging for anatomical targets, which may be partially visible, visually similar, deformable, or defined through surgical context rather than local appearance alone.

The thesis addresses this target-prediction challenge through two complementary modelling directions. The first incorporates weak anatomical context into a strongly supervised instance segmentation architecture. The second separates specialist instrument grounding from semantic interaction prediction, allowing a multimodal large language model (MLLM) to predict ⟨`verb, target`⟩ relationships for already grounded instruments. Together, these approaches investigate how spatial supervision, anatomical information, task factorisation, and temporal context can contribute to grounded surgical interaction understanding.

1.4 Thesis Aim and Research Questions

The aim of this thesis is to advance surgical scene understanding in minimally invasive surgery by developing datasets, benchmarks, and models for spatially grounded interpretation of instrument–tissue interactions in laparoscopic video.

This aim is addressed through the following research questions:

1. How has surgical scene understanding progressed from individual perceptual tasks towards fine-grained interaction recognition, and what limitations prevent surgical action triplets from being spatially grounded to individual instrument instances?

2. Can a large-scale instrument instance segmentation dataset be constructed for laparoscopic cholecystectomy by extending existing public surgical datasets?

3. Can surgical action triplets be grounded to instrument instance masks, enabling spatially interpretable ⟨`instrument, verb, target`⟩ predictions?

4. Can weak anatomical priors improve target prediction and full triplet grounding when integrated into an instance segmentation architecture?

5. Can an MLLM serve as a viable second-stage interaction predictor for grounded instrument instances, and how do alternative task formulations and sources of context affect its predictions?

These questions form a cumulative programme of work, progressing from the identification of the research gap to instance-level spatial supervision, grounded triplet prediction, target-aware modelling, and two-stage multimodal interaction prediction.

1.5 Contributions

The principal contributions of this thesis are summarised below.

1.5.1 A Review-Based Framing of Surgical Scene Understanding

The thesis synthesises the literature on multitask learning in MIS and uses it to examine the wider development of surgical scene understanding. The review organises the field across perceptual, workflow, and activity-understanding tasks, and shows why multiple related outputs are needed to describe a surgical scene. It identifies the movement towards fine-grained action triplets, the limited spatial grounding available to triplet methods, and the relative scarcity of instance segmentation datasets. This analysis motivates the thesis's focus on connecting semantic interaction recognition with instance-level spatial evidence.

1.5.2 CholecInstanceSeg

The thesis introduces CholecInstanceSeg, a large-scale open-access dataset for surgical instrument instance segmentation in laparoscopic cholecystectomy. It contains 41,933 annotated frames from 85 procedures and 64,483 instrument instances. The dataset provides semantic classes and instance identifiers, allowing individual instruments to be distinguished in scenes containing multiple or overlapping tools.

CholecInstanceSeg addresses the shortage of instance-level annotations in routine human laparoscopic surgery and supports the development and evaluation of specialist instrument segmentation models. More importantly for the thesis narrative, it provides the spatial substrate upon which later interaction labels can be grounded.

1.5.3 CholecTriplet-Seg

Building on CholecInstanceSeg and CholecT50, the thesis introduces CholecTriplet-Seg, a dataset and benchmark for spatially grounded surgical action triplet understanding. It aligns instrument instance masks with verb and anatomical target labels across 30,955 frames and 49,866 grounded triplets from 50 laparoscopic cholecystectomy videos.

This contribution changes the triplet task from asking which interactions occur within a frame to asking which instrument instance performs each interaction. It thereby enables direct evaluation of the spatial association between an instrument and its surgical role, while retaining component-level and complete-triplet measures for diagnosing model behaviour.

1.5.4 TargetFusionNet

The thesis proposes TargetFusionNet, a target-aware architecture for grounded surgical action triplet prediction. TargetFusionNet extends a strongly supervised instance segmentation framework with gated fusion of weak anatomical priors. The priors are treated as coarse contextual cues rather than ground-truth anatomical labels, allowing their contribution to target and triplet prediction to be assessed within the grounded benchmark.

Experiments on CholecTriplet-Seg demonstrate the importance of strong instance supervision and show that target-aware fusion improves complete-triplet prediction over the evaluated weakly grounded and strongly supervised baselines. They also establish that target prediction remains the principal bottleneck: improved instrument localisation and component predictions do not automatically yield proportionate gains in complete-triplet performance.

1.5.5 Two-Stage Multimodal Interaction Prediction

Finally, the thesis formulates surgical triplet segmentation as a two-stage problem. A specialist instance segmentation model first grounds the visible instruments, after which an MLLM predicts the corresponding $\langle\texttt{verb, target}\rangle$ interactions. Direct supervised fine-tuning establishes a viable second-stage baseline, while sequential dialogue prediction and short-range temporal context provide promising alternative formulations. Anatomical context, ontology guidance, longer temporal input, and distilled rationale supervision are also examined to determine whether additional structure or context consistently improves interaction prediction.

The complete pipeline is evaluated using predicted instrument instances, rather than only idealised ground-truth grounding. The results establish the viability of specialised grounding followed by multimodal interaction prediction and expose the effect of first-stage errors on the complete system. This chapter represents a forward-looking continuation of the dataset and grounded-modelling contributions, rather than replacing their central role in the thesis.

1.6 Thesis Scope

The thesis focuses on laparoscopic cholecystectomy, for which established public datasets make it possible to develop a coherent sequence of instance-segmentation, triplet-grounding, and interaction-prediction studies. The shared procedural setting allows the contributions to build upon Cholec80, CholecT50, and CholecSeg8k while maintaining consistent definitions of instruments, actions, and anatomical targets.

The thesis does not attempt to solve every aspect of surgical scene understanding. It does not provide exhaustive anatomical segmentation, model the complete temporal workflow of an operation, or validate a clinical decision-support system prospectively. Its focus is the more specific problem of connecting instrument instances to their actions and targets. Cholecystectomy provides the principal test bed; evaluating how well the formulation transfers to other procedures remains an important next step.

Within this scope, the thesis spans dataset construction, benchmark definition, specialist vision models, and multimodal modelling. These elements are unified by spatial grounding rather than by a commitment to any single learning paradigm. Multitask learning, anatomical priors, and multimodal language modelling are used where they help address particular parts of the grounded interaction problem.

1.7 Thesis Structure

The remainder of this thesis is organised as follows.

Chapter 2 reviews multitask learning in surgical scene understanding and examines the progression from individual perceptual tasks towards fine-grained activity recognition. It identifies the absence of instance-level spatial grounding as a key limitation of existing surgical action triplet formulations.

Chapter 3 presents CholecInstanceSeg, the spatial foundation of the thesis. It describes the data sources, annotation protocol, dataset construction, technical validation, and limitations of a large-scale instrument instance segmentation dataset for laparoscopic cholecystectomy.

Chapter 4 introduces CholecTriplet-Seg and TargetFusionNet. It defines triplet segmentation as an instance-grounded extension of surgical action triplet recognition, presents the aligned dataset and evaluation protocol, and assesses target-aware fusion for grounded prediction.

Chapter 5 presents a two-stage formulation comprising specialist instrument grounding followed by MLLM-based interaction prediction. It establishes direct supervised fine-tuning, evaluates several task formulations and contextual adaptations, and assesses the complete pipeline under predicted grounding.

Chapter 6 synthesises the thesis findings across the review, datasets, benchmarks, and models. It discusses the difficulty and evaluation of grounded triplet prediction, identifies the principal limitations of the work, and outlines future research towards robust temporal modelling and evaluation across surgical procedures.

CHAPTER 2

Multitask Learning in Minimally Invasive Surgical Vision: A Review

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This chapter draws on the accepted article *Multitask Learning in Minimally Invasive Surgical Vision: A Review*, published in *Medical Image Analysis* (2025). Full publication details are provided in the List of Publications.

2.1 Introduction

This chapter establishes the literature foundation for the thesis by examining how surgical vision has progressed from individual perceptual tasks towards richer descriptions of operative activity. Multitask learning provides one framework for combining related outputs, but the wider significance of the review is that surgical scene understanding requires complementary information about instruments, anatomy, workflow, and actions. Although the research programme began with this broad multitask perspective, the subsequent thesis focuses more specifically on spatial grounding and instrument–tissue interaction modelling. Within that narrative, the review exposes the gap between recognising scene elements at frame level and spatially associating an interaction with the individual instrument responsible for it.

Minimally invasive surgeries (MIS) have become increasingly popular due to their benefits, such as reduced blood loss, less pain, faster recovery times, and fewer post-surgical complications [Jaffray 2005]. However, MIS utilizes indirect vision with a limited field of view from the endoscope, making it challenging for surgeons to interpret events on the surgical scene accurately [Islam 2019a]. To overcome this, computer-aided intervention techniques that augment the information obtained through minimally invasive cameras have been proposed [Vercauteren 2020; Ward 2021].

Deep learning has demonstrated remarkable success in providing solutions to computer vision tasks for MIS, such as instrument classification [Wang 2017; Mishra 2017; Al Hajj 2018], instrument segmentation [Garcia-Peraza-Herrera 2017; Milletari 2018; Pakhomov 2020; Islam 2019a], and surgical scene depth estimation [Luo 2019; Xiao 2020; Wei 2022; Luo 2022]. Conventionally, separate models would be trained for these related tasks, which can be computationally impractical and inefficient. Therefore, there is a need for deep learning approaches that leverage information from different tasks to improve both performance and efficiency.

Multitask learning (MTL) is a machine learning approach that seeks to improve generalisation performance by leveraging domain-specific information from multiple related tasks, with shared parameters reducing overfitting, memory footprint and improving regularisation [Caruana 1997]. MTL has been successfully applied in various fields such as natural language processing [Collobert 2008], computer vision [Girshick 2015a], and robotics [Kalashnikov 2022]. In the context of MIS, MTL has primarily been utilized to enhance scene understanding. By simultaneously solving multiple tasks and leveraging knowledge from all of them, MTL offers an efficient approach for solving multiple tasks in MIS. To ensure the widespread adoption of computer-aided intervention solutions for MIS, it is crucial to develop systems that can comprehensively understand the surgical scene, rather than addressing individual tasks separately.

In this chapter, we provide a broad survey of multitask learning (MTL) for surgical scene understanding in minimally invasive surgeries (MIS). The review serves as a foundational reference for researchers and practitioners working at the intersection of multitask learning and surgical scene understanding.

We highlight a significant methodological gap between the use of multitask learning in the computer vision and MIS communities, noting that MIS research predominantly relies on basic MTL techniques such as hard parameter sharing and linear scalarization of the loss functions. In contrast, computer vision research has explored a wider array of MTL techniques, such as adaptive task balancing and optimization methods, flexible parameter sharing strategies, and data-efficient learning approaches to handle limited labeled data. This gap presents new opportunities for improved performance in surgical scene understanding tasks through advanced techniques and opens up novel research directions to address domain-specific challenges for combining tasks and utilizing advanced MTL techniques in MIS.

The remainder of this chapter is divided into five sections. Section 2.2 outlines the review methodology employed, including the scope of the review, search criteria, selection process used to identify relevant literature on multi-task learning (MTL) in minimally invasive surgeries (MIS), and a brief comparison to similar review papers. Section 2.3 provides a quick introduction to popular deep MTL techniques commonly employed in MTL research for vision, emphasizing their key features and concepts. Section 2.4 reviews how MTL has been applied to solve multiple tasks for surgical scene understanding, examining each work extensively and identifying the current trends in various research areas. Section 2.5, 'Public datasets for MTL in MIS', presents an overview of publicly available datasets that can be used to advance MTL research in MIS. Section 2.6 presents our insights on the current state of MTL research in MIS, discusses potential directions for future work, and concludes the review by summarising the main findings of this chapter.

2.2 Review methodology and related work

2.2.1 Scope of the review

This review focuses on the application of multi-task learning (MTL) in minimally invasive surgeries (MIS). Specifically, we limit the scope to methods that utilize videos and/or images obtained from MIS cameras to address multiple tasks, where

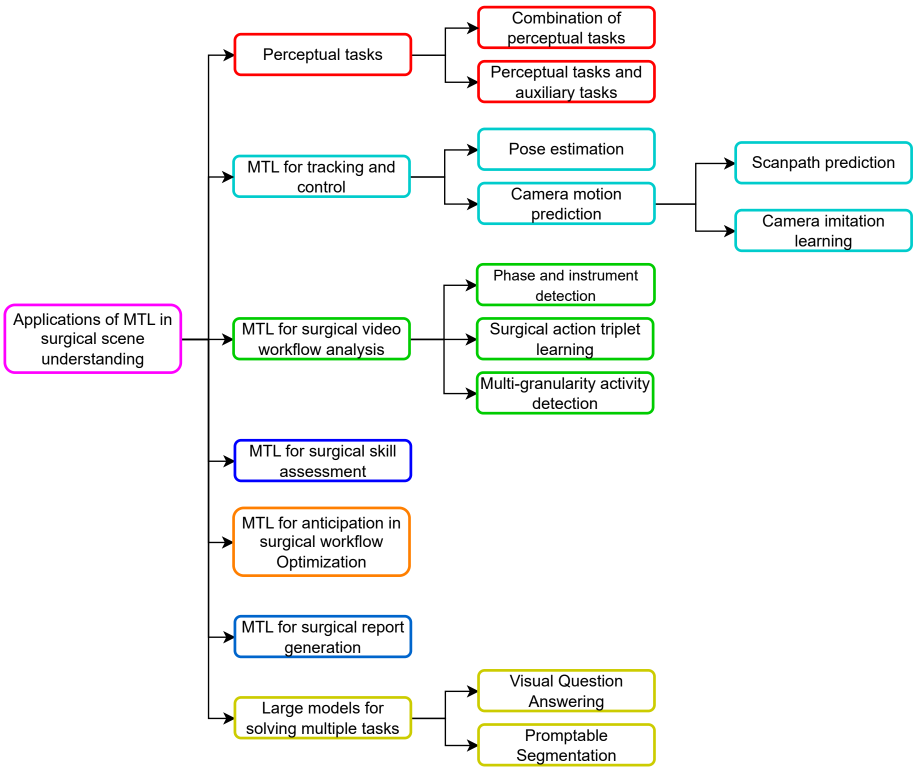

Figure 2.1: Overview of the application areas where multitask learning has been applied in surgical scene understanding.

each task yields meaningful and relevant outputs. Papers that primarily focus on one task, but include auxiliary tasks to guide the learning of the primary task, were also considered for inclusion. With the growing relevance of large models in medical research, we also review studies that leverage such models to address multiple tasks within the context of MIS. To ensure completeness and provide a comparative perspective, we discuss a few seminal multi-task learning papers from the general purpose computer vision and machine learning communities. These works introduce readers to deep multi-task learning techniques from the broader field, helping to highlight which techniques are employed or underutilized in surgical scene understanding.

Our review is structured as a narrative review, rather than a systematic one. This approach allows for greater flexibility in examining the literature, offering a more expansive view than that typically used in systematic reviews.

2.2.2 Search criteria

The works reviewed were identified through an initial search on Google Scholar using keywords "multitask learning surgical vision". Following this, we manually reviewed the references within the initially identified papers to discover additional relevant works. We also conducted author-specific searches to locate further research by key contributors in the field. Finally, thematic analysis was carried out on the studies that tend to have multiple tasks solved. Searches and selections were completed by July 2023.

2.2.3 Selection criteria

In accordance with the scope of this review, we only included publications that presented methods validated on minimally invasive surgical imaging data. Hence, we excluded works that focus on robot kinematics information, and we excluded works that focused on data from open surgeries or external operating room camera setups. MIS typically employs a single, small camera inserted through an incision, offering a narrow field of view on the surgical site to support precision in confined spaces. In contrast, open surgeries and external camera setups involve wide-field imaging, with good natural lighting and fewer distortions. The scene showed by open surgeries and external camera setups have limited visual and semantic similarity to those seen in MIS. These differences make the challenges and research questions in MIS distinct from those in other types of surgery, which is why they were excluded from the review.

Following the search and selection process, 47 studies were chosen for inclusion in this review.

2.2.4 Analysis of selected papers

Fig. 2.1 provides an overview of the selected studies, categorized by their application areas within surgical scene understanding where multitask learning (MTL) has been applied. In perceptual tasks, 11 studies focus on extracting visual information, such as segmentation and depth estimation, and combining tasks to improve accuracy and efficiency. MTL has also been applied to tracking and control of surgical instruments and cameras, with 7 studies addressing this. Surgical video workflow analysis, including phase recognition and action triplet learning, is explored in 13 studies, while 5 studies cover anticipation tasks, such as predicting upcoming instruments, phases and time to surgery end. 2 studies apply MTL to surgical skill assessment by predicting skill-related metrics, and 2 studies focus on surgical report generation using MTL. Finally, 7 studies investigate large models using prompts to solve multiple tasks.

2.2.5 Related reviews

To the best of our knowledge, our review is the first, to focus specifically on multitask learning (MTL) techniques within the context of minimally invasive surgeries (MIS). While existing reviews, such as Vandenhende et al. [2022] and Y. Zhao et al. [2023], provide valuable insights into MTL applications in general computer vision and broader medical imaging, respectively, they do not offer a targeted analysis of the specific challenges and applications of MTL within the MIS domain.

In comparison to related reviews on deep learning in MIS, such as Rivas-Blanco et al. [2021] and Demir et al. [2023], our work delves deeply into each study that applies MTL within MIS. We highlight the multitask parameter sharing approach and optimization method employed in each work, providing a level of detail not covered by these broader reviews. While Rivas-Blanco et al. [2021] and Demir et al. [2023] offer overviews of deep learning applications in MIS, including image analysis and surgical workflow recognition, they do not focus on works that utilize MTL specifically.

2.3 Common deep MTL methodologies in computer vision

This section presents a short introduction to widely used deep MTL techniques in the field of computer vision research. We present a short examination of these techniques and representative papers that showcase the utilization of each technique. By presenting the foundations and applications of MTL techniques, this section aims to equip readers with a basic understanding of the subject, setting the stage for the upcoming sections that analyze methods employed for solving multiple vision tasks in the context of MIS.

2.3.1 MTL concepts

There are different ways of categorizing MTL core concepts. For our purpose, we focus on four concepts that recurrently emerge in the MTL literature for MIS: parameter sharing, optimization and task-balancing, auxiliary objectives, and data-efficient approaches.

Parameter sharing and feature representation

The mainstream multitask paradigm assumes that tasks are related as the knowledge required for solving these tasks is connected. Following this assumption, features produced for the same input sample on multiple tasks which are related should also be related. Hence, common feature representation learnt for multiple tasks would have more informative and robust feature representations as they take into account multiple tasks [Zhang 2022]. Additionally, each task acts as a regularization for other tasks, smoothening out noise, as features which are utilized for multiple different tasks are less likely to overfit to training samples [Zhang 2022].

Historically, the methods for sharing feature representations in deep neural networks are classified into hard and soft parameter sharing [Vandenhende 2022]. A diagrammatic representation of soft and hard parameter sharing can be seen in Fig. 2.2. Hard parameter sharing refers to architectures where tasks are to be jointly learnt utilizing the same weights and biases for some layers. These layers are aptly called *shared layers*. The other layers which are not shared are called *task-specific* layers. Hard parameter sharing is commonly interpreted using an encoder-decoder architecture where the encoder is shared, and the decoder is task-specific. Some network architectures for hard parameter sharing also include methods to facilitate information transfer between different decoders [Xu 2018; Liu 2019; Zhang 2018]. If x_i represents input samples, h denotes hidden layers, and $y_{i,t}$ represent outputs for the i^{th} sample on the t task, $f_{sh}(.)$ represents shared layers and $f_{task}(.)$ represents the task-specific layers, then hard parameter sharing can be summarized as

$$h_i = f_{sh}(x_i) \quad y_{i,t} = f_{task}(h_i) \tag{2.1}$$

On the other hand, soft parameter sharing does not directly share layers for each task. Instead, a separate model is used for each task. Soft parameter sharing instead shares parameters by adjusting the weights and biases of different task models based on information from other task models, leading to model weights and biases, which are functions of representations from different tasks. If $M_t(.)$ is a model for task t and M_t produces features f_t, then soft parameter sharing can be written as

$$M_t = R(f_1, f_2 ... f_t) \tag{2.2}$$

where R is a feature-sharing mechanism which determines how features are shared.

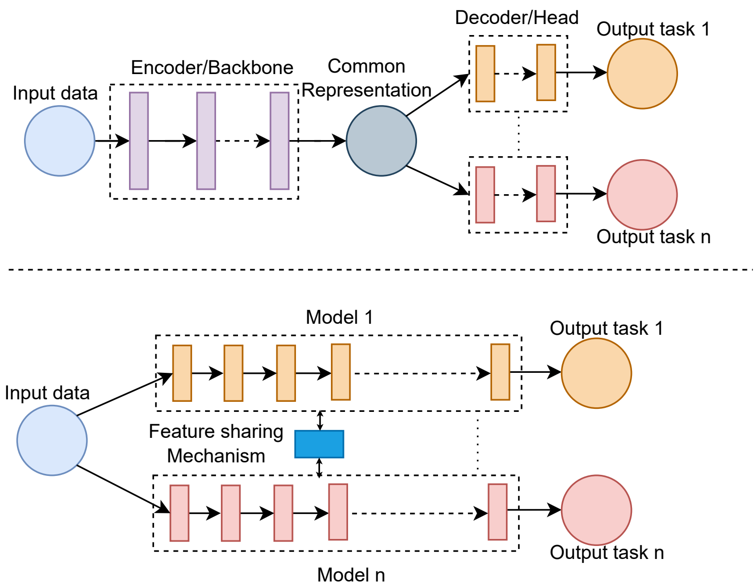

Figure 2.2: Top: hard parameter sharing in deep neural networks for multitask learning, featuring a shared encoder/backbone with a common representation and separate decoders or heads. Bottom: soft parameter sharing with separate models per task and specialized feature-sharing mechanism.

A prototypical example that uses hard parameter sharing is UberNet [Kokkinos 2017]. UberNet is proposed as a general-purpose computer vision model for multiple tasks. The network architecture is a shared multi-resolution VGG [Simonyan 2015] with a branch network from each layer of the VGG for each task. The output of each task branch is fused with branches for the same branch from other layers and finally fused for all resolutions to produce predictions for each task. Other examples of hard parameter-sharing architectures include [Xu 2018; Teichmann 2018; Heuer 2021; Liu 2019; Zhang 2018],

Cross-stitch network [Misra 2016] is a good example of soft parameter sharing. This paper is concerned with the design decision about when to split a network into task-specific networks and general/shared parameters. To solve this, the authors propose a unified splitting architecture, which is a type of soft parameter sharing architecture. Different separate networks are connected with the cross-stitch module across various layers of the separate networks. This is done by learning a linear combination of the input activation maps from both tasks. Other examples of soft parameter sharing architectures include [Gao 2019; Ruder 2019]

Optimization and task balancing

Designing networks with shared parameters and learning these parameters together raises an important question: What is the right way to optimise both tasks jointly?

Fig. 2.3 provides a brief overview of the various methods of optimization methods discussed in this section. LibMTL[Lin 2023] is a python repository with adaptable code implementation for the common multitask architectures and optimization strategies discussed here.

For hard parameter sharing with shared encoders and task-specific decoders, as shown in Fig. 2.2, we can write the baseline equation for the multitask loss while omitting batching considerations for simplicity, as Equation (2.3). The gradient descent equation for the shared parameters can be written as Equation (2.4), and the gradient descent equation for the task-specific parameters as Equation (2.5):

$$L_{total} = \sum_{i=0}^{N} L_i \left(\theta_t, \theta_s, X, Y\right) \tag{2.3}$$

$$\theta_s = \theta_s - \alpha \sum_{i=0}^{N\cdot} \frac{\partial L_i}{\partial \theta_s} \left(\theta_t, \theta_s, X, Y\right) \tag{2.4}$$

$$\theta_{t(a)} = \theta_{t(a)} - \alpha \frac{\partial L_{i(a)}}{\partial \theta_{t(a)}} \left(\theta_t, X, Y\right) \tag{2.5}$$

where N is the number of tasks, X is the input, Y is the ground-truth, α is the learning rate, L_i is the loss for the ith task, a refers to the current task been considered, θ_s denotes the shared parameters while θ_t the task specific parameters.

From Equation (2.4), we can deduce that the network weights in shared layers are affected by multiple supervision signals. Hence, the best way of balancing different task signals in the shared layers has been explored in several works [Sener 2018; Cipolla 2018; Guo 2018].

Equation (2.3) shows the multitask loss for a network called linear scalarization - when the loss functions for N tasks are simply added together. Linear scalarization is the most utilised task-balancing method [Hu 2023]. It involves assigning a scalar weight to each task and optimising the scaled addition as can be seen in (2.3), (2.7), and (2.8)

$$L_{total} = \sum_{i=0}^{N} w_i L_i \left(\theta_t, \theta_s, X, Y\right) \tag{2.6}$$

$$\theta_s = \theta_s - \alpha \sum_{i=0}^{N\cdot} w_i \frac{\partial L_i}{\partial \theta_s} \left(\theta_t, \theta_s, X, Y\right) \tag{2.7}$$

$$\theta_{t(a)} = \theta_{t(a)} - \alpha w_{i(a)} \frac{\partial L_{i(a)}}{\partial \theta_{t(a)}} \left(\theta_t, \theta_s, X, Y\right) \tag{2.8}$$

where w_i is the weight assigned to a particular task, and all other notations remain the same. Despite its simplicity, linear scalarization works surprisingly well, especially when combined with techniques like grid search [Xin 2022].

Another example of a task balancing method is the automatic dynamic tuning of the weight assigned to each task at different points during training. Cipolla et al. [2018] propose a method to automatically weight losses during multitask optimization using the uncertainty in the predictions of each task. The authors reformulate the linear scalarization loss function to include a homoscedastic loss uncertainty term for each task, an uncertainty term independent of the inputs but depends only on the task problem. This method ascribes an uncertainty scalar for each task, and depending on the value of this uncertainty scalar, the gradient magnitudes of each task are weighted. Instead of using uncertainty to automatically determine loss weights, Z. Chen et al. [2018] utilize each task's gradient values to estimate the rate of convergence of each task and corrects this rate of convergence to be equal by adjusting the weights associated with each task in linear scalarization formulation. Guo et al. [2018] introduce the notion of dynamic task prioritization to prioritize more difficult tasks. Guo et al. [2018] observe that imbalance in task difficulty can result in prioritization of the easy tasks during optimization and propose to measure task difficulty as inversely proportional to the task performance. Task performance is measured by key performance metrics (KPIs) like accuracy for classification and IoU for segmentation. Dynamic task prioritization automatically prioritizes more difficult tasks by adaptively adjusting the weight of each task's loss objective based on task KPIs.

A different task-balancing idea is the direct adjustment of the gradients for each task calculated during back-propagation instead of tuning weights to give priority to different tasks. These methods claim to be able to tackle problems with multitask optimization, such as negative interference. Yu et al. [2020] propose the PCGrad method, which uses cosine similarity to check if the directions of gradients

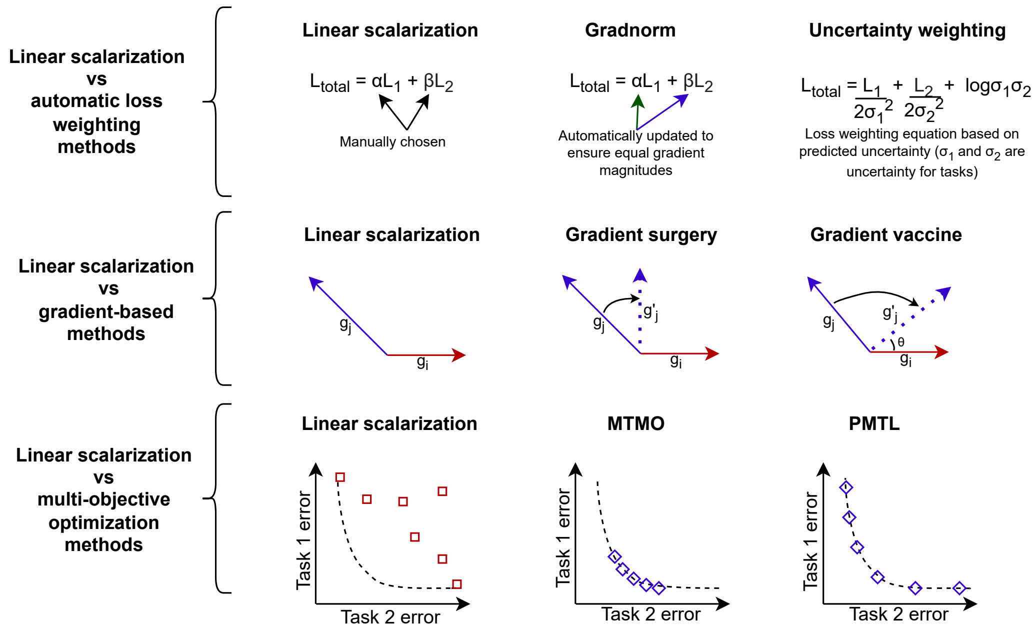

Figure 2.3: An overview of optimization techniques for multitask learning as discussed in Section 2.3.1. The classical method, linear scalarization, involves manually weighting the loss functions of all tasks. The first row illustrates linear scalarization against automatic loss weighting methods that dynamically adjust weights during training, such as updating weights to ensure gradient magnitude consistency [Chen 2018] and defining a loss weighting equation based on predicted uncertainty of each task [Cipolla 2018]. The second row illustrates gradient-based approaches in comparison to linear scalarization. Gradient-based methods directly modify gradients to mitigate negative transfer, achieved by projecting conflicting gradients to the normal plane of the gradient of another task [Yu 2020a] or ensuring gradients are at a target angle to each other [Wang 2021]. The third row illustrates linear scalarization and multi-objective optimization techniques. MTMO [Sener 2018] ensures that solutions are on the Pareto front, while PTML [Lin 2019] enables the selection of Pareto front solutions, favouring specific tasks.

for tasks trained for multitask learning are conflicting, which can lead to negative interference. If the gradients are conflicting, the PCGrad method projects the task gradients of a task to the normal plane of the other task to remove the conflict and optimizes with this projection. Further investigation into conflicting gradients by Z. Wang et al. [2021] reveals that the occurrence of conflicting gradients during training as measured by cosine similarity is very sparse. Hence, Z. Wang et al. [2021] design a more proactive method for ensuring that there are no conflicting gradients called "gradient vaccine". Z. Wang et al. [2021] note that the adjusted gradients using PCGrad project gradients to normals, making a cosine similarity of 0 the target in PCGrad. Gradient vaccine sets a targeted cosine similarity, which is greater than 0. By selecting a targeted cosine similarity, gradient vaccine ensures that both conflicting and non-conflicting gradients that are still very dissimilar can be projected to the target cosine similarity plane, ensuring frequent gradient update.

Another key task-balancing idea is the interpretation of multitask optimization as multi-objective optimization, which requires a Pareto optimal solution [Sener 2018; Lin 2019]. A Pareto optimal solution is reached when one objective function cannot be improved without sacrificing another objective. Pareto optimal solutions lie on a Pareto front, which is a set of optimal solutions in the space of objective functions in multi-objective optimization problems. MTMO [Sener 2018] and PTML [Lin 2019] are examples of papers that try to figure out the best way to optimize multiple tasks together with the Pareto principle. Unlike linear scalarization, which combines multiple loss functions via a system of fixed weights and optimizes the multiple losses as a single loss function, Pareto optimization is a multi-objective optimization problem. Sener et al. [2018] establishes the foundation for a Pareto optimal solution in MTL by mathematically deriving an upper bound for the multi-objective optimization problem. This upper bound is then optimized with a multi-gradient descent algorithm. PTML pushes this idea further by generalizing the mathematical formulations from Sener et al. [2018]. It does so by relaxing the Pareto optimal condition and allows for solutions that are as Pareto optimal as possible while still favouring certain tasks.

Auxiliary objectives

In MTL, auxiliary tasks are additional tasks that are learned simultaneously with the primary task to improve the overall performance of a model on the primary task. The idea behind auxiliary tasks is that they can provide helpful information or constraints to help the model learn a rich and robust representation of the input data, which in turn benefits the primary task [Liebel 2018]. Z. Zhang et al. [2014]

proffer a method that optimizes a main task in their application, facial landmark detection, together with auxiliary tasks like head pose estimation, facial expression recognition and age estimation. The authors propose a deep network to extract shared features with different classifier heads for each task.

Data efficient approaches

Building large-scale computer vision datasets is resource intensive [Liao 2021]. This is even more noticeable for networks that use the MTL paradigm, which requires labelling for multiple different tasks per sample. There are few publically available datasets designed specifically for working with multiple tasks [Zamir 2018; Yu 2020b]. Hence, problems such as limited annotations for multiple tasks and the availability of different task labels in different datasets, are often faced by researchers face when they want to utilize MTL. Consequently, there have been research works to develop MTL techniques to solve these challenges.

A method by which multitask learning can be used to tackle issues with limited annotations for multiple tasks is with better representation learning via multitask learning. Representation learning involves learning the representation of the data that makes it easier to extract useful information for downstream tasks that require a few labelled data [Bengio 2013]. Learning representations that are task-agnostic over a range of tasks would be preferred to learning single-task representations as these representations would be more robust to noise [Nguyen 2022]. Self-supervised MTL methods for representation learning utilize self-supervised tasks which do not require new annotations and learn representations for multiple tasks together. These representations can then be utilized for downstream tasks with fewer amounts of data. An example of such work is from Doersch et al. [2017] who utilize four self-supervised vision tasks (relative position, colourization, the exemplar task, and motion segmentation) for pre-training instead of a single pretraining task. Doersch et al. [2017] demonstrate that the representations learnt using their method are competitive to ImageNet pretrained representations without the need for labelling. Ghiasi et al. [2021] presents Multi-Task Self-Training (MuST), which is a method to harness the knowledge from specialized teacher models for different tasks to create a large multitask dataset with pseudo-labels generated from these specialized teachers. The large pseudo-label dataset is then used to train a multi-task network. The authors show that the representation generated in this multitask network generalizes well to downstream tasks.

There are also examples in the literature for combining information in different datasets to perform MTL. W.-H. Li et al. [2022] proffer a method for combining multiple datasets collected on similar distributions but annotated for different tasks. The authors propose a method that leverages task relations between task pairs to supervise MTL task pairs jointly. They map a task pair to a supervised joint space to enable information sharing between two tasks. Then, they propose a supervised learning loss for the task with known labels and a consistency loss in the joint task space to train tasks with unknown labels. Dorent et al. [2021] tackles the problem of learning from domain-shifted datasets, each with single task-specific annotations. Specifically, for the brain tissue segmentation task, there are datasets with segmentation of brain structures in healthy brains, and there are datasets which segment pathologies like brain lesions and tumours, but there are no datasets with both types of labels. The authors derive an upper bound of the loss for the joint probability problem.

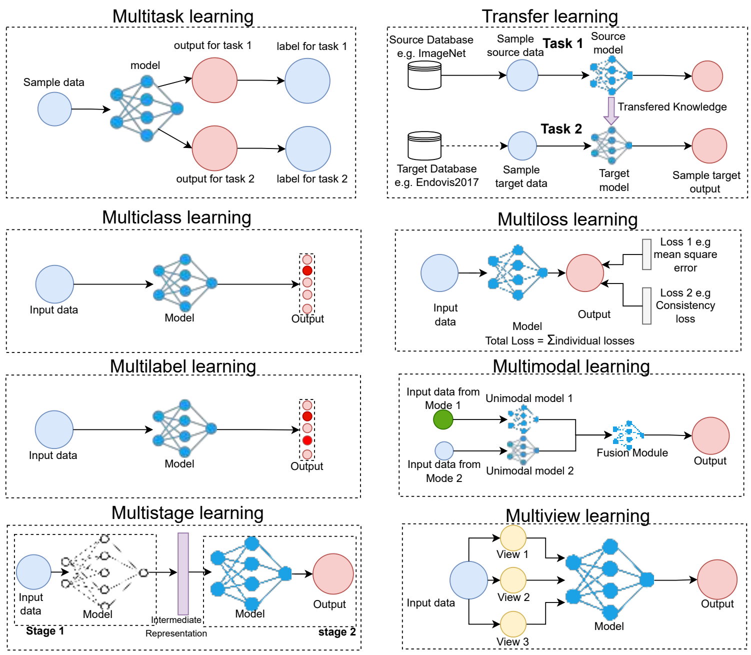

Figure 2.4: Diagram illustrating the key differences between multitask, transfer, multiclass, multiloss, multilabel, multimodal, multistage, and multiview learning.

2.3.2 MTL and other learning paradigms

MTL is a learning paradigm that trains a single model to perform multiple tasks simultaneously by sharing information across them, leading to better representations useful for all tasks.

The idea of using information from a different task or different outputs to improve representation is not unique to MTL and shares similarities with other learning paradigms, which can make it challenging to differentiate between them. Related learning paradigms include transfer learning [Oquab 2014],multiview learning [Andrew 2013], multiloss learning [Johnson 2016], multilabel learning [Wang 2016], multiclass learning [Krizhevsky 2012], multistage learning [Girshick 2015b], multimodal learning [Frome 2013].

A diagrammatic representation of the paradigms discussed in this section can be seen in Fig. 2.4. Despite their similarities, each paradigm has its unique features and objectives, and understanding their differences and similarities is essential for their proper application.

Transfer learning enhances target task performance using knowledge from a related source task, while MTL trains multiple tasks concurrently. *Multi-loss learning* employs multiple loss functions for neural network training. This approach finds use in both MTL and single-task learning. *Multilabel learning* refers to problems where single data instances can have multiple class labels. *Multiclass learning* classifies input data into one of several classes, whereas MTL concurrently optimizes multiple related tasks, sharing the same input data. *Multiview learning* optimizes multiple data representations (views) with the same output. *Multistage learning* processes inputs through multiple stages. Multitask networks can have single or multiple stages based on the design. *Multimodal learning* involves learning over diverse data modalities. Multimodal learning is usually done with a separate encoder for each different modality and a fusion module. Multitask learning can have inputs of different modalities.

2.4 Applications of MTL in surgical scene understanding

This section delves into the applications of the multitask paradigm in the context of surgical scene understanding, where it tackles the challenge of simultaneously learning multiple subtasks to enhance both efficiency and accuracy. Additionally, it explores the utility of auxiliary tasks to augment primary tasks in surgical scene understanding. In the subsequent subsections, we describe the deployment of MTL for surgical scene understanding across seven distinct categories: MTL for perceptual tasks, MTL for tracking and control, MTL for surgical workflow analysis, MTL for anticipation in surgical workflow optimization, MTL for surgical skill assessment, MTL for report generation, and large models for solving multiple tasks.

Fig. 2.1 shows an overview of the application areas where multitask learning has been applied in surgical scene understanding.

Table 2.1: Methods that utilize multitask learning for perceptual tasks. *HP - hard parameter sharing.

	Publication	Tasks	Optimization	Architecture	Speed(fps)
Combination of perceptual tasks	B. Huang et al. [2022]	binary tool segmentation unsupervised depth estimation	linear scalarization	shared encoder with multiple task branches (HP)	172
	Sanchez-Matilla et al. [2021]	tool classification tool segmentation tool detection	linear scalarization	shared encoder with multiple task branches (HP)	22.4
	Islam et al. [2020]	tool detection tool segmentation	sequential training	shared encoder with multiple task branches (HP)	18
	Baby et al. [2023]	tool classification tool detection tool segmentation	sequential training	modified Mask2former (HP)	-
	Z. Zhao et al. [2022]	tool classification tool detection tool segmentation	linear scalarization	modified DETR (HP)	23
	Das et al. [2023]	landmark detection landmark spatial relationship	linear scalarization	shared encoder with multiple task branches (HP)	10
	Psychogyios et al. [2022]	depth estimation tool segmentation	linear scalarization	shared encoder with multiple task branches (HP)	22
Perceptual tasks and auxiliary tasks	F. Qin et al. [2020]	tool segmentation contour detection	linear scalarization	shared encoder with multiple task branches (HP)	-
	Bhattarai et al. [2023]	tool segmentation HoG prediction	linear scalarization	shared encoder with main task branch and deep supervision auxiliary heads (HP)	-
	B. Huang et al. [2022]	depth estimation 3D left-right consistency	linear scalarization	multistage network (HP)	105
	J. Wang et al. [2022]	landmark detection landmark spatial relationship	linear scalarization	shared encoder with multiple task branches (HP)	37

2.4.1 MTL for perceptual tasks

In the context of this report, *perceptual tasks* refer to tasks such as image segmentation, object detection, motion estimation, and depth estimation. These tasks are focused on extracting essential visual information from images or videos, with an emphasis on revealing the spatial layout, motion, semantic, and depth relationships of objects within the scene. Perceptual tasks provide valuable information about spatial layout, motion, and depth relationships in the scene. The outcomes of perceptual tasks play a critical role in providing insights into the surgical scene, serving as the foundational elements for more advanced computer vision processes, including action recognition and object tracking. By enhancing the performance of perceptual tasks, researchers can better prepare input data in a way that significantly eases and enhances the accuracy of subsequent higher-level tasks. Table 2.1 summarizes the papers discussed in this subsection, highlighting the multiple tasks addressed, the optimization approaches adopted, the architectural strategies used for multitask learning, and the reported speeds for each study.

Combination of perceptual tasks

MTL for perceptual tasks in minimally invasive scenes seeks to identify task combinations that can improve problem-solving accuracy [Huang 2022a; Islam 2020c; Sanchez-Matilla 2021; Qin 2020a; Bhattarai 2023; Wang 2022a] or efficiency, taking into account factors such as inference speed [Sanchez-Matilla 2021; Islam 2020c], memory usage [Sanchez-Matilla 2021; Islam 2020c], and annotation requirements [Sanchez-Matilla 2021]. Furthermore, the existing body of literature demonstrates the presence of various distinct auxiliary tasks that can provide valuable guidance for optimizing perceptual tasks [Qin 2020a; Bhattarai 2023; Wang 2022a].

In their work, B. Huang et al. [2022] present an approach that concurrently tackles the depth estimation and binary tool segmentation tasks in laparoscopic images. The methodology employs a U-Net-like architecture with a shared encoder and two separate decoders, one dedicated to each task. Laparoscopic stereo images serve as the model's input. In this framework, the segmentation decoder follows a conventional U-Net decoder style [Ronneberger 2015]. On the other hand, the depth estimation decoder generates disparity maps using the left-right consistency unsupervised depth estimation method, as outlined in Godard et al. [2017]. Notably, the unsupervised depth estimation approach is advantageous for endoscopic surgical datasets, which often lack ground-truth depth labels. The results of this study indicate improvements in both instrument binary tool segmentation and unsupervised depth estimation tasks. This study suggests that binary tool segmentation and depth estimation could be effectively learned together within the MTL framework. However, additional datasets are required to generalize this hypothesis.

Sanchez-Matilla et al. [2021] introduce an efficient and memory-friendly approach to MTL, which simultaneously addresses classification, instrument-type segmentation, and bounding box detection. The authors recognize the challenges associated with obtaining segmentation labels and, as a result, devise a solution that supports weak supervision. This method harnesses the power of an EfficientNet backbone [Koonce 2021]. Unlike B. Huang et al. [2022], which advocates for distinct decoders for each task, the proposed model employs a single backbone with straightforward task heads for each objective. During training, the emphasis is placed on leveraging bounding box and classification labels, with only limited instrument segmentation labels. In cases where segmentation labels are unavailable, class activation maps are employed for guidance and supervision. The outcomes reported by the authors are notable, demonstrating that even with just one per cent of the segmentation labels, they achieve commendable results in the surgical tool segmentation task which indicates that multitask learning could be used to reduce annotation costs. However, the results were evaluated on a single dataset EndoVis2018 [Allan 2020], and more experiments are required to prove generalization.

In their work illustrated in Fig. 2.5, Islam et al. [2020] introduce a real-time network designed for solving instrument detection and instrument-type segmentation in robotic surgery, referred to as the Attention pruned MTL network (AP-MTL). AP-MTL adopts a U-Net-like architecture with shared encoders and task-specific decoders. The decoder for object detection follows the design of the multi-box single shot detector (SSD) [Liu 2016], while the segmentation network is a variant of the standard U-Net segmentation decoder. It incorporates custom squeeze excitation modules [Hu 2018] known as the spatial channel squeeze excitation module (scSE). Additionally, Islam et al. [2020] presents a custom optimization method known as Asynchronous Task-aware Optimization (ATO). It is a form of sequential training that optimizes different parts of the proposed network separately to ensure that correlated tasks converge at the same point, even if they do so at varying speeds. Following the optimization step, ATO introduces regularization to promote a more generalized network and ensures the smooth flow of gradients. To implement the ATO algorithm, gradients for each task are calculated and optimized separately, simulating an *attach, optimize, and detach* mechanism for the task decoders. This study stands out by proposing a custom sequential training MTL optimization technique specifically tailored to their use case.

In their work, Baby et al. [2023] introduce a modification to the standard Mask-RCNN and Mask2former framework [He 2017; Cheng 2022] for addressing instance segmentation in surgical instruments. The authors identify an issue where the classification of predicted masks often produces inaccurate results, despite the bounding box detection and mask segmentation tasks generally being performed correctly. To address this challenge, the authors propose the incorporation of a dedicated classification module to decouple the classification task from the region proposal and mask prediction processes. This new module takes as input multiscale features extracted from the feature extractor and the predicted instance masks. These predicted instance masks are employed for multiscale masking, and the results at different scales are subsequently merged and passed to a classification head. Sequential training is used where each stage of the proposed model is optimized separately with a training scheme. The authors report improvements compared to standard segmentation models and popular instrument segmentation models. However, with its three-stage approach, and extra parameters included compared to a standard Mask-RCNN and Mask2former, the inference speed and memory requirements would be increased.

Z. Zhao et al. [2022] propose TraSeTR, a method for leveraging tracking cues to enhance surgical instrument segmentation. This approach employs a transformer-based architecture that bears similarities to the DETR architecture [Carion 2020] for predicting instrument class, bounding box, and binary segmentation classes. Z. Zhao et al. [2022] improve on the standard DETR architecture by utilizing queries from prior frames for the instrument detection in the current frame. These queries are encoded with previous instrument information and serve as a form of tracking signal. These queries are applied to the transformer decoder, and identity matching between the previous queries and current queries are used as tracking cues. In addition, the authors also apply a contrastive query learning strategy to reshape the query feature space and alleviate difficulties in identity matching. The authors report improved performance on instrument segmentation using their approach. This work builds upon ideas from DETR [Carion 2020] and Trackformer [Meinhardt 2022]. TraSeTR can also be applied to non-surgical domains and compared to seminal works such as DETR and Trackformer, which would provide valuable insights.

Multitask learning for perceptual tasks has also been used for pituitary surgery. Das et al. [2023] present Pituitary Anatomy Identification Network (PAINet) and a newly introduced dataset for endoscopic pituitary surgery. They tackle the challenging task of identifying ten anatomical structures, with two prominent ones addressed through semantic segmentation and the other eight through centroid prediction. Their approach employs a U-Net architecture with two different task heads for each task for joint learning of the semantic segmentation and centroid prediction task. The authors report improvements in IoU and mean percentage of correct key points (MPCK) compared to standard non-multitask models.

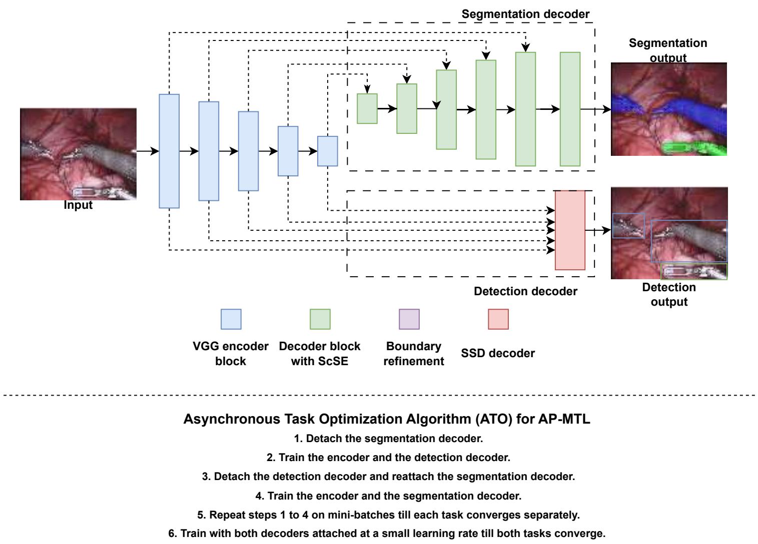

Figure 2.5: Illustration of the Attention Pruned Multitask Learning (AP-MTL) Network and the optimization method used for training this network [Islam 2020c]. The top image shows an encoder-decoder network with skip connections for its segmentation and detection decoders. A summary of the Asynchronous Task Optimization (ATO) for obtaining convergence for both tasks in the AP-MTL network is provided at the bottom.

These papers [Islam 2020c; Sanchez-Matilla 2021; Huang 2022a; Baby 2023; Zhao 2022; Das 2023] demonstrate cases where the utilization of multiple tasks leads to improvements in the performance of all tasks. They show that multitask learning can be highly efficient, achieving real-time performance while predicting multiple tasks. The use of weak supervision with multitask learning highlights the potential for efficient and cost-effective labelling strategies [Sanchez-Matilla 2021]. However, it is important to note that these works do not report on negative transfer.

Psychogyios et al. [2022] highlights that MTL can face challenges as well. The authors present a method for jointly learning both disparity estimation and surgical instrument segmentation, but the results show a decrease in performance. The architecture employed in Psychogyios et al. [2022] is U-Net-like, featuring a shared encoder and separate decoder heads for each task, including segmentation. The model undergoes pretraining using the Flythings3D dataset [Mayer 2016] for both the encoder and the disparity decoder. Following pretraining, Psychogyios et al. [2022] present results from various training schemes. These include MTL training alone, training on one task and fine-tuning on the other, and training with MTL followed by fine-tuning on a single task. Interestingly, the authors observed decreased performance when utilizing plain MTL for their two tasks. Instead, their experiments revealed that for marginal improvements in the segmentation task, it is necessary to first train the model using MTL and then fine-tune solely on segmentation. Conversely, for training the disparity task, it proves more effective to train the disparity task alone without segmentation. Due to the absence of a multitask dataset for depth and segmentation, multiple single-task datasets from different sources, were used for multitask training. Multitask learning not improving the disparity task, could also be because of the domain shifts.

Perceptual tasks and auxiliary tasks

In the previous papers, the focus is on enhancing the performance of two or more tasks by training them together. However, an alternative approach is to prioritize one primary task and introduce another task as an auxiliary task [Qin 2020a; Bhattarai 2023; Huang 2022b; Wang 2022a]. The role of the auxiliary task is to guide the training of the primary perceptual task. These auxiliary tasks can take different forms. They can be real tasks with specific objectives [Huang 2022b], or derived tasks constructed from existing information in the image or labels [Wang 2022a; Bhattarai 2023; Qin 2020a]. In some cases, auxiliary tasks are used to inject domain-specific or prior knowledge into neural networks, enriching their capacity to learn and generalize [Wang 2022a; Huang 2022b].

In their work, F. Qin et al. [2020] introduce the concept of contour supervision, a form of boundary prediction to serve as an auxiliary task for semantic segmentation. Contour supervision involves the creation of object outlines or contours for class instances within a semantic segmentation, with the network tasked with predicting these contour maps. The authors argue that contour prediction is valuable for localizing precise edges of the segmentation mask and providing information about the outer shape of objects. Different decoders are used to predict the contours and image segmentation. Contour supervision as an auxiliary task can be seen as a technique for improving the accuracy of image segmentation at object boundaries similar to loss functions that aim to improve models learning border pixels better Ronneberger et al. [2015].

In a different approach illustrated in Fig. 2.6, Bhattarai et al. [2023] suggest employing the prediction of the Histogram of Gradients (HoG) of the image as an auxiliary task to guide the learning of the segmentation task. The authors contend that leveraging image features as pseudo-labels in image classification is a well-established practice, and the histogram of gradient is a widely used handcrafted feature for object detection. Therefore, learning HoG can serve as valuable supervision for the segmentation task. To implement this approach, the authors propose a network that utilizes either the U2-Net [Qin 2020b] or a U-Net with a single encoder and single decoder with deep supervision [Lee 2015]. Auxiliary heads are integrated into the architecture to predict the histogram of gradients at various decoder levels. This work provides further evidence that auxiliary tasks generated from the image can enhance the features extracted by a model.

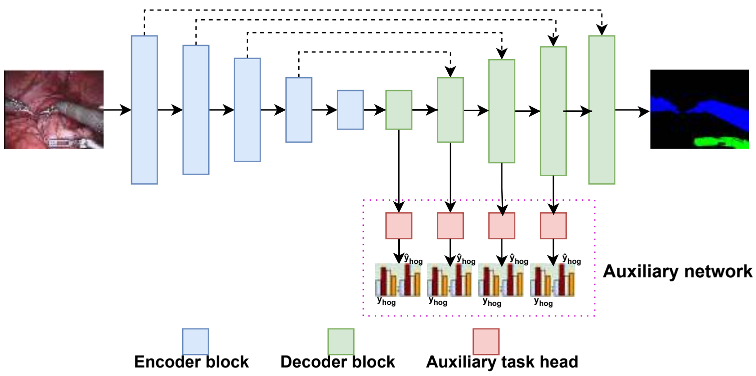

Figure 2.6: Histogram of gradient multitask learning (HoG-MTL) [Bhattarai 2023] demonstrates the use of an unconventional auxiliary task of predicting the histogram of gradients of input images image as an auxiliary task for semantic segmentation.

In the paper B. Huang et al. [2022], the authors focus on depth estimation and adopt the standard left-right consistency methodology, as introduced in Godard et al. [2017]. However, they introduce a novel auxiliary task by considering the 3D left-right consistency of 3D point clouds. The authors argue that while left-right consistency in disparity images is valuable, an additional source of information can

Table 2.2: Methods that utilize multitask learning for tracking and control. *HP - hard parameter sharing.

	Publication	Tasks	Optimization	Architecture	Speed(fps)
Pose estimation	Laina et al. [2017]	tool segmentation 2D pose heatmaps	linear scalarization	shared encoder with multiple task branches (HP)	18
	Hasan et al. [2021]	tool presence detection tool segmentation geometric primitive detection	linear scalarization	shared encoder with multiple task branches (HP)	-
	B. Li et al. [2022]	tool segmentation tool number keypoint detection	linear scalarization	shared encoder with multiple task branches (HP)	-
	B. Li et al. [2021]	unsupervised depth estimation tool-tip segmentation	linear scalarization	multistage network (HP)	50
Camera motion prediction	Islam et al. [2019]	tool segmentation saliency prediction	sequential training	shared encoder with multiple task branches (HP)	127
	Islam et al. [2020]	tool segmentation saliency prediction	sequential training	shared encoder with multiple task branches (HP)	42
	B. Li et al. [2022]	future optical flow future tool segmentation future camera action prediction	linear scalarization	multistage network (HP)	23

be derived from the left-right consistency of point clouds, generated using predicted disparity images and information about laparoscopic stereo cameras. Their method follows a two-stage MTL approach. In the first stage, a standard depth estimation architecture predicts 2D left and right disparity images. Subsequently, these disparity images, along with the focal length, the stereo distance, and predetermined blind masks for outlier removal, are used to generate 3D point clouds. The 2D depth loss functions include an appearance matching loss, a smoothness loss, and a left-right disparity consistency loss. In addition, the authors employ the iterative closest point (ICP) algorithm [Besl 1992] to create a loss function for 3D left-right consistency by using the final residual registration error after ICP minimization. These losses are combined using linear scalarization. Incorporating the 3D auxiliary task is a key feature, as it enables the integration of information about the stereo camera into the optimization process of the neural network.

In endoscopic submucosal dissection (ESD), dissection landmarks are crucial for marking the boundary between a lesion and normal tissues. ESD involves creating landmarks around the lesion to label the boundary between the lesion and normal tissues [Ono 2021]. J. Wang et al. [2022] present a neural network designed for detecting dissection landmarks. This approach stands out by introducing an auxiliary task for capturing the spatial relationship between each dissection landmark. Essentially, the authors seek to enhance the detection of dissection landmarks by incorporating domain knowledge that dictates alignment along a curve. To represent the spatial relationships between the nearest landmark neighbours on the curve, an edge map is generated. This edge map is then transformed into a heatmap that preserves these spatial relationships. The authors propose a shape-aware relation network based on the U-Net architecture featuring multiple decoders. This network functions as an MTL system that predicts both landmark positions and the heatmap of landmark spatial relationships. Similar to B. Huang et al. [2022], auxiliary tasks were used to incorporate domain knowledge into the network, yielding positive results.

2.4.2 MTL for tracking and control

In this subsection, we explore multitask learning (MTL) applications in the context of tracking in the surgical scene and the control of instruments or minimally invasive cameras. We focus on two main areas within this context: pose estimation and camera motion prediction. Pose estimation involves determining the position and orientation of surgical instruments or anatomical structures within the surgical field. Camera motion prediction, on the other hand, deals with forecasting the correct movement of the minimally invasive camera based on the current and past states of the surgical environment.

Table 2.2 summarizes the papers discussed in this subsection, highlighting the multiple tasks addressed, the optimization approaches adopted, the architectural strategies used for multitask learning, and the reported speeds for each study.

Pose estimation

Pose estimation is the process of determining the spatial configuration of objects or entities within a given scene, with a focus on estimating their positions and orientations. Pose estimation can be categorized into two branches based on the coordinate system in which the pose is measured: 2D pose estimation and 3D pose estimation. 2D pose estimation is particularly well-suited for scenarios where depth information is not essential or available. The goal is to accurately localize key points or landmarks that define the orientation of the object of interest within the image. On the other hand, 3D pose estimation involves estimating the full three-dimensional spatial configuration of objects in the scene. This estimation includes both the 2D position and depth information, along with orientation, making it a more challenging task. Accurate pose estimation is of paramount importance in various applications related to endoscopic surgeries, including instrument tracking, automatic surgical camera control, augmented reality applications, and robotics. It is worth noting that the literature in the field of endoscopic surgeries often uses the 2D coordinate space [Li 2022b; Laina 2017; Hasan 2021; Islam 2019b; Islam 2020b; Li 2021]. Still, it may use algebraic and projective geometry techniques or depth information to derive 3D pose estimation if needed [Hasan 2021; Li 2021].

In their paper, Laina et al. [2017] introduce a 2D instrument pose estimation method, employing the MTL paradigm. The two tasks trained for this purpose are segmentation and a variant of the instrument keypoint detection task. To achieve this, the authors reframe the 2D pose landmark detection task as a regression problem focusing on instrument heatmaps. The model utilized in this approach is a variant of the U-Net-like architecture with two different decoders: a segmentation decoder and a heatmap regression decoder. The output from the segmentation task is combined with the feature maps in the heatmap regression branch, a design choice that contributes to the improvement of predicted regression maps. This work suggests that keypoint heatmap prediction and tool segmentation might be complementary tasks. However, more datasets and experiments are needed to generalize this finding.

Hasan et al. [2021] present an MTL method for instrument presence detection, segmentation, and 2D pose estimation. The 2D pose estimations are represented as 'geometric primitives', which in their work refer to heat maps of critical components of the instruments, such as edge lines, mid-lines, and the tool-tip, as depicted in Fig. 2.7. The geometric primitives approach is chosen as they easily facilitate the calculation of 3D instrument poses. Unlike conventional 2D pose estimation methods that predict landmark positions [Li 2022b] or heatmaps [Laina 2017], this approach predicts a geometric primitive map of the relevant tool parts, including the tool edge, shaft mid-line, and tool-tip. The network architecture employed is U-Net-like, featuring a single encoder, direct instrument presence detection connected to the encoder, and four decoders for segmentation, edge-line primitive mapping, shaft mid-line primitive mapping, and tool-tip primitive mapping. The 2D geometric primitive map, once predicted, is then used in combination with prior information about the instrument, such as the radius of the tool shaft and the length of the tool head, to calculate the required 3D pose using algebraic and projective geometry techniques, followed by refinement. With multiple decoders for each geometric primitive, the network was very slow and unsuitable for real-time applications. To address this, the authors used depth-wise separable convolutions [Chollet 2017]. Exploring ways to reduce the number of decoders while preserving geometric primitives could further enhance the speed of this model.

B. Li et al. [2022] introduce an alternative method for predicting 2D pose estimation while leveraging the MTL paradigm. The authors highlight a common challenge in the field - the abundance of datasets for tasks like surgical instrument segmentation, contrasted with the scarcity of datasets for pose estimation. To address this issue, they propose a multitask semi-supervised approach to pose estimation. Their method predicts three different outputs: instrument segmentation, instrument number and instrument landmark keypoint detection. The instrument segmentation task is used as a spatial constraint, and the instrument number pre-

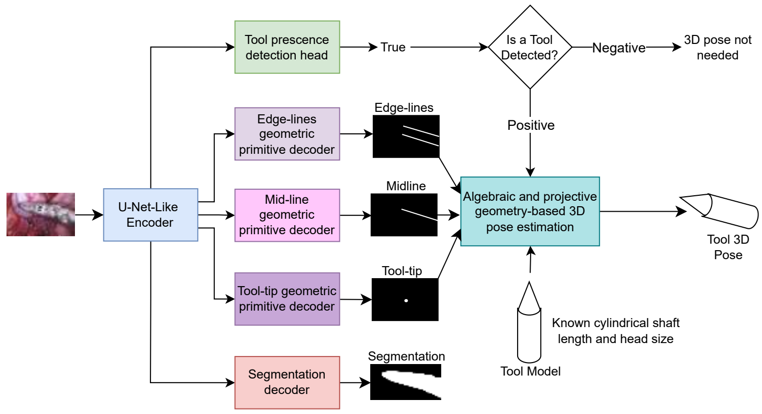

Figure 2.7: The Augmented Reality Tool Network (ART-NET) [Hasan 2021] presents a system to produce 3D pose estimations of instruments for augmented reality and 3D length measurement applications. As seen in the diagram, ART-Net simultaneously learns to predict tool presence, edge-lines, mid-lines, tool-tip and segmentation. The geometric primitives predicted are combined with prior information to produce the required Tool 3D pose.

diction task as a global constraint for the instrument landmark keypoint detection task. The semi-supervised technique employed here is the mean-teacher framework for semi-supervised learning [Tarvainen 2017]. Both student and teacher models take the form of feature pyramidal networks [Lin 2017], each with task-specific heads for the three tasks trained in conjunction.

B. Li et al. [2021] introduce a three-stage framework for optimizing laparoscopic field of view (FOV) control during minimally invasive surgery (MIS) through tracking and 3D localization. The framework begins with 3D localization, using multitask learning with a shared encoder and task-specific decoders for simultaneous surgical tool-tip segmentation and depth estimation. The predicted tool-tip segmentation and depth, along with physical constraints like tool diameter, generate the 3D pose of the tool-tip. In the optimal view control stage, the framework tracks the localized tool-tip and focuses on a data-driven 2D image region to perform laparoscopic control. The final stage includes an affine mapping-based Minimize Rotation Constraint (MRC) method to correct visual misorientation from the Remote Center of Motion (RCM) constraint, and a null-space controller to optimize 2D and 3D tool positions relative to the laparoscope. This work uses a heuristic of tracking the tool-tip for camera control. However, the camera motion becomes reactive, and this heuristic can fail in complex scenes with multiple tools moving in and out of the operating field.

As evident from the literature on instrument/camera tracking methods in MIS, such as [Hasan 2021; Li 2022b; Laina 2017; Li 2021], the predominant approach involves predicting some form of 2D pose estimation rather than 3D pose estimation. This predicted 2D pose is usually used to facilitate other tasks such as tracking by prediction of 2D poses in a video sequence, as exemplified in Laina et al. [2017], generating 3D poses to be used for laparoscope automation via tracking [Li 2021] or enabling augmented reality overlays, as demonstrated in Hasan et al. [2021].

Another closely related and prevalent task to surgical instrument pose estimation is motion prediction for cameras, which also plays a significant role in the context of tracking and control in minimally invasive surgeries.

Camera motion prediction

The motion prediction task for cameras has been explored along two primary avenues: scanpath prediction and camera imitation learning. Scanpath prediction involves forecasting the potential camera paths based on the most captivating elements within the camera's current field of view. It draws inspiration from a theory in human gaze fixation, which posits a direct connection between human gaze scanpath and the attentional priority of objects in an image, often referred to as object saliency. In simpler terms, humans tend to focus on the most vital information first before shifting their gaze to less significant details. In the context of MIS, scanpath prediction tasks are centred around predicting the most salient objects in a scene and plotting a path from the most salient to the least salient areas. Foulsham [2019] provide a good reference for further insights on scanpaths, saliency, and their correlations.

Islam et al. [2019] introduce a method for estimating saliency in surgical scenes, specifically by training in conjunction with segmentation as an auxiliary task. A notable challenge in this context is the absence of ground truth saliency datasets for MIS. To address this issue, the authors propose a novel method for generating saliency maps. The approach assumes that the most intriguing parts of surgical instruments are the wrist and claspers, and fixation points are located on these instrument segments. Additionally, the movement of instruments is utilized as a key indicator of saliency. Instruments with more significant movement are assigned higher saliency values than those with minimal movement. A scanpath is then defined as the movement from the most salient to the least salient instrument. Fig. 2.8 illustrates the process of generating a scanpath from images as presented in Islam et al. [2019]. The architectural framework employed is based on a U-Net structure featuring an encoder and two decoders, one dedicated to saliency map prediction and the other to instrument class segmentation. Attention modules are incorporated in the saliency map prediction branch to suppress irrelevant regions and highlight salient features. The authors utilize a two-phase learning strategy to address the challenge of converging both tasks in the same epoch. In the first phase, equal loss weights are assigned to both tasks. In the second phase, the model fine-tunes the loss weights based on the performance of the converged task. Islam et al. [2019] is unique in MTL for MIS it diverges from the standard linear scalarization optimization strategy.

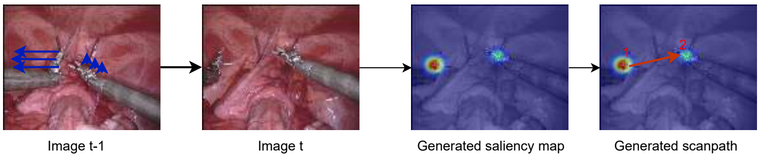

Figure 2.8: Illustration of the heuristic used for the generation of saliency maps and scanpaths in Islam et al. [2019]. A saliency map is generated (using the code provided by Islam et al. [2019] and images from EndoVis2017 [Allan 2019]), which correlates to the motion of fixation points and the size of the fixation points, which are assumed to be located on the instrument wrist and claspers. The scanpath is then assumed to be the movement between the most salient to the least salient instrument.

Table 2.3: Methods that utilize multitask learning for surgical video workflow analysis. *HP - hard parameter sharing.

	Publication	Tasks	Optimization	Architecture	Speed(fps)
Phase and instrument detection	Twinanda et al. [2016]	phase recognition tool presence detection	linear scalarization	multistage network (HP)	-
	Twinanda et al. [2016]	phase recognition tool presence detection	linear scalarization	multistage network (HP)	-
	Czempiel et al. [2020]	phase recognition tool presence detection	linear scalarization	multistage network (HP)	-
	Mondal et al. [2019]	phase recognition tool presence detection	linear scalarization	multistage network (HP)	-
	Sanchez-Matilla et al. [2022]	phase recognition scene segmentation	sequential training	multistage network (HP)	-
	Y. Jin et al. [2020]	phase recognition tool presence detection	sequential training	multistage network (HP)	3.3
Surgical action triplet learning	Nwoye et al. [2020]	surgical action triplet recognition	linear scalarization	shared encoder with multiple task branches (HP)	-
	Nwoye et al. [2022]	surgical action triplet recognition	uncertainty weighting	shared encoder with interacting task branches (HP)	28.1
	Sharma et al. [2023]	surgical action triplet recognition	linear scalarization	shared encoder with interacting task branches (HP)	-
	Yamlahi et al. [2023]	surgical action triplet recognition phase recognition	linear scalarization	encoder with multiple branches (HP)	-
	Sharma et al. [2023]	surgical action triplet detection	linear scalarization	multistage network (HP)	-
Multi-granularity activity detection	S. Ramesh et al. [2021]	phase recognition step recognition	linear scalarization	multistage network (HP)	-
	Valderrama et al. [2022]	phase recognition step recognition Action recognition Instrument detection	linear scalarization	multiple encoders with with multiple task branches (HP)	-

Building upon the foundation laid by Islam et al. [2019], the subsequent paper on scanpath prediction with MTL, known as ST-MTL [Islam 2020b], maintains a similar architectural framework. However, the authors assert that the correct prediction of saliency maps requires information from the current frame and insights from previous frames. To address this temporal relationship, they introduce a convolutional Long Short-Term Memory (ConvLSTM) [Shi 2015] into the saliency prediction decoder. The utilization of ConvLSTM modules enhances the model's ability to capture temporal dependencies and leverage information from past frames, contributing to more accurate saliency predictions. Moreover, Islam et al. [2020] incorporate a sequential training method for a spatiotemporal model similar to ATO optimization method developed in Islam et al. [2020], as discussed in Section 2.4.1.

Scanpath prediction, despite its merits, has several limitations in minimally invasive surgery (MIS). It currently focuses solely on surgical instruments, ignoring critical tissues and organs, which can lead to suboptimal camera movements. The dynamic nature of surgical procedures, with instruments frequently entering and exiting the field of view, poses challenges for scanpath models to adapt quickly. Additionally, the complexity of the surgical environment, involving multiple instruments and varying tissue types, can be oversimplified by saliency prediction. These models also lack contextual awareness, failing to consider the surgeon's intentions or the stage of the operation, which can result in misalignment with the surgeon's preferred view and potential inefficiencies.

Another promising avenue in the realm of camera motion prediction is camera imitation learning. This approach emphasises emulating the camera movements observed during surgeries performed by surgeons or their assistants. This technique is particularly relevant in the context of MIS, which often follows predefined procedural steps. These surgeries typically adhere to a fixed sequence of steps, and under normal circumstances, camera movements should exhibit similar patterns. Camera imitation learning aims to address the reactive nature of instrument tracking for camera control, as seen in B. Li et al. [2021]; Gruijthuijsen et al. [2022], by adding contextual awareness and emulating the behaviour of surgical assistants in various scenarios.

B. Li et al. [2022] introduce a method for the proactive adjustment of the camera's field of view, achieved by modifying the camera's position in the x, y, and z coordinates. To replicate the motion patterns of a laparoscope camera, the authors employ a ConvLSTM for sequence modelling, predicting feature motions on a per-frame basis. A unique aspect of this research is the generation of ground truth laparoscope motion from laparoscopic videos. This process involves dynamic camera motion estimation to infer camera pose. The authors utilize the Neural-Guided Random Sample Consensus (NG-RANSAC) method [Brachmann 2019] to match stereo-images under dynamic conditions. Additionally, they apply the remote centre of motion constraint to optimize the pose estimation. The ground truth motion in the x, y, and z directions is derived from different sequential poses. The inputs to the ConvLSTM model comprise estimated segmentation and optical flow obtained from off-the-shelf models. The model optimizes for correct outputs of the subsequent N optical flow and segmentation results. The optical flow outputs are then passed through a laparoscopic action head to predict the camera's movements in the x, y, and z coordinates. B. Li et al. [2022] improve on their previous work [Li 2021] by focusing on proactive camera motion and learning the motion strategy from clinical videos instead of reactive motion prediction and instrument tracking. Additionally, B. Li et al. [2022] directly predict in 3D space instead of generating results in 2D and scaling to 3D, making their approach unique among the methods reviewed.

2.4.3 MTL for surgical video workflow analysis

Surgical video workflow analysis is the systematic examination of videos recorded during surgical procedures, aiming to extract valuable insights into various facets of the surgery. This analysis serves multiple critical purposes, including providing context-aware intraoperative assistance, enhancing surgeon training, facilitating procedure planning, supporting research endeavours, and enabling retrospective analysis [Lalys 2014].

The analysis process involves breaking down a surgical procedure video into distinct segments, which are categorized based on the surgeon's various activities. Different surgeries can be divided into specific activities, which can be examined at multiple levels of granularity.

We adopt a representation for surgical video workflow analysis from Valderrama et al. [2022], where the holistic analysis of a surgical video and its workflow is divided into four granularities. The initial granularity level involves identifying *instruments* present in the surgical scene, which are features of the surgical scene directly responsible for the surgical workflow. This stage entails understanding the instruments within the field of view as these instruments movements and interactions with different tissues give rise to specific *actions*, such as grasping or cutting, which represents the second granularity level. The third granularity stage involves the sequence of actions performed in pursuit of a particular surgical objective, which is called a *step*. Multiple steps are executed together to accomplish a portion of the surgery, termed a surgical *phase*, which is the fourth and final granularity of a surgical procedure. Fig. 2.9 provides a visual representation of the different levels of granularity.

Addressing the problem of surgical video workflow analysis is a critical step towards enhancing computer-aided interventions in surgical procedures. However, recognizing and distinguishing between different phases, steps, or actions within surgical videos remains a formidable challenge. This challenge is attributed to several factors, including the limited availability of publicly accessible data, the

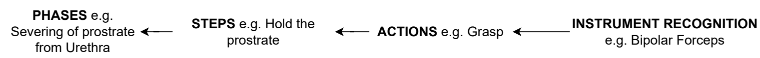

Figure 2.9: A visual representation of the different levels of granularities in surgical video workflow analysis. This representation for surgical video workflow analysis is adopted from the framework for holistic analysis of surgical videos in Valderrama et al. [2022].

high resemblance between long-range sequences belonging to different phases or steps, the substantial variability within sequences associated with a single phase or step, and the extended duration of surgical procedures necessitating thorough analysis.

In this section, we explore the literature which applies MTL to solve the surgical activity recognition problem and divide it into three subsections, namely: auxiliary tasks for surgical activity recognition, surgical action triplet learning, and multi-level activity recognition. Table 2.3 summarizes the papers discussed in this subsection, highlighting the multiple tasks addressed, the optimization approaches adopted, the architectural strategies used for multitask learning, and the reported speeds for each study.

Phase and instrument detection

Early research in surgical activity recognition with MTL primarily centred on phase recognition [Twinanda 2016a; Czempiel 2020; Sanchez-Matilla 2022; Jin 2020]. phase recognition was usually predicted alongside instrument recognition specifically tool presence detection. The rationale for focusing on instrument recognition is rooted in the understanding that surgical instruments are the means by which surgeons interact with the surgical scene. Assuming standard surgical practices are followed, a substantial correlation exists between the choice of specific tools and the corresponding activities undertaken during a surgical procedure.

One of the pioneering papers that introduces the application of MTL and CNNs to surgical phase recognition is EndoNet [Twinanda 2016a]. This method follows a two-stage prediction approach. The first stage jointly learns tool presence and phase prediction for each frame. The second stage refines the phase predictions generated in the initial stage. To implement this, EndoNet employs a pretrained AlexNet [Krizhevsky 2012] as the encoder for its first stage, which is fine-tuned by simultaneously predicting tool presence and phase detection using dedicated task heads. In the second stage, EndoNet combines the features for phase detection in the current frame with information from the previous frames and feeds this combined data into a Hidden Markov Model (HMM) [Padoy 2009]. An architectural overview of

EndoNet is illustrated in Fig. 2.10. In another approach, Twinanda et al. [2016] replaces the HMM in EndoNet with an LSTM [Hochreiter 1997] to improve temporal modelling. Incorporating a temporal model, such as HMM in EndoNet and LSTM in Twinanda et al. [2016], significantly improved phase recognition, validating the intuition that temporal features are crucial for accurate phase prediction. [Twinanda 2016a] released the Cholec80 dataset for benchmarking phase recognition and tool presence detection using accuracy, precision, and recall as metrics.

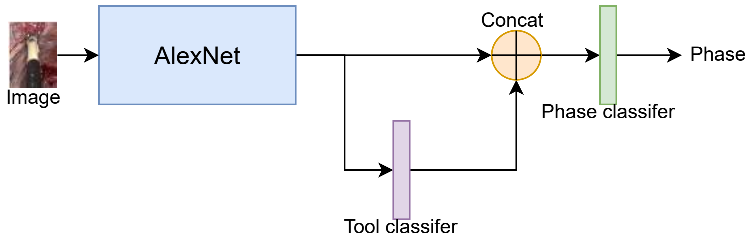

Figure 2.10: One of the earliest works that attempts to solve the phase recognition problem is the EndoNet [Twinanda 2016a]. It uses an Alexnet as a feature extractor with two task heads - A tool classifier head and a phase classifier head. Logits from the tool classifier head are concatenated with features from Alexnet before predicting phase.

In a manner similar to the approach employed by Twinanda et al. [2016], Czempiel et al. [2020] also leverage the tool presence detection task along with a multistage network for phase recognition with different temporal modelling and optimization. Instead of utilizing the LSTM/HMM models as seen in previous works, they opt for the Temporal Convolutional Network (TCN), as described by Lea et al. [2017], for the refinement phase. The multitask optimization method used involves implementing the median frequency balancing technique [Eigen 2015]. As for the input to the TCN, it comprises a concatenation of features extracted from the current frame and the N preceding frames. TCN is preferred to LSTM/HMM approaches because it is notably faster, less resource-intensive and gives competitive results.

In a similar vein, Mondal et al. [2019] delve into the intricate relationship between tool presence detection and phase detection. The authors introduce a two-stage network identical to previous approaches, but Mondal et al. [2019] differ in how they link the phase detection task with the tool presence detection task. Unlike the methodologies found in [Czempiel 2020; Twinanda 2016b; Twinanda 2016a], where refinement primarily concerns phase detection, this approach extends refinement to both tool and phase detection. Key to their methodology is the introduction of a joint probability loss function, which serves as the binding agent between the two tasks at each stage. The joint probability loss function is a product of the probabilities associated with tool and phase detection, weighted by the inverse of the tool's appearance frequency in a given phase. This approach seeks to establish a more integrated and interdependent relationship between the tasks, offering a fresh perspective on the problem. A training procedure was designed that sequentially trains different parts of the multistage network with different losses and then jointly optimises the whole network.

Sanchez-Matilla et al. [2022] take a distinct approach by strongly emphasising enhanced temporal modelling by adopting a different task from tool presence detection for their two-stage network. In this paper, the authors opt for semantic segmentation and phase recognition, deviating from the conventional choice of tool presence detection. The first stage is a multitask network incorporating two distinct decoders, one for segmentation and another for phase prediction. The second stage of their network implementation employs a Temporal Convolutional Network (TCN) for the refinement phase. A notable finding of their research is the potential for performance enhancement with as little as 5% of the data labelled for segmentation.

In their research, Y. Jin et al. [2020] offer an end-to-end trained network that capitalizes on the strong correlation between tool presence and surgical phases, much like previous works. However, it distinguishes itself by adopting a single-stage network design. The authors introduce a novel network known as MTRCNet-CL, featuring a shared encoder with two branches. One branch is dedicated to tool presence detection, while the other is focused on phase recognition. The phase recognition decoder branch incorporates an LSTM for temporal modelling. An innovative aspect of their approach is formulating a correlation loss that models the intricate relationship between tool presence and the predicted phase in the decoders. Their hypothesis posits that given the correlation between these two tasks, the logit values of the two tasks should also exhibit a certain degree of correlation. To measure this correlation, they calculate the Kullback-Leibler (KL) divergence between the phase logit values and the tool presence logit values.

Twinanda et al. [2016]; Y. Jin et al. [2020]; Mondal et al. [2019]; Czempiel et al. [2020]; Sanchez-Matilla et al. [2022] were all benchmarked using the Cholec80 dataset and similar metrics, facilitating direct comparisons between these works. This highlights the benefits of using a multitask dataset with predefined tasks and widely accepted metrics, as shown in Table 2.4.

As evident from the studies presented in this section, early research in surgical activity detection primarily concentrated on the surgical phase granularity. Researchers explored various architectural and loss function strategies to leverage the

Table 2.4: Phase recognition and instrument detection results of various multitask learning methods on Cholec80. Results are retrieved from references. The mean of the published results are reported.

	Phase Recongition			**Tool presence detection**
Publication	Avg. precision	Avg. Recall	Accuracy	Avg. precision
EndoNetTwinanda et al. [2016]	81.7	73.7	79.6	81.0
Sanchez-Matilla et al. [2022]	-	-	89.51	-
TeCNOCzempiel et al. [2020]	88.56	81.64	85.2	-
Mondal et al. [2019]	85.7	83.5	92.0	93.53
MTRCNet-CL Y. Jin et al. [2020]	86.9	88.0	89.2	89.1

relationship between phase recognition and tool presence detection. However, it is important to note that tool presence detection has inherent limitations. It can introduce noise into the analysis and lacks the precision required to pinpoint where the surgical action is unfolding within the video. Moreover, beyond the tools, the specific tissues that surgical tools interact with during a procedure also offer valuable insights into the ongoing surgical activity. Additionally, the identification of the instrument, tissue, and the actions taking place in a scene can provide valuable clues about the ongoing surgical steps and phases [Valderrama 2022]. Each step and phase requires very specific actions performed on specific tissues, in a specific order, and with specific instruments.

Surgical action triplet learning

More recent literature has compellingly argued against the exclusive prediction of surgical workflow either directly or solely based on tool presence. It has been contended that such an approach proves inadequate for a comprehensive understanding of surgical scenes. Instead, these works shift their focus towards the more granular action recognition task, striving to establish connections between three crucial elements before attempting to predict activities with longer sequences: the specific instrument in use, the action being performed, and the target anatomy undergoing the procedure, often collectively represented as three distinct labels, <instrument, action, target>. This predictive task illustrated in Fig. 2.11 and involving these three distinct labels is often referred to as the 'surgical triplet prediction task' [Katić 2014]. The surgical action triplet task fundamentally represents a multilabel (instrument, action, target) multiclass classification problem, with the understanding of the relationship between the labels being paramount. Typically, the problem is framed with the input as a single frame [Nwoye 2020; Nwoye 2022b], or multiple consecutive frames [Sharma 2023a], with the objective of predicting the <instrument, action, target> triplet. The benchmark datasets for surgical triplet prediction are

CholecT45 and its more recent version, CholecT50, which includes five additional sequences [Nwoye 2022b]. The surgical action triplet tasks are evaluated using IVT metrics:

1. Component Average Precision: This evaluates the correct recognition of the instrument (AP_I), verb (AP_V), and target (AP_T) components of the triplets.

2. Triplet average precision: This assesses the correct recognition of interactions between tools, actions, and targets, including instrument-verb interaction (AP_{IV}), instrument-target interaction (AP_{IT}), and the main metric, instrument-verb-target (AP_{IVT}), for surgical action triplet recognition

For more information about the metrics behind the CholecT45 and CholecT50 datasets, we refer readers to Nwoye et al. [2022]. All the works in this subsection use the CholecT45, CholecT50, or extensions of these datasets with additional information (such as instrument bounding boxes). The results of the studies on surgical action triplet prediction are presented in Table 2.5.

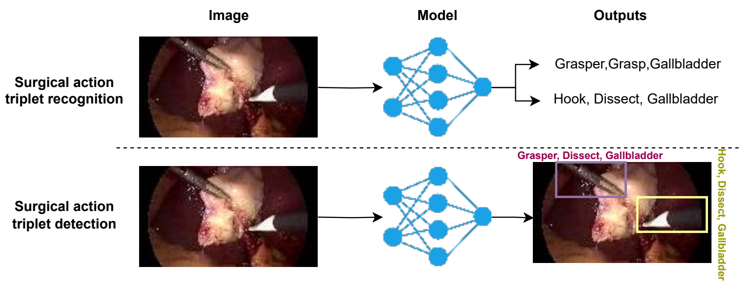

Figure 2.11: The triplet task of recognizing the <instrument, action and target> is traditionally cast as surgical action triplet recognition, which is a multi-label multi-classification recognition problem [Nwoye 2020; Nwoye 2022b; Sharma 2023a] as shown in the top image. Recently, there has been a progression in the task difficulty that cast this problem as surgical action triplet detection, which involves localizing all instruments and associating the corresponding instrument triplet to all localized instruments [Sharma 2023b].

In their study, Nwoye et al. [2020] train a multitask network tailored specifically for the surgical action triplet task. They proposed the Tripnet architecture, which employs an encoder to extract joint features and three dedicated decoders for each of the task components. The first decoder operates as a convolutional-based unit, with a dual purpose: it predicts instrument classes in the image and generates class activation maps. These class activation maps are essential for the functioning of the other two decoders. The second and third decoders are *class-activation-map-guided* convolutional units that extract features relevant to action recognition and target recognition, respectively. They use the class activation maps from the instrument decoder as a guide.

The class activation maps are concatenated with the features in the action and tissue target decoders. This approach is underpinned by the hypothesis that the instrument plays a pivotal role in interacting with the surgical environment to initiate actions on a desired target. By incorporating a weak localization guide indicating the instrument's current location, the authors expect to enhance the performance of both the action recognition and tissue target recognition tasks. A visual representation of the Tripnet structure can be found in Fig. 2.12.

Furthermore, the authors observe that not all triplet combinations are feasible, and a data association problem arises as these components (predicted instruments, actions, and tissues) are interconnected. Instead of predicting each label separately, the authors opt for a joint prediction approach. The logits produced from each encoder (I for Instrument, V for action verb, and T for tissue target) are used to create a 3D interaction space volume (Y) through an outer product operation:

$$Y = \alpha I \otimes \beta V \otimes \gamma T$$

Here, α, β, and γ represent learnable weights. The 3D volume is quantized, with values above a chosen threshold accepted as valid triplets, while spaces in the 3D volume that can never form triplets are masked out. The loss function for this approach comprises the standard cross-entropy loss for all tasks, combined into a linear combination of losses.

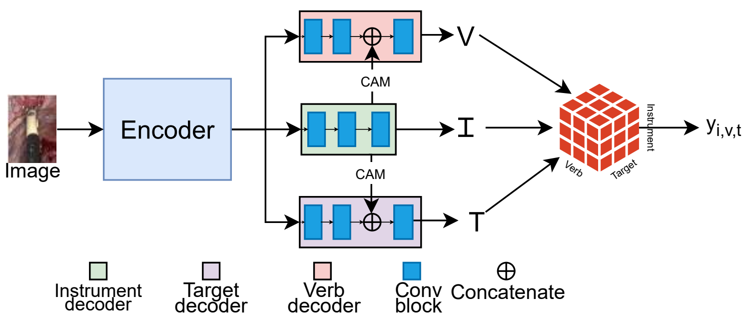

Figure 2.12: The proposed Tripnet architecture for surgical action triplet recognition [Nwoye 2020]. The model contains an encoder and three separate branches for predicting instrument verb and targets. The class activation maps from the instrument branch are used as weak localization guides for action and target prediction.

Nwoye et al. [2022], also known as RDV, is a sequel to the earlier work [Nwoye

2020]. This paper introduces a noteworthy improvement by incorporating repeated attention mechanisms to facilitate inter-task knowledge transfer, drawing inspiration from the transformer architecture. Similar to Nwoye et al. [2020], RDV employs a single encoder with three decoders. However, it distinguishes itself by employing attention-based decoders, referred to as the *Class Activation Guided Attention Mechanism* (CAGAM).

The CAGAM mechanism replaces class-activation-map-guided convolutional units that extract features relevant to action recognition and target recognition used in the Tripnet architecture [Nwoye 2020]. Similar to the Tripnet architecture, It leverages class activation maps (CAM) from the Instrument decoder as a source of weakly supervised spatial guidance for the action and tissue target recognition tasks. But instead of just concatenating, the CAM is used to produce Key and Query vectors to form an attention matrix, which is used to scale the features from the Targets and Actions. Following the CAGAM Mechanism, RDV yields instrument, action, and target features, which are further processed through another attention mechanism. This mechanism is a variant of the multi-head attention models used in transformers. This process is integral to combining the features and is notably the second application of the attention mechanism. This dual-attention mechanism strategy is what the authors aptly refer to as a *rendezvous*. The rendezvous attention mechanism employs self-attention and cross-attention to capture complex semantic features in the instrument, verb, and target features. In contrast to the volume-based predictions in Nwoye et al. [2020], RDV opts for a simpler classifier for prediction tasks. To fine-tune the model and optimize its performance, RDV employs uncertainty weighting [Cipolla 2018] to automatically determine the hyperparameters for each loss function.

Table 2.5: Results of methods on the CholecT45 dataset. Results should be compared with caution as each reported work sometimes reported a different training/testing split. The mean of the published results are reported.

	Component average precision			**Triplet average precision**		
Method	AP_I	AP_V	AP_T	AP_{IV}	AP_{IT}	AP_{IVT}
Nwoye et al. [2020]	89.9	59.9	37.4	31.8	27.1	24.4
Nwoye et al. [2022]	89.3	62.0	40.0	34.0	30.8	29.4
Sharma et al. [2023]	88.6	64.0	43.4	38.3	36.9	29.7
Yamlahi et al. [2023]	-	-	-	-	-	36.1

Surgical action triplets therefore provide a substantially more fine-grained semantic description than phase recognition or tool-presence detection. Nevertheless, the conventional formulation remains primarily frame-level: it predicts which interactions occur in an image or short temporal window without necessarily assigning each interaction to the individual physical instrument instance responsible for it. Class activation maps and later detection extensions introduce spatial cues, but do not provide the strong instance-level association required to distinguish multiple instruments of the same class reliably. This separation between semantic interaction recognition and explicit spatial grounding is a central limitation addressed by the subsequent technical chapters of this thesis.

The third instalment in this series of papers, Sharma et al. [2023] called Rendezvous in Time (RIT), follows the preceding works. While RDV focuses solely on single-frame features for triplet recognition, RIT incorporates temporal modelling into the RDV model. In particular, Sharma et al. [2023] introduce the Class Activation Guided Temporal Attention Module (CAGATAM), designed to enhance verb prediction and build upon the foundation of CAGAM. The Temporal Attention Module (TAM) plays a role in the temporal fusion of verb features extracted from the current and past frames, weighted by attention scores.

Yamlahi et al. [2023] apply the principle of self-distillation to solve the problem of class imbalance and label ambiguity in surgical action triplet recognition task. The method utilizes MTL and ensemble models as regularization to improve performance. A single teacher model with a Swin Transformer [Liu 2021] backbone and a classifier are trained on hard labels with binary cross entropy loss for the three tasks of instrument, action and target detection separately. Then, the Sigmoid probabilities from the teacher model are used to train three self-distilled student models as an ensemble to minimize a distillation loss.

In the latest addition to the RDV series of papers, Sharma et al. [2023] introduce the surgical action triplet detection task: the localization of surgical instruments along with the recognition of surgical action triplets. The authors address a challenge in datasets designed for binary triplet label recognition when used for detection tasks. Specifically, there may be multiple instruments in an image from the CholecT50 dataset [Nwoye 2022b], but only the primary instrument that is most prominent is annotated. This leads to a data association problem. To tackle this issue, the authors propose a two-stage network named the Multi-class Instrument-aware Transformer - Interaction Graphs (MCIT-IG). The first stage is referred to as a multi-class instrument-aware Transformer (MCIT) that performs target prediction sub-task by using the knowledge of the current instruments and their classes. It does this by first detecting all the instruments in an image using a deformable DETR model [Zhu 2020] trained on an annotated Cholec80 dataset, then extracting global features from the same image by using a feature extractor chained with a small transformer to provide global features, and finally combining the instrument detection information (from DETR) and image global feature information (from feature extractor) with a lightweight transformer to predict targets for each instrument. The second stage (IG) utilizes the instrument and target features from the first stage to construct a bipartite graph with action relationships as interaction edges. Although the supervision for the second stage is weak, a heuristic is employed to address the data association problem. The authors note that the better the instrument localization prediction, the better the accuracy score for surgical action triplet prediction is.

In their work, Y. Chen et al. [2023] point out the challenges of jointly optimizing three distinct classification problems as a single multiclass multilabel problem. They highlight that this problem is unbalanced, as positive results are only achieved when all three components of the triplet are predicted correctly. Moreover, the presence of multiple instruments in a single image and the lack of annotations for some instruments further introduce ambiguities to the triplet recognition task. To address the complexities of triplet recognition, the authors propose a solution involving five smaller subnetworks. The first subnetwork is responsible for counting the number of tools and predicting the presence of key triplets or irrelevant triplets, including null actions and null targets. The second subnetwork predicts the tool classes in the image, and similar to the Tripnet approach [Nwoye 2020], it utilizes class activation maps. However, in this case, the Inflated 3D Convolutional Network (I3D) [Carreira 2017] is used, which allows for the generation of class activation maps, which the authors found to be more accurate. The third and fourth subnetworks jointly predict verbs and targets. Finally, the fifth subnetwork serves as both a fine-tuning and masking network, removing impossible triplets. It takes the logits predicted by the previous subnetworks and the current video clip as input to predict fine-tuned triplet logits and perform classification. The authors emphasize that they train each subnetwork in different stages, as attempting to address multiple auxiliary tasks simultaneously can lead to task distraction and negative transfer.

The surgical action triplet task is evolving in complexity. Initially framed as a multilabel multiclassification task for $\langle$instrument, verb, target$\rangle$ by Nwoye et al. [2020]; Nwoye et al. [2022], it now includes bounding box detection and segmentation of instruments [Sharma 2023b] and video-based analysis [Nwoye 2020]. The multilabel multiclassification task and its progression is similar to fields such as human-object interaction [Kim 2021] and panoptic scene generation [Yang 2022] in computer vision, reflecting an increasing emphasis on spatial tasks and comprehensive scene understanding.

Multi-granularity activity detection

A comprehensive understanding of temporal relationships in MIS hinges on a network's ability to grasp the concept that multiple actions are required to complete a step and that multiple steps collectively constitute phases. This interconnection of granularities in surgical activities can serve as valuable signals for training deep neural networks in surgical activity recognition. Despite the limited available datasets, there is a growing interest in multi-granularity surgical activity learning, spurred by the recent introduction of multiple datasets providing labels for various granularities [Wang 2022b; Huaulmé 2021; Valderrama 2022].

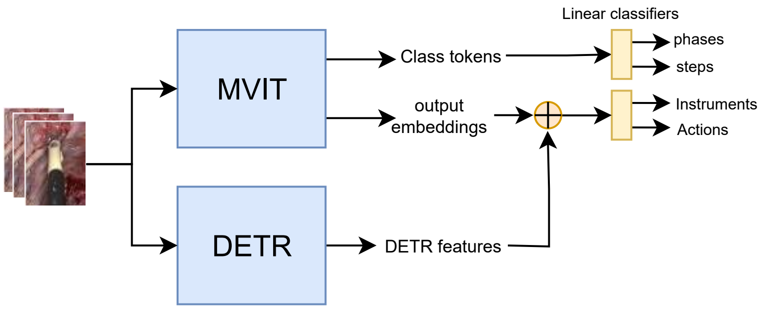

Figure 2.13: The TAPIR architecture from Valderrama et al. [2022] for the prediction of phases, steps, instruments, and actions. It utilizes two separate encoders, a video feature extractor (MViT), and an instrument detector (DETR), along with various classification heads.

In an early endeavour to simultaneously predict phase and step activities, S. Ramesh et al. [2021] offer an innovative approach. Their proposed network adopts a two-stage design similar to the architecture of EndoNet [Twinanda 2016a]. The initial stage comprises a feature extractor with a classifier, where two distinct linear classifier heads are responsible for predicting phase and step for N frames. In the subsequent stage, the vectors extracted in the first stage for multiple consecutive frames are concatenated and injected into a temporal convolutional neural network for temporal modelling, which leads to improved predictions for both phase and step activities. This work introduced the Bypass40 [Ramesh 2021] dataset and evaluated it using separate classification metrics for phase and step prediction, demonstrating improvements over off-the-shelf models. The authors experimentally show that joint training of phase and step recognition is beneficial.

The authors of Valderrama et al. [2022] aim to completely address the temporal scene understanding problem in MIS. They introduce a multi-level activity detection model illustrated in Fig. 2.13 named Transformer for Action, Phase, Instrument, and Steps Recognition (TAPIR) to achieve this goal. TAPIR leverages

Table 2.6: Methods that utilize multitask learning for anticipation in surgical workflow optimization. *HP - hard parameter sharing, GAN - generative adverserial network

	Publication	Tasks	Optimization	Architecture	Speed(fps)
Anticipation in surgical workflow optimization.	Twinanda et al. [2018]	remaining surgery duration progress estimation	linear scalarization	multistage network (HP)	-
	Rivoir et al. [2020]	instrument anticipation time anticipated instrument state	linear scalarization	multistage network (HP)	-
	Yuan et al. [2022]	instrument anticipation time phase anticipation time	linear scalarization	multistage network (HP)	34
	Ban et al. [2022]	phase anticipation phase recognition	linear scalarization	GAN network (HP)	-
	Y. Jin et al. [2022]	phase anticipation phase recognition	linear scalarization	multistage network (HP)	74.63

two distinct backbones to capture various types of information. The first, the Multiscale Vision Transformer (MViT) [Fan 2021], for extracting global and temporal features from sequential video frames. The second, Deformable Transformers for End-to-End Object Detection (Deformable DETR), focuses on capturing spatial features relevant to instrument detection and box proposals. The authors use a concatenation approach to combine the insights from these two backbones. The concatenated features from the MVIT and DETR are passed to linear classifiers to predict the action and instrument detection tasks. As for the prediction of phases and steps, this is solely carried out using the class tokens from the MViT backbone and linear classifiers. This work introduced the PSI-AVA dataset and evaluated it using mean average precision for phase, steps, actions, and instrument detection. The authors demonstrate improvements over off-the-shelf models and show that multitask frameworks outperform their single-task counterparts.

Valderrama et al. [2022]; S. Ramesh et al. [2021] demonstrate the benefits of jointly learning multiple activities. Further research investigating the relationships between different surgical workflow granularities using the provided datasets would be valuable to the community. Additionally, metrics that measure the joint prediction of multiple granularities together, thereby assessing the accuracy of relationship prediction between granularities, would also be useful.

2.4.4 MTL for anticipation in surgical workflow optimization

Optimizing surgical workflows is critical for enhancing the efficiency and safety of procedures in the Operating Room (OR) [Yuan 2022]. Multitask learning (MTL) has been employed to improve various aspects of surgical workflow, such as predicting future instrument usage, phase transitions or time till the end of the ongoing surgery. This subsection reviews four studies that utilize MTL for predicting future instruments and phases to optimize surgical workflows in MIS. Table 2.6 summarizes the papers discussed in this subsection, highlighting the multiple tasks addressed, the optimization approaches adopted, the architectural strategies used for multitask learning, and the reported speeds for each study.

Twinanda et al. [2018] propose RSDNet, a deep learning pipeline to estimate remaining surgery duration (RSD) and surgery progress using visual information from laparoscopic videos. They employ a two-stage network with a ResNet feature extractor and an LSTM to predict the remaining time until the end of the surgery and the percentage if surgery completed. The authors demonstrate that their approach outperforms naive strategies.

Rivoir et al. [2020] introduce a method to anticipate surgical instrument usage with sparse annotations using Bayesian deep learning [Kendall 2017]. They employ a Bayesian AlexNet-style network [Krizhevsky 2012] and a Bayesian LSTM [Hochreiter 1997] to handle uncertainty and predict the remaining time until an instrument is needed (regression) and the current state of an instrument (whether it will be used soon, is being used, or is not needed). Experiments on the Cholec80 dataset [Twinanda 2016a] reveal that prediction uncertainty varies by instrument, with some instruments strongly related to specific phases showing low uncertainty, while others are more difficult to predict.

Yuan et al. [2022] propose the Instrument Interaction Aware Anticipation Network (IIA-Net) to predict surgical phases and instrument usage. IIA-Net consists of two stages: a spatial feature extractor and a temporal model. The spatial feature extractor, a multitask model with a ResNet50 backbone, extracts visual features, predicts tools and current phase, and captures geometric and semantic features of instrument interactions through an Instrument Interaction Module. The temporal stage is a causal dilated multi-stage temporal convolutional network (MS-TCN) for temporal pattern recognition, which uses predictions from the first stage to anticipate phases and instrument usage.

Ban et al. [2022] and Y. Jin et al. [2022] use multitask learning to perform workflow recognition as well as workflow anticipation.Ban et al. [2022] present SUPR-GAN, a generative adversarial network (GAN) designed to predict future surgical phase trajectories in laparoscopic surgery. SUPR-GAN generates possible future phase trajectories with a generator and uses a discriminator to ensure phase accuracy. The generator, a CNN-LSTM with an encoder-decoder architecture, predicts future phase trajectories based on past video frames, while the discriminator, an LSTM, distinguishes between real and fake trajectories. Although primarily focused on future phase prediction, SUPR-GAN employs a multitask learning framework by integrating current phase recognition and future phase prediction tasks. This multitask approach allows the model to utilize shared temporal features, enhancing predictive accuracy and robustness. The authors use per-transition accuracy to evaluate phase transitions and Levenshtein distance [Levenshtein 1966] to measure the similarity between predicted phase sequences and the ground truth.

Table 2.7: Methods that utilize multitask learning for skill assessment. * HP - hard parameter sharing.

	Publication	Tasks	Optimization	Architecture	Speed(fps)
skill assessment	Jian et al. [2020]	surgical skill level classification surgical skill score prediction	linear scalarization	shared encoder with multiple task branches (HP)	-
	T. Wang et al. [2020]	surgical gesture recognition surgical skill level classification(global) surgical skill score prediction(global) skill score prediction(intermediate) gesture evaluation(intermediate)	uncertainty weighting	multistage network (HP)	-

Y. Jin et al. [2022] presents Trans-SVNet, a novel hybrid embedding aggregation model using Transformers and multitask learning to address surgical workflow analysis. It leverages both spatial and temporal embeddings to improve the performance of workflow recognition and workflow anticipation. The paper introduces a Transformer-based model that combines spatial and temporal embeddings, enabling better preservation of spatial details and improved temporal information extraction.The authors experimentally show that their model improves performance in both tasks on multiple surgical video datasets.

These five methods Twinanda et al. [2018]; Rivoir et al. [2020]; Yuan et al. [2022]; Ban et al. [2022]; Y. Jin et al. [2022] highlight several key benefits of multitask learning, such as enhanced predictive accuracy through shared representation, robustness against noise and sparsity [Rivoir 2020; Jin 2022], and real-time inference capabilities without compromising accuracy [Ban 2022; Yuan 2022]. Despite their methodological differences, all five studies underscore the importance of integrating multiple tasks to improve predictive performance and robustness. While phase anticipation is valuable, it would also be beneficial to test these phase anticipation techniques on other public datasets, particularly those with finer granularity than phases. For instance, the PSI-AVA dataset [Valderrama 2022] with 21 steps or the MultiBypass140 dataset [Lavanchy 2024] with 46 steps could provide further insights.

2.4.5 MTL for surgical skill assessment

Traditionally, surgical skill assessment involves senior surgeons observing and evaluating less experienced counterparts using standardized rating checklists such as the Objective Structured Assessment of Technical Skills (OSATS) [Martin 1997]. However, the demand for training more doctors exceeds the number of experienced surgeons, making an automated system for scoring a surgeon's skill invaluable for trainees. In this subsection, we discuss works that utilized multitask learning for surgical skill assessment. The JHU-ISI Gesture and Skill Assessment Working Set (JIGSAWS) dataset [Gao 2014] is the standard benchmark for training and evaluating surgical gestures and skills [Lam 2022] in minimally invasive surgeries. Common metrics used in these evaluations are accuracy for gesture recognition and Spearman's correlation (ρ_{GRS} and ρ_{OSATS}) for skill scores. Table 2.7 summarizes the papers discussed in this subsection, highlighting the multiple tasks addressed, the optimization approaches adopted, the architectural strategies used for multitask learning, and the reported speeds for each study.

In Jian et al. [2020], the authors introduce a multitask network designed to tackle the dual objectives of overall surgical skill level classification (categorizing surgeons as expert, intermediate, or novice) and attribute score regression (modified OSATS attributes from datasets like JIGSAWS [Gao 2014]). This network architecture revolves around a shared encoder featuring two heads: one for classification and the other for the regression task. The core of this encoder is a variation of I3D [Carreira 2017], a proven feature extractor in the domain of action recognition. To enhance the I3D capabilities, the encoder incorporates attention modules that enable it to focus on crucial segments of the video clips. To facilitate efficient computation, the authors split the input videos into K equal parts, and clip snippets are sampled within these segments. Subsequently, a classification head is employed to predict the overall surgical skill assessment score, while a regression head predicts individual attribute scores. The classification and regression scores for each snippet are then aggregated, with the average of output features forming the final solution.

Table 2.8: Methods that utilize multitask learning for surgical report generation. *HP - hard parameter sharing.

	Publication	Tasks	Optimization	Architecture	Speed(fps)
report generation	Seenivasan et al. [2022]	tool segmentation tool-tissue interaction	sequential training	shared encoder with interacting task branches (HP)	-
	Seenivasan et al. [2023]	tool-tissue interaction scene caption optimization	sequential training	shared encoder with multiple task branches (HP)	-

In T. Wang et al. [2020], the focus is on achieving interpretable skill assessment inspired by how senior clinicians assess other surgeons' performance. The authors observe that senior clinicians assess how well each surgical gesture is executed. The authors propose a two-stage network to replicate this assessment method. The first stage primarily aims to detect surgical gestures in different clips along with predicting the skill level classification and skill level score as auxiliary tasks. The surgical video input is initially split into C clips, designed to streamline the computational process. These clips are then processed by a shared C3D encoder adapted from D. Tran et al. [2015]. The features obtained from all C clips are concatenated and used as the input to the decoder. The surgical gesture recognition decoder adopts a

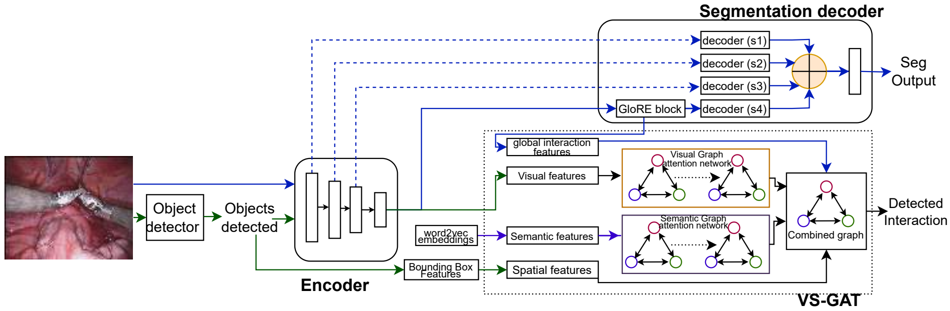

Figure 2.14: The proposed architecture for globally-reasoned multi-task surgical scene understanding in Seenivasan et al. [2022] for performing instrument segmentation and tool-tissue interaction. The model features a feature encoder, global and local reasoning for instrument segmentation and a VS-GAT [Liang 2021] model for interaction detection.

multi-stage temporal convolutional network. In parallel, the skill level classification and skill level prediction are executed through a shared LSTM, with classification and regression heads. The second stage focuses on the task of predicting skill assessment for the gestures identified in the first stage. To achieve this, the predicted gestures from the initial stage serve as a basis for segmenting the original surgical video into distinct gesture clips. These clips are then utilized as input for a C3D network, initialized with the weights from the C3D encoder employed in the first stage. Similarly, the decoder for predicting gesture level immediate skill score is implemented using an LSTM, featuring a dedicated head for gesture level immediate skill score prediction. Loss weighting with uncertainties Cipolla et al. [2018] was utilized for multitask optimization. On the dataset, the authors extend the JIGSAWS dataset by having surgeons annotate each gesture with a binary label of good(1) or bad(0).

Both T. Wang et al. [2020] and Jian et al. [2020] demonstrate improvements over single-task surgical skill assessment models. However, the JIGSAWS dataset, released in 2014, remains the standard benchmark for surgical skill assessment [Lam 2022]. Given its relatively small size compared to contemporary deep learning datasets, there is a need for larger and more comprehensive datasets in the field.

2.4.6 MTL for surgical report generation

The automatic generation of surgical reports can free surgeons and nurses from the tedious task of document entry, allowing them to focus more on patients and post-operative interventions [Xu 2021; Lin 2022]. Currently, in the existing literature, the approach to the surgical report generation task primarily revolves around frame-by-frame scene captioning [Xu 2021; Lin 2022; Seenivasan 2022b; Seenivasan 2023b]. The increasing interest in surgical report generation is closely tied to advancements in scene captioning and scene graph generation within the broader context of scene captioning research [Liang 2021; Cornia 2020] Notably, MTL has been applied to this field, incorporating aspects like image captioning and scene graph generation alongside other tasks to further enhance the overall capabilities of the system. The benchmark dataset, an extension of the EndoVis2018 segmentation dataset to include tool-tissue interactions, is used to form captions via sentence templates [Islam 2020a]. We term this dataset EndoVis2018-with-interactions. The metrics used include tool-tissue interaction accuracy (ACC), mean average precision (mAP), recall (RE), and other metrics depending on the predicted tasks.

Table 2.8 summarizes the papers discussed in this subsection, highlighting the multiple tasks addressed, the optimization approaches adopted, the architectural strategies used for multitask learning, and the reported speeds for each study.

In Seenivasan et al. [2022], a new approach is introduced for optimizing scene graphs to facilitate object-to-object interactions, focusing on the surgical report generation automation task. The authors frame this challenge as a multitask problem involving two core tasks: instrument segmentation and tool-tissue interaction. The authors propose a network with a single encoder and two decoders, as visualized in Fig. 2.14. The process commences with a preprocessing step, which generates bounding boxes and their corresponding classifications. These bounding boxes serve as a foundation for relationship modelling. The encoder, based on ResNet18, is responsible for shared feature extraction. The first decoder concentrates on local and global reasoning for instrument segmentation, configured as a U-net-like decoder with two noteworthy differences. The first is that the outputs of each lower-resolution decoder block do not feed into the next higher-resolution. Instead, each decoder block produces an output directly, which is concatenated together and passed through a conv block to produce the final segmentation prediction. Additionally, a GloRE block [Chen 2019b] is integrated into the output features of the encoder, enriching the global reasoning capabilities in the feature space. The features obtained from the GloRE block, known as Global Reasoned features, are also leveraged in the tool-tissue interaction decoder. The tool-tissue interaction decoder is structured as a scene graph from which the surgical report is generated. The authors employ the visual-semantic graph attention network (VS-GAT) [Liang 2021]. This network essentially comprises two components: a visual graph attention network and a semantic graph attention network. Both networks are harmoniously combined to create a unified graph. Notably, the authors introduce global interaction features from the GloRE block into this combined graph before instrument-tissue interaction prediction. The multitask optimization is a variation of the ideas of sequential training from Islam et al. [2019].

Table 2.9: Methods that utilize large models for multiple tasks. *PE - post encoding i.e. speed assuming image encodings provided.

	Publication	Tasks	Optimization	Architecture	Speed(fps)
Visual Question Answering	Seenivasan et al. [2022]	VQA	single loss	modified VisualBERT	-
	Seenivasan et al. [2023]	VQA	single loss	modified GPT-2	-
Promptable Segmentation	Zhou et al. [2023]	promptable segmentation image reconstruction	linear scalarization	modified CLIP	22(PE)
	A. Wang et al. [2023]	promptable segmentation	zero-shot inference	SAM	20(PE)
	Ma et al. [2023]	promptable segmentation	combined loss	SAM	-
	Yue et al. [2024]	promptable segmentation	combined loss	modified SAM	2
	Paranjape et al. [2023]	promptable segmentation	combined loss	modified SAM	-

Seenivasan et al. [2023] explore surgical report generation as a two-task problem of scene graph optimization for object-to-object relationships in a scene and frame-by-frame image captioning. The network architecture developed for this purpose is built around a shared encoder and two decoders. Based on the standard ResNet-18, the shared encoder forms the foundational feature extraction component. The object-to-object scene graph optimization decoder leverages the visual-semantic graph attention network from Liang et al. [2021]. This aspect of the network focuses on enhancing the understanding of complex relationships between objects within the surgical scene. In parallel, the frame-by-frame image captioning task is addressed through the utilization of a meshed-memory transformer from Cornia et al. [2020]. The optimization process is facilitated through a sequential training method similar to ATO [Islam 2020c]. Recognizing the challenge of model generalization to target domains and the need to accommodate factors like new instruments, the authors introduce a loss function inspired by continual learning, termed class incremental contrastive loss.

The surgical reporting task is still in its early stages. While the EndoVis2018-with-interactions dataset, which incorporates sentence templates and classification questions, has been useful, it may not fully capture the complexity of natural language used in real-world surgical reports. Consequently, the relevance of captioning metrics and tasks based on this dataset can be questioned. Developing a dataset based on actual surgical reports generated by surgeons in day-to-day procedures would likely provide more valuable insights and improve the robustness of surgical report generation systems.

2.4.7 Large models for solving multiple tasks

Large models pretrained on extensive datasets, such as GPT-2 [Radford 2021] and SAM [Kirillov 2023], have demonstrated remarkable generalization capabilities.

They exhibit emergent properties, such as the ability to solve multiple tasks by utilizing task-promptable engineering [Radford 2019; Kirillov 2023]. Task-promptable engineering allows these models to adapt to various tasks based on given prompts, making them highly versatile and generalizable backbones capable of zero-shot learning or fine-tuning for different downstream applications [Su 2024]. In the current literature, large models exhibit this capability of solving multiple tasks in surgical scene understanding, with fields such as task-promptable segmentation, and visual question answering being explored.

Table 2.9 summarizes the papers discussed in this subsection, highlighting the multiple tasks addressed, the optimization approaches adopted, the architectural strategies used for multitask learning, and the reported speeds for each study.

Visual question answering

Seenivasan et al. [2022] introduce the problem of visual question answering (VQA) in the context of surgery. They develop a large model capable of providing textual answers to a wide range of questions, including tasks such as tissue presence recognition, instrument localization, tool presence recognition, and action recognition. To achieve this, they expand the EndoVis2018 and Cholec80 datasets by adding sentence-based and classification-based answers to a predefined set of questions. The authors propose two models based on the VisualBERT architecture [Li 2020] for classification-based and sentence-based answering in surgical VQA. They improve the VisualBERT encoder by introducing cross-token and cross-channel submodules to enhance the interaction between visual and text tokens. For the classification-based model, they utilize linear classifiers, with initial sentence classification determining the appropriate linear layer. The sentence-based model employs a standard transformer decoder.

In subsequent work, SurgicalGPT [Seenivasan 2023a] for the surgical VQA task, the authors replace the VisualBERT architecture with GPT-2 [Radford 2019], enhancing it with a vision encoder. They concatenate tokens from the text and vision encoders before feeding them to the GPT decoder, which they refer to as LV-GPT. Additionally, they change the unidirectional attention to a bidirectional attention model in the GPT decoder. Notably, they report improved performance compared to Seenivasan et al. [2022].

Seenivasan et al. [2022]; Seenivasan et al. [2023] generated multiple VQA datasets by converting existing interaction-based minimally invasive surgery datasets, such as Endovis-with-interaction, Cholec80, and PSI-AVA, into Visual Question Answering (VQA) datasets. They demonstrated improvements over state-of-the-art VQA models for medical applications using these datasets. Models trained on classification-based VQA datasets perform tasks such as counting, interaction recognition, localization, and classification by utilizing text prompts. However, it would be interesting to see these models applied to more complex VQA datasets with more complex answers, that require a really deep understanding of the surgical scene.

Promptable segmentation

Another application of the concept of generalizable tasks is *promptable segmentation*, which primarily focuses on binary segmentation for ideas represented in various forms, including text [Zhou 2023], points [Kirillov 2023], bounding boxes [Kirillov 2023], and reference images [Lüddecke 2022]. This approach allows for the execution of binary segmentation, instance segmentation, part segmentation, and instrument-type segmentation, provided that suitable prompts are provided.

An example of promptable segmentation in surgical scene understanding can be found in the work of Zhou et al. [2023]. Their work addresses the challenge of distinguishing and segmenting diverse surgical instruments using textual prompts. To achieve this, they leverage pretrained CLIP text and vision encoder foundational models and introduce a novel text promptable mask decoder. The authors report capabilities to generalize text promptable segmentation to tissue segmentation, instrument-part, and instrument-type segmentation. Notably, their method demonstrates strong generalization capabilities, as evidenced by robust performance in cross-dataset evaluations.

Recent advancements in promptable segmentation have seen remarkable progress through methods based on the Segment Anything Model (SAM) [Kirillov 2023]. SAM serves as a foundational model that accommodates different prompting strategies, namely: points, bounding box, segmentation mask, and text.

SAM has demonstrated exceptional generalization abilities and has found applications in surgical segmentation tasks. An empirical study conducted in A. Wang et al. [2023] examines the robustness and generalizability of SAM when applied to the EndoVis2017 and EndoVis2018 datasets. Their findings reveal that pretrained SAM excels in terms of generalizability, especially when used with bounding box prompts, achieving state-of-the-art results. However, it is important to acknowledge that comparing bounding box prompts to class-based segmentation techniques might not be entirely fair.

A. Wang et al. [2023] also observe that SAM does not perform well for surgical instrument segmentation with point-based prompts, and its generalizability can be compromised under conditions of data corruption commonly encountered in surgical segmentation tasks. In summary, while SAM exhibits strong generalization capabilities, it relies on users to supply real-time bounding boxes for each image. In addition, SAM may not be robust in the presence of noise and data corruption. To address these challenges, several innovative approaches have been proposed.

MEDSAM [Ma 2023] is a foundational model designed for universal medical image segmentation, curated from a dataset comprising over one million images, including CT scans, histology images, and surgical scenes. This model aims to bridge the gap between SAM for natural images and SAM for medical images. It adopts bounding box prompts as input, with a key distinction being the freezing of the SAM encoders, while only training the SAM decoder.

SurgicalSAM [Yue 2024] proposes using the concepts of a prototype image, a real image of a particular instrument that captures the interested image, as a prompt instead of bounding boxes per image as prompts or directly using class names. More specifically, their method builds a memory bank of prototype images then they use the prototype-based class prompt encoder to exploit similarities between images in the dataset and class prototype images to create prompts, which they call prompt embeddings. In addition, the authors also propose a contrastive prototype learning loss to ensure that during training, the feature space for each different prototype is far from each other.

AdaptiveSAM [Paranjape 2023] addresses the bounding box requirement by utilizing text as prompts. It employs text embeddings from pretrained CLIP and passes them through a trainable affine layer before applying them to the SAM prompt decoder. A notable feature of AdaptiveSAM is the introduction of *bias tuning* as a more memory-efficient method to adapt the SAM encoders. This approach trains the bias of the multi-head attention layers in the SAM image encoder and the normalization layers, achieving adaptation with higher efficiency.

All these methods [Ma 2023; Yue 2024; Paranjape 2023] report significant improvement over the vanilla SAM approach when applied to surgical segmentation datasets over various segmentation tasks.

2.5 Public datasets for MTL in MIS

Table 2.10: A summary of publically available multitask learning datasets for minimally invasive surgeries with information on the dataset characteristics and annotation characteristics. The column N reports the number of procedures. Datasets annotated for a single task that may have been used in MTL research such as Endovis2017, Endovis2018 and Cholec80 are omitted from the table for clarity.

Dataset	Brief description	N	Task	Input	Annotations	Paper
AutoLaparo	A dataset for image-guided surgical automation in laparoscopic hysterectomy comprising three sub-datasets: surgical workflow recognition, laparoscope motion prediction, instrument and anatomy segmentation.	21	Phase recognition task	1388 minutes of surgical videos	1388 minutes of phase labels	[Wang 2022b]
			Laparoscope motion prediction	300 video clips	300 next motion mode labels	
			Instrument part segmentation	1800 keyframes	1800 part segmentation maps	
			Key anatomy segmentation	1800 keyframes	1800 tissue segmentation maps	
HeiSURF	A laparoscopic cholecystectomy designed for surgical activity recognition, full scene segmentation, and skill assessment. It is a parent dataset to the HeiChole dataset	33	Phase recognition	23.3 hours of surgical videos	23.3 hours of phase labels	[Bodenstedt 2021]
			Full scene segmentation	827 keyframes	827 full scene segmentation maps	
			Action recognition	5514 instances in frames	5514 action labels	
			Tool presence prediction	6980 instances in frames	6980 tool presence labels	
			Surgical skill-score prediction	99 video clips	495 skill scores	

Continued on next page

Table 2.10: A summary of publicly available multitask learning datasets for minimally invasive surgeries (continued).

Dataset	Brief description	N	Task	Input	Annotations	Paper
PSI-AVA	A dataset of robot-assisted radical prostatectomy designed for research into the complementary nature of surgical activity recognition tasks.	8	Phase recognition	20.45 hours of surgical videos	20.45 hours of phase labels	[Valderrama 2022]
			Step recognition	20.45 hours of surgical videos	20.45 hours of step labels	
			Action recognition	5804 instances in frames	5804 action labels	
			Instrument detection	5804 instances in frames	5804 bounding boxes with labels	
HeiCo	A dataset of three different colorectal procedures (proctocolectomy, rectal resection, and sigmoid resection procedures) with emphasis on dataset generalization and diversity.	30	Phase recognition	9.45 hours of surgical videos	9.45 hours of phase labels	[Maier-Hein 2021]
			Instrument instance segmentation	10,040 keyframes	10,040 instance segmentation maps	
MISAW	A micro-surgical anastomosis dataset with a focus on evaluating the impact of learning multiple surgical activity recognition tasks together.	27	Phase recognition	1.5 hours of surgical videos	1.5 hours of phase labels	[Huaulmé 2021]
			Step recognition	1.5 hours of surgical videos	1.5 hours of step labels	
			Action recognition	1.5 hours of surgical videos	1.5 hours of action labels	
			Tool presence prediction	1.5 hours of surgical videos	1.5 hours of tool presence labels	

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Table 2.10: A summary of publicly available multitask learning datasets for minimally invasive surgeries (continued).

Dataset	Brief description	N	Task	Input	Annotations	Paper
			target tissue prediction	1.5 hours of surgical videos	1.5 hours of tool target labels	
PE-TRAW	A dataset containing the peg transfer task in laparoscopic surgery training. It is designed for multiple surgical activity recognition tasks. This dataset is collected from a virtual reality simulator	150	Phase recognition	5.86 hours of surgical videos	5.86 hours of phase labels	[Huaulmé 2023]
			Step recognition	5.86 hours of surgical videos	5.86 hours of step labels	
			Action recognition	5.86 hours of surgical videos	5.86 hours of action labels	
			target tissue prediction	5.86 hours of surgical videos	5.86 hours of tool target labels	
			Full scene segmentation	5.86 hours of surgical videos	5.86 hours of scene segmentation	
Multi-Bypass 140	A phase and step recognition dataset, particularly focusing on datasets collected from different hospitals (multi-centric) and the importance of multi-centric datasets for generalization.	140	Phase recognition	214hrs of surgical videos	214hrs of phase labels	[Lavanchy 2024]
			Step recognition	214hrs of videos	214hrs of step labels	
SAR-RARP50	A dataset of suturing segments of robotic assisted radical prostatectomy. The dataset focuses on tool segmentation and surgical action recognition.	50	Phase recognition	3 hours of surgical videos	3 hours of action labels	[Psychogyios 2023]
			Instrument segmentation	1000 keyframes	1000 Instrument segmentation maps	

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Table 2.10: A summary of publicly available multitask learning datasets for minimally invasive surgeries (continued).

Dataset	Brief description	N	Task	Input	Annotations	Paper
SARAS-MESAD	A dataset containing both real and virtual human prostatectomy procedures for research into surgical activity recognition and instrument detection with a focus on cross-domain learning	9	Instrument detection	59k frames	59k frames with bounding boxes	[Cuzzolin 2021]
			Action recognition	59k frames	59k frames with action labels	
Cholec T50	The CholecT50 is designed for fine-grained action recognition and tool-tissue interaction in laparoscopic cholecystectomy surgeries. CholecT50 is the latest in the Cholec series of datasets.	50	Tool presence recognition	100.9k frames	100.9k frames with tool presence labels	[Nwoye 2022b]
			Action recognition	100.9k frames	100.9k frames with action labels	
			Tissue target recognition	100.9k frames	100.9k frames with target labels	
ART-Net	A non-robotic laparoscopic hysterectomy dataset designed for 3D graphics applications like 3D measurement and Augmented Reality (AR).	29	Tool presence prediction	1500 frames	1500 frames with tool presence labels	[Hasan 2021]
			Binary instrument segmentation	635 keyframes	635 binary instrument segmentation maps	
			Geometric map prediction	635 keyframes	635 geometric map labels	

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Table 2.10: A summary of publicly available multitask learning datasets for minimally invasive surgeries (continued).

Dataset	Brief description	N	Task	Input	Annotations	Paper
EndoVis-18-VQA	A dataset designed for advancing research in visual question answering using the Endovis2018 dataset. It contains classification-based questions about tissues, actions and tool locations.	14	VQA	11.7k questions	2k frames	[Seenivasan 2022a]
Cholec 80-VQA	A Surgical VQA dataset based on the Cholec80 dataset. It contains classification-based questions about surgical phases and instrument presence.	40	VQA	43k questions	21.6k frames	[Seenivasan 2022a]
PSI-AVA-VQA	A Surgical VQA dataset based on the PSI-AVA dataset . It contains classification-based questions about the surgical phase, step, and location of surgical tools.	8	VQA	10.3k questions	2.2k frames	[Seenivasan 2023a]
SSG-VQA	A Surgical VQA dataset comprising images from the CholecT45 dataset. It contains complex questions about tool location and presence, tissue location and presence, instrument action, instrument targets, object colours, and their relationships	45	VQA	960k questions	25k frames	[Yuan 2024]

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Table 2.10: A summary of publicly available multitask learning datasets for minimally invasive surgeries (continued).

Dataset	Brief description	N	Task	Input	Annotations	Paper
JIG-SAWS	An automatic gesture recognition and surgical skill assessment dataset containing videos of surgeons performing suturing, knot-tying and needle-passing on a bench-top model with a da Vinci.	104	Surgical gesture classification	208 minutes of surgical videos	208 minutes of surgical gesture labels	[Gao 2014]
			Surgical skill-score prediction	104 video clips	104 global surgical skill scores	
Endoscapes	Endoscapes dataset was designed for advancing research on the Critical View of Safety (CVS), segmentation and object in Laparoscopic Cholecystectomy surgeries.	201	CVS prediction	58.8k frames	58.8k frames with CVS labels	[Murali 2023]
			Scene object detection	11k frames	11k frames with bounding boxes	
			Scene instance segmentation	422 frames	422 frames with instance seg masks	

In this section, we explore the public datasets curated to support MTL for MIS. These datasets offer a valuable foundation for researchers to experiment, innovate, and address real-world MIS challenges.

Due to the inherent difficulties in producing surgical video datasets, the availability of public datasets suitable for MTL in this domain is currently limited. While there are some datasets designed specifically for MTL, such as the *multi-granularity surgical activity recognition datasets*, the number of datasets catering to other applications of MTL in surgical vision remains scarce. Nevertheless, it is worth noting that some authors have successfully explored approaches to leverage single-task datasets for MTL by generating self-supervised auxiliary tasks from the existing data. While we acknowledge that some single task datasets such as Endovis2017 [Allan 2019], Endovis2018 [Allan 2020], Cholec80 [Twinanda 2016a], and m2cai [Jin 2018] have played a key role in some MTL research, we do not detail these single task datasets in this section. Moreover, there are other multitask datasets that have received less attention in the literature, which we aim to highlight. We present a comprehensive list of datasets designed to address multiple tasks in surgical vision.

For a quick overview of datasets designed for multiple tasks, along with information such as the amount of of images and labels, and other relevant information, readers can refer to Table 2.10. We do not include private datasets (e.g. ByPass40 [Ramesh 2021]). For more information about surgical tool datasets, including non-multitask learning datasets, we recommend the following online resource: https://github.com/luiscarlosgph/list-of-surgical-tool-datasets.

2.5.1 Integrated Multi-tasks for Image-guided Surgical Automation in Laparoscopic Hysterectomy Dataset (AutoLaparo)

The AutoLaparo dataset [Wang 2022b] supports image-guided surgical automation in laparoscopic hysterectomy, offering three sub-datasets: phase recognition, laparoscope motion prediction, and instrument segmentation with key anatomy annotation. The 21 procedures are recorded at 25 fps with a resolution of 1920x1080 pixels. These procedures are annotated with phase labels (7 phases). The laparoscope motion prediction sub-dataset consists of 300 clips, extracted from phases 2-4 of recorded procedures, each annotated with one of seven motion modes (up, down, left, right, zoom-in and zoom-out). The segmentation sub-dataset includes instruments (4 instruments) and key anatomy (1 anatomy) segmentation for keyframes in the motion prediction clips.

2.5.2 Augmented Reality Tool Network dataset (ART-Net)

The ART-Net dataset [Hasan 2021] is tailored for non-robotic laparoscopic hysterectomy, emphasizing 3D graphics applications. Annotations cover tool presence detection, binary tool segmentation, and 2D pose estimation. Extracted from 29 procedures, the dataset provides frames with and without instruments, annotating these frames for tool presence. Keyframes are annotated for binary tool segmentation and 2D pose in the form of geometric primitives (tool-tip, midline and instrument shaft heatmaps).

2.5.3 HeiChole Surgical Workflow Analysis and Full Scene Segmentation dataset (HeiSURF)

The HeiSURF dataset [Bodenstedt 2021] for day-to-day laparoscopic cholecystectomy includes annotations for phase recognition, full scene segmentation, action recognition, instrument presence detection, and skill assessment. Each procedure in the dataset is annotated for phase (7 phases), and keyframes are annotated for full scene segmentation and actions. Three videos per procedure are extracted and annotated with 5 skill annotations. The HeiSURF dataset is a parent dataset of an earlier dataset - the HeiChole dataset (This did not contain segmentation information) [Wagner 2023].

2.5.4 Phase, Step, Instrument, and Atomic Visual Action recognition dataset (PSI-AVA)

The PSI-AVA dataset [Valderrama 2022] focuses on robot-assisted radial prostatectomy for phase recognition, step recognition, instrument presence, instrument bounding boxes, and action recognition. The whole dataset is labelled for phase (11 phases) and step (21 steps) recognition. Keyframes annotated with bounding boxes provide detailed instrument detection (7 instruments) and corresponding actions (16 actions).

2.5.5 The Heidelberg Colorectal Dataset for Surgical Data Science in the Sensor Operating Room dataset (HeiCo)

The HeiCo [Maier-Hein 2021] dataset features colorectal procedures for instrument segmentation and phase recognition. The dataset, derived from 30 procedures, covers proctocolectomy, rectal resection, and sigmoid resection. Phase recognition (14 phases) annotations are provided for the whole dataset. Keyframes with instance segmentation labels are also provided. A 10-second video snippet preceding each annotated keyframe is provided for temporal context. HeiCo contains data on surgical workflow analysis from the sensorOR challenge [Maier-Hein 2021] and data from the Robust-MIS segmentation dataset [Ross 2020].

2.5.6 The John Hopkins University - Intuitive Surgical Inc. Gesture and Skill Assessment Working Set dataset (JIGSAWS)

JIGSAWS dataset [Gao 2014] contains annotations for manipulator gestures and surgical skills collected from recorded trials of eight medical doctors with varying skill levels performing suturing, knot-tying, and needle-passing task with the da Vinci Robot Surgical System (DSS). The dataset includes kinematic data (76D vector capturing the position, orientation, velocity and angle of the manipulator and camera), synchronized stereo video recordings (each at 30fps for approximately 2 minutes), and annotations for automatic gesture recognition (15 gestures) and skill assessment (a modified form of OSATS [Martin 1997]).

2.5.7 The MIcro-Surgical Anastomose Workflow recognition on training sessions dataset (MISAW)

MISAW dataset [Huaulmé 2021] focuses on multi-granularity surgical activity recognition in micro-surgical anastomosis. The dataset consists of 27 sequences recorded using a stereo-microscope, along with annotations for phase (2 phases), step (6 steps), action (17 actions), instrument presence (1 instrument), and instrument tissue targets (9 targets) for each frame. Synchronized kinematic data are also provided.

2.5.8 PEg TRAnsfer Workflow recognition by different modalities dataset (PETRAW)

The PETRAW dataset [Huaulmé 2023] is designed for workflow recognition in peg transfer training sessions. The dataset comprises 150 sequences of peg transfer sessions recorded on a virtual reality simulator, kinematic data, videos, and annotations for semantic segmentation (2 targets and 1 instrument), phase (2 phases), step (12 steps), and action annotations (6 actions).

2.5.9 Surgical Instrumentation Segmentation and Action Recognition on Robot-Assisted Radical Prostatectomy dataset (SAR-RARP50)

The SAR-RARP50 dataset [Psychogyios 2023] is designed to address data-scarcity challenges in surgical action recognition and tool segmentation for in vivo Robotic Assisted Radical Prostatectomy. The dataset comprises of 50 suturing segments acquired with a DaVinci Si Robot that features a stereo endoscope. The dataset is annotated for instrument segmentation (9 instruments) at keyframes and action (8 actions) for the whole dataset.

2.5.10 The CholecT50 dataset

CholecT50 [Nwoye 2022b] is a dataset for furthering the research of tool-tissue interactions, formalized as action triplets ⟨instrument, action, target⟩ for laparoscopic cholecystectomy procedures. It consists of 50 laparoscopic cholecystectomy procedures annotated with action triplets. As it is a subset of the Cholec120 dataset, it also contains relevant annotations from Cholec120 such as phase recognition, and tool presence labels for the relevant procedures. CholecT50 is the most comprehensive iteration of action triplet datasets released by the CAMMA research group which also includes CholecT40 [Nwoye 2020], and CholecT45. Nwoye et al. [2022]

gives in-depth information about the CholecT50 and related datasets, the benchmarks, metrics, and other relevant information.

2.5.11 Multi-centric Multi-activity Dataset laparoscopic Roux-en-Y gastric bypass (LRYGB) dataset(MultiBypass140)

The MultiBypass140 dataset [Lavanchy 2024] is designed for phase and step recognition, particularly focusing on datasets collected from different hospitals (multi-centric) and the importance of multi-centric datasets for generalization. The dataset comprises 140 sequences of laparoscopic Roux-en-Y gastric bypass (LRYGB) surgeries performed at two medical centres and annotations for phase (12 phases), and steps (46 steps). MultiBypass140 is an extension of the Bypass40 [Ramesh 2021] dataset.

2.5.12 The Endoscapes Dataset

The Endoscapes dataset was designed for advancing research in automated assessment of surgical scenes, particularly focusing on the Critical View of Safety (CVS), segmentation and object in Laparoscopic Cholecystectomy surgeries [Murali 2023]. The dataset comprises 201 videos and annotations for CVS(3 labels), bounding boxes for anatomy(5 classes) and a surgical instrument (1 class), as well as segmentation masks for these bounding boxes (6 classes).

2.5.13 SARAS challenge on Multi-domain Endoscopic Surgeon Action Detection dataset (SARAS-MESAD)

SARAS-MESAD dataset [Cuzzolin 2021] facilitates surgical activity recognition research and cross-domain learning. It includes MESAD-Real and MESAD-Phantom sub-datasets, offering annotated frames for instrument detection and action recognition for human prostatectomy and phantom surgeries. This dataset builds on the previous SARAS-ESAD dataset [Bawa 2021].

2.5.14 The EndoVis-18-VQA dataset

The EndoVis-18-VQA dataset [Seenivasan 2022a] was designed for advancing research in visual question answering using the Endovis2018 dataset. The dataset comprises images from the 14 sequences in the Endovis2018 dataset and contains classification-based questions and answers about tissues, actions and tool locations from the Endovis2018 annotations. This dataset is the more recent version of Endovis2018-with-interactions discussed in Section 2.4.6

2.5.15 Cholec80-VQA

The Cholec80-VQA dataset was designed for advancing research in visual question answering using the Cholec80 dataset [Seenivasan 2022a]. The dataset comprises images from 40 sequences in the Cholec80 dataset and contains classification-based questions on surgical phase and instrument presence from the Cholec80 annotations.

2.5.16 PSI-AVA-VQA

The PSI-AVA-VQA dataset was designed for advancing research in visual question answering using the PSI-AVA dataset [Seenivasan 2023a]. The dataset comprises images from 8 sequences in the PSI-AVA dataset and contains classification-based questions on surgical phase, step, and location annotations from the PSI-AVA annotations.

2.5.17 Surgical Scene Graph-based dataset (SSG-VQA)

The SSG-VQA dataset was designed for advancing research in visual question answering, in laparoscopic cholecystectomy surgeries [Yuan 2024]. The dataset comprises images from the 45 sequences of the CholecT45 dataset [Nwoye 2022b], interaction labels(⟨instrument, action, target⟩) from the CholecT45 dataset, bounding boxes generated by pseudo-labelling using models trained on m2cai16-tool-locations [Jin 2018] and CholecSeg8k [Hong 2020], and questions for various tasks such as questions on spatial locations of instruments and tissues, relationships between objects in a scene, generated by a well-curated question template engine with strategies to address the class imbalance and remove poorly formulated questions that are not challenging. Questions were designed to be more complex than other surgical VQA datasets available.

2.6 Discussion and conclusion

In this section, we consolidate our insights and observations from the MIS-related papers in Section 2.4 and connect them with generic MTL techniques for natural images introduced in Section 2.3. We also draw parallels with recent trends in the deep learning community.

2.6.1 Learning multiple tasks together in the MIS domain

A prominent observation from the reviewed papers in Section 2.4 is that there are successful applications of MTL in minimally invasive surgeries (MIS). . We have observed several methodologies for learning perceptual tasks together in this context [Huang 2022a; Sanchez-Matilla 2021; Islam 2020c; Baby 2023; Zhao 2022; Das 2023; Psychogyios 2022]. While most reported positive outcomes [Huang 2022a; Sanchez-Matilla 2021; Islam 2020c; Baby 2023; Zhao 2022; Das 2023], others indicate negative transfer effects or negligible improvements [Psychogyios 2022], demonstrating the need for in-depth investigations on which tasks assist each other or regularly cause negative transfer, similar to the work of Standley et al. [2020]. Furthermore, we found a lack of studies focusing on learning perceptual tasks like motion flow, and normal estimation, suggesting an unexplored research direction or that these tasks may not work well in the multitask learning framework.

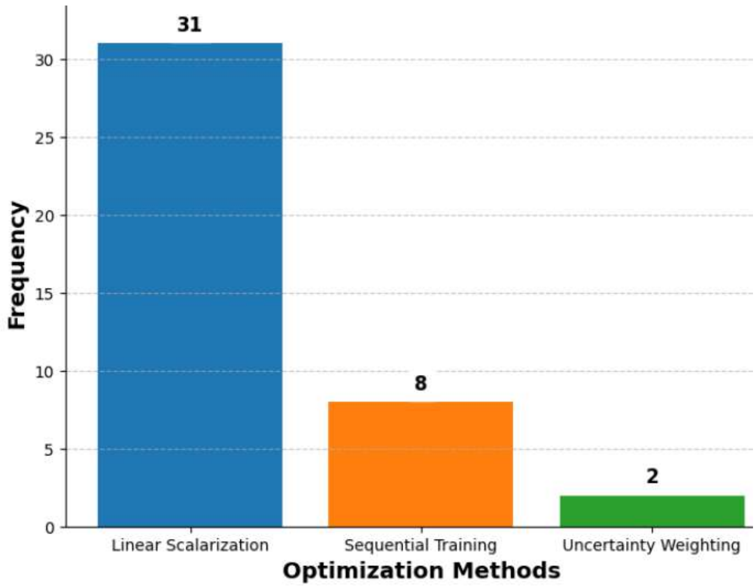


Figure 2.15: Prevalence of linear scalarization, sequential training, and uncertainty weighting optimization approaches among reviewed surgical vision works.

The most common optimization method which was utilized in the reviewed surgical vision works was linear scalarization. Custom sequential training schemes were also utilized, followed by the uncertainty weighting approach [Cipolla 2018]. Fig. 2.15 illustrates the prevalence of these three methods in the reviewed literature. Hence, an interesting future research question would be to systematically compare the efficiency of various multitask optimization techniques for learning multiple tasks in MIS compared to standard linear scalarization with grid search similar to the works of Kurin et al. [2022] and Xin et al. [2022]. There is a noted rarity of multitask optimization methods in MIS, which should be investigated.

The architecture of how multiple tasks are learned together in the context of MIS presents an interesting observation. In all surgical vision MIS papers reviewed, the standard approach involves the use of hard parameter sharing. The architectures used in these surgical vision works can be categorized into four main types: 'shared encoder with multiple task branches',' multistage network', 'shared encoder with interacting task branches', and 'unique'. The 'unique' category includes architectures with distinctive configurations, such as multiple encoders or transformers, where the task structure deviates significantly from conventional designs. Details on these architecture types for each reviewed work can be found in Table 2.1, Table 2.2, Table 2.3, Table 2.6, Table 2.7 and Table 2.8.

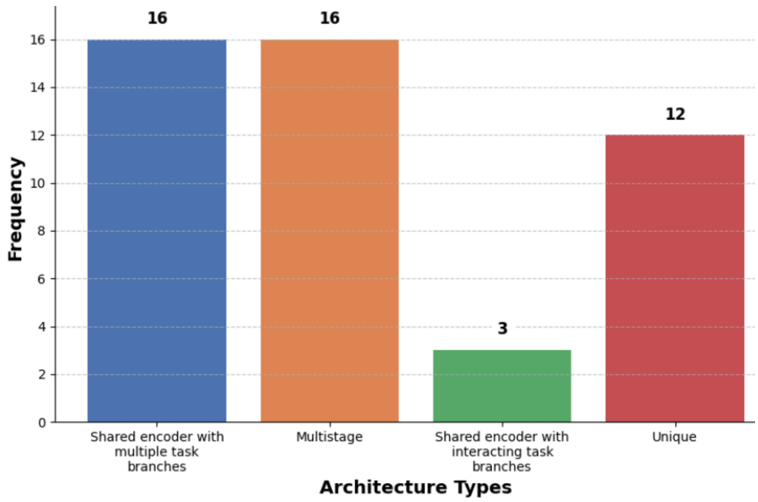


Figure 2.16: Prevalence of the architecture types in the reviewed surgical vision works.

Fig. 2.16 illustrates the prevalence of each architecture type among the reviewed works, with 'shared encoder with multiple tasks branches' and 'multistage' architectures as the most common architectures. However, one noticeable absence is the use of soft parameter sharing in MTL for MIS. Soft parameter sharing can allow tasks to share information at different levels and determine how to share information as discussed in Section 2.3. This could be because of the complicated nature of soft parameter sharing networks or that hard parameter sharing is just good enough in most MIS scenarios. An area of exploration in future research could be to determine if soft parameter sharing can provide advantages over hard parameter sharing in the MIS context by comparing and contrasting the performance of both architectural styles on minimally invasive surgeries.

Another observation pertains to the impact of auxiliary tasks on learning accuracy. While many studies have shown the advantages of auxiliary tasks in enhancing task performance [Lin 2020; Bhattarai 2023; Huang 2022b], it is essential to remain critical and consider possible negative transfers. The choice between directly incorporating domain-specific information into the primary network or predicting this information as an auxiliary task should be further explored. For instance, using contour prediction as an auxiliary task [Qin 2020a] or applying boundary loss [Kervadec 2019] may yield different outcomes in terms of both efficiency and informativeness.

Another noteworthy point is that only a few works among the reviewed surgical vision papers incorporate information flow between decoders in the MIS field, specifically through the 'shared encoder with interacting task branches' architecture [Nwoye 2022b; Sharma 2023a; Seenivasan 2022b]. This trend is illustrated in Fig. 2.16 , which displays the prevalence of different architectures among the reviewed surgical vision works. The additional connectivity between decoders can facilitate inter-task relationship learning, which is a unique advantage of MTL. By allowing tasks to influence and inform each other, models can develop a deeper understanding of the underlying relationships between tasks and potentially improve overall performance. While learning better representations through MTL is valuable, it is essential to acknowledge that there are other learning paradigms, such as self-supervised learning, which excel in representation learning. However, the specific advantages of MTL in MIS, including inter-task relationship learning, make it a compelling choice for solving complex surgical tasks.

2.6.2 Multitask learning for automatic camera control

The discussion on automatic camera control in the context of MIS raises some interesting points. While the scanpath prediction task has been a significant development, it may not provide a comprehensive solution for automating camera movements in these surgeries. The approach of focusing primarily on surgical instruments [Islam 2019b; Islam 2020b], while valuable, may overlook the importance of capturing the surgical field comprehensively, including tissues and organs. Surgical procedures often involve dynamic movements where instruments enter and exit the field of view, and the camera's focus may need to adapt to these changes, as detailed by Brigham et al. [2019]. This highlights the complexity of the task, where understanding the surgical context and deciding what to focus on is crucial.

An emerging approach to automating camera control is camera imitation and surgical intent learning [Huber 2023; Li 2022a]. While this field is still relatively new, and there are only a handful of papers exploring it, it holds promise for improving camera control in minimally invasive surgeries. The release of datasets like AutoLaparo is expected to drive more interest and research in this area. Incorporating MTL into camera control, where tasks include predicting future segmentation and motion for the camera, is great. The future development of this field may involve adding more tasks, such as predicting ongoing and future actions, surgical steps and phases, to provide a holistic solution for camera control.

As the field of automatic camera control in MIS matures, it will be fascinating to see how researchers tackle the challenges of understanding and target domain generalization of surgical scenes to enhance camera movements and, ultimately, improve the surgical experience and outcomes.

2.6.3 Multitask learning for surgical video workflow analysis

The evolution of surgical video workflow analysis from phase detection to action triplets and multi-granular activity detection is a notable development. This progression underscores the importance of understanding the intricate relationships between different aspects of surgical activities, leading to improved recognition of complex surgical procedures and their contextual understanding. Table 2.11 presents the reviewed studies in surgical video workflow analysis (along with the publication year), dataset used, tasks addressed, and architectural choice.

Table 2.11: Evolution of Surgical Activity Recognition.

Study	Tasks	Dataset	Architecture
Twinanda et al. [2016]	phase recognition tool presence detection	Cholec80	multistage
Twinanda et al. [2016]	phase recognition tool presence detection	Cholec80	multistage
Mondal et al. [2019]	phase recognition tool presence detection	Cholec80	multistage
Czempiel et al. [2020]	phase recognition tool presence detection	Cholec80	multistage
Y. Jin et al. [2020]	phase recognition tool presence detection	Cholec80	multistage
Sanchez-Matilla et al. [2022]	phase recognition scene segmentation	Cholec80 CholecSeg8k	multistage
Nwoye et al. [2020]	surgical action triplet recognition	CholecT40	encoder with multiple branches
Nwoye et al. [2022]	surgical action triplet recognition	CholecT50	shared encoder with interacting task branches
Sharma et al. [2023]	surgical action triplet recognition	CholecT45	shared encoder with interacting task branches
Yamlahi et al. [2023]	surgical action triplet recognition	CholecT45	encoder with multiple branches
Sharma et al. [2023]	surgical action triplet detection	CholecT50	multistage
Valderrama et al. [2022]	phase recognition step recognition Action recognition Instrument detection	PSI-AVA	multiple encoders with multiple task branches

In the early stages, much of the focus was on phase detection, which was partially influenced by the availability of datasets like Cholec80. Early models were primarily two-stage systems, incorporating temporal modelling to capture the dynamics of surgical activities. A significant shift has been observed in recent research, where surgical activity detection has been framed as the prediction of surgical action triplets, which include instrument, verb, and target. This approach provides a more detailed and informative way to describe surgical activities. It also highlights the need for understanding complex interactions between these components during surgical procedures.

A complementary direction is to move beyond recognising that an interaction occurs and determine which physical instrument instance performs it. This requires spatially grounding each predicted ⟨`instrument, verb, target`⟩ in an individual instrument rather than associating interactions only with the frame or instrument class. The subsequent technical chapters pursue this progression: Chapter 3 establishes instrument-instance supervision, Chapter 4 connects those instances to verb and target labels, and Chapter 5 investigates specialist instrument grounding followed by multimodal interaction prediction.

The exploration of multi-granularity research, as exemplified by the PSI-AVA and MISAW datasets, offers valuable insights into recognizing different levels of surgical activities. It would be interesting to see how these methods that are surgery-specific and models that are trained specifically for multiple granularities of this specific procedure are generalized to other surgical procedures, surgeon styles, and surgical contexts.

As the field of surgical activity recognition continues to evolve, addressing the challenge of adapting models to different surgeries and achieving broader generalization is essential. Future research may focus on creating more versatile models that can understand and recognize surgical activities across various surgical procedures, improving the practical utility of these techniques in clinical settings.

2.6.4 Multitask learning for report generation

The application of MTL to surgical report generation is a promising avenue for improving the documentation and record-keeping aspects of minimally invasive surgeries. This complex task requires the generation of detailed and organized information about the events and procedures that occur in the intra- or pre-operative period.

Two approaches have been explored as discussed in Section 2.4.6 : one involving training scene graphs with segmentation models to obtain reports [Seenivasan 2022b] and the other incorporating scene graphs and frame captioning techniques to generate reports [Seenivasan 2023b]. These approaches are initial steps towards addressing the challenge of surgical report generation, and they have shown potential for producing structured information from surgical data.

However, a more sophisticated dataset that captures the intricacies of real-life surgical reports and allows for a transition from frame-to-frame inferences to full video inferences is essential for advancing this field, as the currently used dataset, EndoVis2018-with-interactions, is limited by its reliance solely on pixel-level information from the images in EndoVis2018, as described in Section 2.5.14

2.6.5 Datasets, Unification and Ethics.

For multitask problems like surgical action triplets, there exists a standard benchmark dataset—the CholecT50. Designed for solving multiple tasks together, CholecT50 is large, encompassing over 100k frames across 50 surgeries, and includes metrics for evaluating both single tasks and relationships between tasks. It also provides a metrics library , facilitating easy comparison between methods and lowering entry barriers. The CholecT50 dataset is part of the Cholec series, which features similar annotation styles, shared images, and metrics, making it easily extendable.

As large models that solve multiple tasks become more prevalent in the field, the creation of more extensive multitask datasets will be essential. We encourage collaboration within the community to build comprehensive, unified datasets, benchmarks, and metrics. This collaborative effort will enhance the robustness and comparability of research outcomes.

Creating larger datasets involves challenges such as data collection, privacy, patient safety, and regulatory concerns. Collaboration with surgeons, institutions, regulatory bodies and the general public is crucial to address these issues. More initiatives that inform the public about data usage, its potential to improve healthcare, and the associated ethical concerns, and solutions need to be raised. Addressing regulatory concerns involves ensuring compliance with data protection laws and ethical guidelines, which can be facilitated through robust anonymization processes and transparent collaboration with regulatory bodies to establish clear protocols for data usage and sharing. Programs like the SAGES video donation program, which promotes the donation and anonymization of surgical videos and fosters collaboration among researchers, surgeons, and hospitals, are recommended.

2.6.6 Large models in surgical scene understanding

The trend towards large models with the capacity to solve multiple problems by predicting a universal task is an intriguing development in the field of computer vision. Language models like the GPT series [Radford 2019] have demonstrated the potential of understanding the nature of language through a single pretext task (next-token prediction) and subsequently applying that understanding to various language-related tasks. This notion of having a universal pretext task that can be leveraged for solving diverse problems is both promising and efficient.

While the concept of universal pretext tasks has gained traction in natural language processing, it is noteworthy that the vision domain is yet to have a similarly unified and versatile framework. Recent efforts, such as the SAM [Kirillov 2023] and models derived from SAM, bridge this gap for segmentation, but this is just a single visual task. Looking forward, it is possible that developing a universal vision task and devising efficient methods for querying this universal task for specific applications could become a foundational approach for visual models. This paradigm would allow for a more streamlined and versatile way of addressing multiple vision-related challenges and potentially lead to breakthroughs in the field and, by extension, the computer vision for the MIS.

The generalizability of large models for multiple tasks is great. However, a main issue with large models in the surgical scenario is that for the networks to be utilized in Operating rooms (ORs), they need to perform inference in real-time and fit into devices and computers in the OR, which would be a cost-inefficient solution.

2.6.7 Real-time deployment

The deployment of artificial intelligence (AI) in surgery is in its nascent stages compared to fields like radiology [Maier-Hein 2022]. Despite the potential benefits, there have been relatively few clinical trials exploring AI for intraoperative assistance [Varghese 2024; Auloge 2020; Mascagni 2024].

The primary challenges hindering the widespread adoption of AI in surgery are multifaceted. Technically, collecting and managing large volumes of surgical data, ensuring real-time processing, and achieving low-latency inference are significant hurdles. Integrating these AI systems into existing surgical workflows without disruption is also a considerable challenge [Auloge 2020]. Moreover, regulatory and ethical challenges, including ensuring compliance with stringent medical standards, addressing ethical concerns, and maintaining data privacy, require careful consideration. Operational challenges, such as training surgical staff, managing costs, and upgrading infrastructure, further complicate the deployment of AI in surgery.

Multitask learning (MTL) presents a promising solution to some of these challenges, particularly in addressing the need for low-latency inference. Several MTL models reviewed in this survey report speeds exceeding 20 frames per second for multiple tasks while also reducing the number of parameters required by sharing encoders, as shown in Table 2.1, Table 2.2, Table 2.3, Table 2.6, Table 2.7, Table 2.8, and Table 2.9.However, more work is required to fully realize the potential of MTL in real-world surgical applications. Unification in datasets and metrics, continued research, development, and collaboration are necessary to address these challenges and leverage AI and MTL to enhance surgical outcomes and efficiency.

2.6.8 Conclusion

MTL has established itself as an important paradigm across minimally invasive surgical vision. This chapter introduced its principal objectives and methods, reviewed six application areas, surveyed the datasets that support these tasks, and identified opportunities arising from more advanced task-sharing and optimisation strategies. More broadly, the review shows that no individual perceptual or semantic task provides a complete description of a surgical scene: instrument, anatomy, workflow, and action predictions offer complementary information, while surgical action triplets provide a particularly useful representation of fine-grained instrument–tissue interactions. The thesis therefore carries forward the review's broader lesson that these complementary forms of information must be connected, while narrowing its technical focus from multitask learning in general to spatial grounding and instrument–tissue interaction modelling.

The literature also exposes a central gap addressed by the remainder of this thesis. As discussed in the main review of surgical action triplets (Section 2.4.3) and the subsequent workflow-analysis discussion (Section 2.6.3), the dominant triplet formulation can identify an interaction at frame level without reliably associating it with the individual instrument instance responsible for it, while large-scale instance-level annotations of laparoscopic instruments have historically been limited. Spatially grounded triplet understanding therefore first requires a reliable means of distinguishing and localising individual instruments. Chapter 3 addresses this data bottleneck through CholecInstanceSeg, establishing the instance-segmentation foundation for the subsequent grounded interaction models.

Chapter 3

CholecInstanceSeg: A Tool Instance Segmentation Dataset for Laparoscopic Surgery

Table of contents

This chapter draws on the accepted article *CholecInstanceSeg: A Tool Instance Segmentation Dataset for Laparoscopic Surgery*, published in *Scientific Data* (2025). Full publication details are provided in the List of Publications.

3.1 Introduction

The literature review in Chapter 2 identifies a missing spatial link in fine-grained surgical activity understanding. Frame-level instrument and action labels can describe what is present, while semantic segmentation can localise an instrument class, but neither necessarily distinguishes multiple physical instances of that class. Without this distinction, actions and anatomical targets cannot be assigned unambiguously to the individual instruments responsible for them. This chapter addresses that data limitation through CholecInstanceSeg, providing the spatial supervision needed for the subsequent progression from instrument localisation to grounded $\langle$`instrument, verb, target`$\rangle$ interaction prediction.

Within the constrained view provided by laparoscopic cameras, computer-aided intervention systems require precise identification and localisation of the instruments present in the surgical scene [Islam 2019a; Vercauteren 2020; Ward 2021]. Instrument instance segmentation supplies this information by distinguishing individual physical tools, providing a necessary basis for assistive and semi-autonomous systems and, in this thesis, for assigning interactions to the instruments responsible for them.

Current publicly available datasets for surgical tool segmentation have several notable limitations. Many focus on porcine or ex-vivo surgeries rather than routine human procedures, limiting their applicability to day-to-day clinical settings [Bodenstedt 2018; Allan 2019; Allan 2020]. Others are multitask datasets with limited tool segmentation data [Wang 2022b], or they provide only binary instance segmentation for a single tool class [Ross 2020]. Additionally, available surgical tool segmentation datasets lack instance-specific information [Bodenstedt 2018; Allan 2019; Allan 2020; Wang 2022b; Psychogyios 2023; Hong 2020], and most offer a relatively small number of annotated segmentation masks.

To overcome these challenges, we introduce the CholecInstanceSeg open-access dataset, in which we meticulously annotate 41,933 frames from the existing Cholec80 [Twinanda 2016a] and CholecT50 [Nwoye 2022b] datasets with instance segmentation labels and class labels for 7 instrument classes: Grasper, Bipolar, Hook, Clipper, Scissors, Irrigator, and Snare. CholecInstanceSeg contains frames from 85 videos of Laparoscopic cholecystectomy - a routine surgical procedure for gallbladder removal, commonly performed to treat gallstones or cholecystitis with over 300,000 surgeries performed annually in the United States [Hassler 2023]. To the best of our knowledge, CholecInstanceSeg represents the most extensive surgical tool instance segmentation dataset. Table 3.1 compares popular datasets used in tool segmentation research and CholecInstanceSeg. We acknowledge a concurrent effort, the PhaKIR dataset, which is being developed for the upcoming MICCAI2024 PhaKIR-challenge. The PhaKIR dataset is expected to include 15 videos and approximately 30,000 instance annotations. However, as this dataset is only partially released and not yet available for research, we have not included it in our comparisons.

Dataset	Patient	Environment	Procedures	Annotations	Semantic	Instance
Endovis2015 [Bodenstedt 2018]	Human	Ex-vivo and In-vivo	8	10k	3 classes	No
Endovis2017 [Allan 2019]	Porcine	In-vivo	10	2.4k	10 classes	No
Endovis2018 [Allan 2020]	Porcine	In-vivo	16	2.4k	10 classes	No
CholecSeg8k [Hong 2020]	Human	In-vivo	17	8k	12 classes	No
AutoLaparo (Multitask) [Wang 2022b]	Human	In-vivo	21	1.8k	7 classes	No
ROBUSTMIS2019 [Ross 2020]	Human	In-vivo	30	10k	2 classes	Yes✓
SAR-RARP50 (Multitask) [Psychogyios 2023]	Human	In-vivo	50	10k	10 classes	No
CholecInstanceSeg (Ours)	Human	In-vivo	85	41.9k	8 classes	Yes✓

Table 3.1: Comparison of popular datasets used in tool segmentation research, indicating the type of surgeries, the number of surgical procedures, the number of frames annotated, the semantic classes annotated, and whether each dataset contains instance information.

Furthermore, we provide technical validations of our dataset, demonstrating our comprehensive quality control procedures and showcasing CholecInstanceSeg's capability for training instance segmentation models.

CholecInstanceSeg is ideally suited for training models on instance and semantic segmentation for laparoscopic surgeries. Additionally, it can be effectively integrated with other datasets that utilize surgical procedure videos from Cholec80 and CholecT50. These datasets, derived from Cholec80 and CholecT50 share frames with CholecInstanceSeg but are annotated for different tasks, such as phase recognition (Cholec80 [Twinanda 2016a]) instrument action and target detection (CholecT50 [Nwoye 2022b]), and Critical View of Safety (CVS) assessment (Cholec80-CVS [Ríos 2023]).

3.2 Methods

This section outlines the methodology for creating the CholecInstanceSeg dataset, divided into four main topics: data sources, labelling protocol, annotation tool and techniques, and the annotation workflow and procedures.

First, we describe the data sources and how the frames were extracted. Next, we outline the principles guiding our instance segmentation dataset, emphasizing the labelling protocol. We then discuss the tools and techniques used to overcome annotation challenges. Finally, we provide a detailed account of the annotation process, including information about the annotators and the time spent on creating CholecInstanceSeg.

3.2.1 Data Sources

The CholecInstanceSeg dataset is curated using frames extracted from three existing laparoscopic cholecystectomy datasets: Cholec80 [Twinanda 2016a], CholecT50

[Nwoye 2022b], and CholecSeg8k [Hong 2020], containing a total of 85 image sequences. These datasets are publicly available subsets of the in-house Cholec120 dataset [Twinanda 2018] from the CAMMA research group which was recorded at the University Hospital of Strasbourg [Twinanda 2016a; Nwoye 2022b] . The interrelation between these datasets is illustrated in Fig. 3.1.

When discussing the source datasets, we distinguish between videos and image sequences. Videos are multimedia compressed files provided in formats such as .mp4, while image sequences refer to extracted sequential frames from videos.

The frames from Cholec80 and CholecT50 used in CholecInstanceSeg are primarily at a resolution of 854x480 pixels, consistent with the original Cholec80 and CholecT50 datasets. However, three sequences from CholecT50 are provided at a higher resolution of 1920x1080 pixels. CholecT50 contains pre-extracted image sequences and we used these as is, while frames from Cholec80 were extracted at 1 fps using the FFMPEG library.

CholecInstanceSeg includes frames from publicly available datasets that already have the necessary consents and ethical considerations. Additionally, the licenses for the source datasets permit the enhancement of the data with further annotations (Cholec80 license, CholecT50 license, CholecSeg8k license).

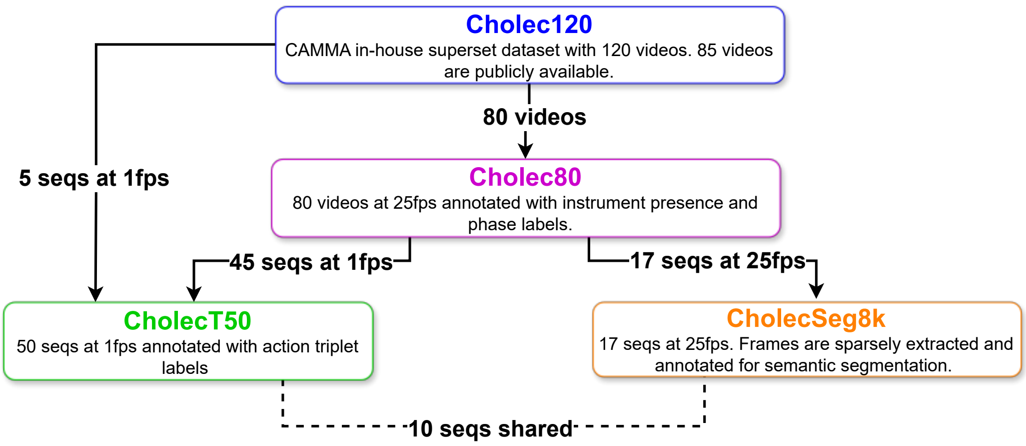


Figure 3.1: CholecInstanceSeg contains frames extracted from CholecT50, CholecSeg8k, and Cholec80. These three datasets are sub-datasets of the CAMMA in-house dataset called Cholec120. There are also some shared image sequences (referred to as *seqs* in the figure for brevity) between CholecT50 and CholecSeg8k.

CholecSeg8k

CholecSeg8k comprises 8,080 laparoscopic cholecystectomy image frames extracted from 17 surgical procedure video clips within the Cholec80 dataset. This dataset provides 80 consecutive images sparsely annotated for semantic segmentation (but not instance segmentation). To enhance CholecSeg8k, instance IDs are required for the existing semantic segmentation labels. As CholecSeg8k already contains semantic segmentation labels, it serves as a good starting point for further annotation. However, it is important to note that CholecSeg8k includes only two tool classes: Grasper and Hook, which is fewer than the number of tool classes in the parent datasets.

CholecT50

CholecT50 is a publicly available dataset comprising 50 real-world videos of laparoscopic cholecystectomy surgeries. This dataset integrates 45 videos from the Cholec80 dataset along with 5 additional videos from the in-house Cholec120 dataset. The image sequences in CholecT50 are extracted at a rate of 1 frame per second (fps). Each frame in CholecT50 is annotated with fine-grained tool-tissue interaction labels, specifying the `<instrument, action, target>`. Although CholecSeg8k and CholecT50 share 10 image sequences, we confirmed that the video extraction methods used for these datasets are different. We checked the similarity between extracted frames in the shared sequences from both datasets and found no exact matches.

The CholecT50 dataset consists of 100.9k frames distributed across 50 image sequences. Given the high cost of annotating every single frame, we adopted a strategy to balance thoroughness and feasibility. Consequently, we divided CholecT50 into two partitions:

1. CholecT50-full: This partition was chosen to feature comprehensive annotations, including instance IDs and segmentation masks for every frame in the image sequence. This thorough annotation provides detailed data for in-depth analysis and model training, allowing for the exploration of ideas like video segmentation.

2. CholecT50-sparse: In this partition, annotations are provided sparsely in time, yet they cover the entire sequence. This approach reduces annotation costs while still offering valuable insights into the location and identification of tools for each sequence. We sample one in every 30 frames ($\frac{1}{30}$ fps) and add more frames if the resulting sample contains fewer than 50 frames per sequence.

Cholec80

Cholec80 comprises 80 videos of cholecystectomy surgeries. It is labelled with phase information at 25 fps and tool presence annotations at 1 fps. The tool annotations include instruments where at least half of the tool-tip is visible. There are six tools available in the provided tool presence annotations.

Given the overlap between Cholec80, CholecSeg8k, and CholecT50, we focus on annotating the 28 videos in Cholec80 that are not present in CholecSeg8k and CholecT50. We extract frames from these 28 videos at a rate of $\frac{1}{30}$ fps and provide annotations for these frames. To maintain consistency with the extraction protocol of CholecT50 and CholecSeg8k, we blacked out frames that showed the exterior of the body via the laparoscope. Additionally, if the resulting sample contained fewer than 50 frames per sequence, more frames were included to ensure adequate coverage.

Summary of CholecInstanceSeg Partitions

CholecInstanceSeg, which contains 41.9k frames from 85 unique image sequences, can be partitioned into four distinct sections based on the data source: Instance-CholecSeg8k, Instance-CholecT50-full, Instance-CholecT50-sparse, and Instance-Cholec80-sparse. Our process for selecting the frames and sequences for each partition is as follows:

1. Instance-CholecSeg8k: We selected all the 8,080 frames from all the 17 sequences provided in CholecSeg8k, we utilized all the images in CholecSeg8k.

2. Instance-CholecT50-full: We randomly selected 15 sequences within CholecT50 (CholecT50-full), we utilized all frames provided by CholecT50 for these 15 sequences - 28,317 frames.

3. Instance-CholecT50-sparse: For the remaining 35(50-15) sequences in CholecT50(CholecT50-sparse), we sampled these sequences by selecting one frame out of every 30 frames ($\frac{1}{30}$ fps), ensuring a minimum of 50 frames in each sequence - 2,681 frames .

4. Instance-Cholec80-sparse: After selecting sequences for Instance-CholecSeg8k, Instance-CholecT50 (both full and sparse), there remained 28 videos from Cholec80, which are not included in CholecT50 and CholecSeg8k, and these videos were sampled at $\frac{1}{30}$ fps, ensuring a minimum of 50 frames in each sequence - 2,855 frames .

A visual representation of dataset partitions in CholecInstanceSeg can be seen in Fig. 3.2.

The differences in frame rates for each dataset partition reflect both the design of the parent datasets – Cholec80 provides videos at 25fps which we then extract at 1fps and CholecT50 provides image sequences at 1fps directly – and practical considerations for annotating instance segmentation on a large number of frames. Annotating every sequence densely at 1fps is prohibitively expensive. Thus, we adopted a mixed approach, converting existing semantic segmentation to instance segmentation, annotating some sequences *fully* (using all frames available from the parent datasets), and others sparsely (sampling frames at reduced frame rates of $\frac{1}{30}$ fps). This approach allows us to balance annotation density and dataset diversity, ensuring coverage of both densely annotated sequences and a wider range of surgical scenes.

Fig. 3.7 presents the exact sequences in each partition, represented as VID-XX, where XX is the sequence number.

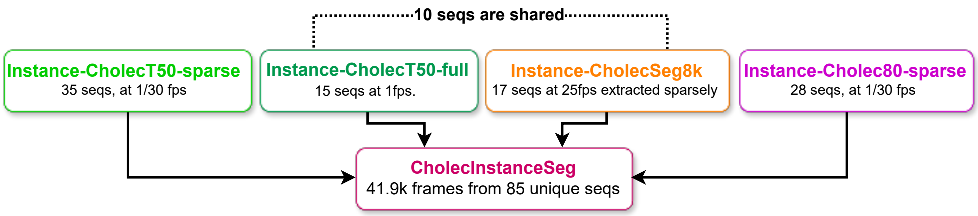


Figure 3.2: CholecInstanceSeg is composed of frames from four dataset partitions: Instance-CholecT50-sparse, Instance-CholecT50-full, Instance-CholecSeg8k, and Instance-Cholec80-sparse. The diagram illustrates the number of image sequences (denoted *seqs* in the figure for brevity) and extraction rates for each partition. Note that 10 sequences are shared between Instance-CholecT50-full and Instance-CholecSeg8k.

3.2.2 The Labelling Protocol

To facilitate the creation of a high-quality instance segmentation dataset for laparoscopic cholecystectomy videos, we established specific rules to ensure accurate and consistent identification and segmentation of surgical instruments. These rules are discussed under three main topics: Tool Categories, Annotation Scope, and Annotation Guidelines.

Tool Categories

In this section, we detail the tool categories selected for annotation and discuss the objects that were deliberately excluded, along with the rationale for these decisions.

We identified seven tool categories for annotation based on our observations from the extracted CholecInstanceSeg sequences:

1. Grasper: Standard grasper used for holding tissues.
2. Bipolar: Bipolar grasper used for grasping and coagulation.
3. Hook: Monopolar hook used for dissection and coagulation.
4. Clipper: Clip applier used for applying clips.
5. Scissors: Monopolar scissors used for cutting tissues.
6. Irrigator: Suction/irrigation device used for fluid management.
7. Snare: Laparoscopic snare used for ligation of tissues.

The inclusion of the snare category deviates from the standard six categories (grasper, bipolar, hook, clipper, scissors, irrigator) defined by Cholec80 and CholecT50. We determined that the snare's distinct visual characteristics and unique functionality warranted its classification as a separate tool category.

Annotation Scope

The annotation scope defines the entities included and excluded to maintain a focused and high-quality dataset. While we annotated all laparoscopic tools, some objects used in conjunction with these tools were excluded to streamline the annotation process:

- Excluded Entities: Individual clips from the clipper, surgical needles, surgical sutures, the specimen bag, the camera port, and gauze.
- Instrument Ports: We did not annotate the instrument port when no instrument extended from it. However, when an instrument was seen extending from the port, we annotated the instrument port as part of the instrument. This approach aligns with the CholecSeg8k annotation style and addresses the difficulty of distinguishing the exact point where the port ends and the instrument begins.

Annotation Guidelines

Our annotation guidelines ensure consistency and accuracy in the labelling process. These guidelines cover various scenarios and provide specific instructions for annotators:

- Annotation Quality: We aimed to annotate every visible instrument while maintaining high segmentation quality. Although achieving pixel-perfect annotation for every instance was impractical due to time and budget constraints, we allowed minor imperfections in instrument boundaries, provided that the fundamental characteristics of the instrument and its end-effectors were accurately outlined. Holes within instruments were considered integral parts of the instrument itself, following the standard annotation conventions in popular tool segmentation datasets like EndoVis2017 [Allan 2019] and EndoVis2018 [Allan 2020].

- Annotation of Hard Cases: CholecInstanceSeg captures routine surgical procedures, numerous challenging scenarios were encountered. We accounted for twelve specific situations: motion blur, presence of smoke, occlusion by soft transparent tissues or blood, tissues attached to instruments, instruments at the edge of the frame, instruments far from the camera, light reflection, low lighting conditions, bright lighting conditions, lens dirtiness, camera in camera port, and instruments in liquid. Specialized instructions and examples were provided to annotators, including checking neighbouring frames for clarification, utilizing the expected shape of the instrument when in doubt, annotating as much of the instrument as possible, and annotating even under low visibility conditions.

3.2.3 Annotation Tool and Techniques

To provide annotations for CholecInstanceSeg, we needed an annotation tool for performing manual annotations, a method to convert existing semantic segmentation to instance segmentation, and a way to efficiently label a large number of images. This section discusses how we addressed these needs: the annotation tool, converting semantic segmentation to instance segmentation, and semi-automatic annotation.

Annotation Tool

Data labelling was primarily executed using a minimally customized version of the Segment Anything Annotator tool [Wang 2023b], which is open-source software based on the LabelMe annotation project [Wada 2016]. This tool integrates instrument segmentation functionalities and interactive segmentation capabilities, leveraging the Segment Anything Model (SAM) [Kirillov 2023]. Our customizations included enhancements for navigation, reduction in the size of annotation label files, and the addition of a quality control mode. The customized version is available on github.

We predominantly utilized point-based SAM prompts for efficient and quick interactive annotation. However, point-prompt-based interactive segmentation often encounters challenges in complex scenarios, such as when instruments and surrounding backgrounds are similar in color, low lighting, glare, blurs, dirty lenses, reflections, and the presence of smoke. In such instances, manual annotation can also be performed using the same tool. Further discussion of challenging scenarios is provided in a later section.

Converting Semantic to Instance Segmentation

One of our data sources, CholecSeg8k, already had semantic segmentation labels. However, we aimed to assign instance IDs to distinguish each instrument instance within an image. To add instance labels and convert existing semantic segmentation labels to instance segmentation, we followed two main steps:

1. **Class-Aware Connected Component Analysis**: We employed connected component analysis for each class separately to identify and segregate different instances within the frames. This technique effectively assigned instance IDs to instruments from different classes and correctly distinguished instances of instruments belonging to the same class that do not overlap in pixel space.

2. **Identifying and Addressing Failure Cases**: Class-aware connected component analysis encountered several challenges. A total of 892 frames with these issues were carefully identified and documented. To rectify these problems, we developed and applied several pipelines:

    (a) **Occlusions**: When pixels of the same instrument are not connected, class-aware connected component analysis assigns multiple instance IDs to a single instrument. This issue, mainly caused by occlusions at the frame's edge or by tissues, was addressed by combining multiple instances that were actually the same using a Python pipeline.

    (b) **Overlap**: A single instance ID was assigned to different instances of the same instrument class that overlapped in pixel space. We reannotated these frames using our annotation tool.

(c) **Dataset Noise**: Stray erroneous pixels, incorrect instrument class labels, and missing instrument annotations led to incorrect instance ID assignments. We built pipelines to remove instances caused by dataset noise and correct class labels. Missing instruments were also annotated as required.

Semi-Automatic Annotation

For the CholecT50-full dataset, which requires annotations for full sequences (nearly 30,000 frames from 15 sequences extracted at 1 fps), we opted for a semi-automatic approach with a human-in-the-loop, utilizing a deep learning model to assist in producing instance segmentation annotations.

1. Initial Annotation: We began with initial annotations from the Instance-CholecSeg8k dataset. Since CholecSeg8k only contains two tool categories (Grasper and Hook), which is fewer than the seven tool categories identified for annotation, we performed additional annotations using the annotation tool. We focused on images that included the new instruments. The combined dataset of Instance-CholecSeg8k and these initial annotations provided a foundation to start the semi-automatic pipeline for generating labels.

2. Model Training: A deep neural network, the Real-Time Models for Object Detection and Instance Segmentation Large (RTMDet-Ins-l) [Lyu 2022], was trained using the available annotated datasets (Instance-CholecSeg8k and Instance-CholecT50-sparse).

3. Annotation Proposal Generation: Leveraging the trained model, we generated annotations for a larger portion of the unannotated data.

4. Annotation review and correction: Human annotators reviewed the model-generated annotations and corrected any errors or inaccuracies, ensuring the quality and accuracy of the annotations. Minor errors were corrected with the capabilities of the LabelMe labeling tool, while major errors requiring new annotations utilized the point-prompt method of the SAM model.

5. Augment training dataset: The machine learning model was retrained using the expanded training dataset, incorporating the corrected annotations. This iterative process enhanced the model's performance. The model was trained twice: initially, with an annotation set of approximately 11k images and their corresponding labels. The trained models was used to generate labels for

about 10k images. Human annotators corrected labels for 5k of these images. The model was then retrained with 21k images, generating labels for another 17k images, with labels for 4k images requiring human annotators for correction.

A visual representation of our semi-automatic approach to annotation can be seen in Fig. 3.3.

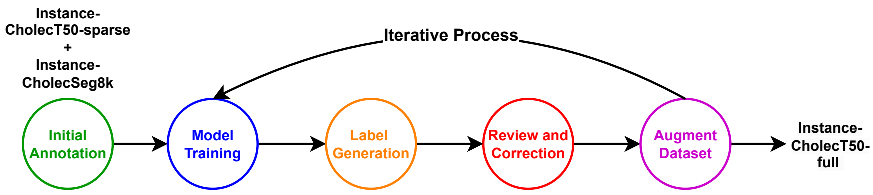


Figure 3.3: Semi-automatic annotation pipeline. The process begins with initial annotations using Instance-CholecT50-sparse and Instance-CholecSeg8k, followed by model training. The trained model generates labels, which are then reviewed and corrected by human annotators. The corrected annotations augment the training dataset, iterating through the cycle until the final Instance-CholecT50-full dataset is produced.

3.2.4 Annotation Workflow and Procedures

This section provides an overview of the step-by-step procedure used to create the CholecInstanceSeg dataset, along with detailed insights into the human effort involved in the annotation process. We organize the discussion into two main topics: the step-by-step procedure, and Annotators.

Step-by-Step Procedure

1. Annotation of Instance-CholecSeg8k (2 weeks): Converted the CholecSeg8k dataset, which initially contained semantic segmentation labels, into Instance-CholecSeg8k with instance IDs. Quality checks were performed to ensure accuracy. This dataset partition only had two classes and lacked many challenging scenarios present in other datasets.

2. Creating the Labelling Protocol (3 weeks): Extensive discussions among the annotation team addressed the challenges encountered during the initial annotation attempts of the Instance-CholecT50-sparse partition. These discussions led to the development of a comprehensive labelling protocol to handle hard cases and unique scenarios effectively and consistently.

3. Annotation of Instance-CholecT50-Sparse (3 weeks): Annotated the Instance-CholecT50 dataset partition using the annotation tool. This occurred in tandem with the creation of the labelling protocol. Quality checks ensured annotations were accurate and consistent with the newly established guidelines.

4. Annotation of Instance-CholecT50-Full (3 weeks): Using Instance-CholecT50-Sparse and Instance-CholecSeg8k as initial annotations, we employed a semi-automatic pipeline to generate the Instance-CholecT50-Full dataset. This process included iterative model training and corrections to enhance annotation quality.

5. Annotation of Instance-Cholec80 (1 week): To ensure comprehensive coverage and enhanced diversity for CholecInstanceSeg, we utilized the annotation tool to manually annotate the Instance-Cholec80 partition.

6. Final Quality Control (2 weeks): Quality control was conducted after the creation of each dataset and again upon finalizing CholecInstanceSeg. This process included several steps to ensure accuracy and consistency. Initially, we performed manual quality control, manually reviewing each image to ensure correct annotations. A second annotator then reviewed a subset of CholecInstanceSeg and performed additional corrections.

   We also leveraged specific properties of CholecInstanceSeg and its relationships with other datasets to enhance quality control. For instance, we used existing tool presence labels from related datasets for cross-validation. Detailed steps for quality control, utilizing these properties and external dataset labels, are extensively discussed in the quality control section.

Annotators

The annotation process for creating the CholecInstanceSeg dataset involved a primary annotator responsible for the majority of the annotation tasks and quality control. A secondary annotator, with greater experience and medical qualifications, assisted with annotation and quality control. Both annotators were supported by an expert team to ensure accuracy and consistency in the annotations.

3.3 Data Records

The CholecInstanceSeg dataset [Alabi 2025] is publicly available. Cholec80 and CholecT50 video frames used to generate the dataset can be obtained from the CAMMA research group website(http://camma.u-strasbg.fr/datasets). CholecSeg8k can be obtained at (https://www.kaggle.com/datasets/newslab/cholecseg8k).

The dataset is provided as a single archive, *CholecInstanceSeg.zip*, which contains the instance segmentation annotation files in *cholecinstanceseg.zip* and a metadata file *cholecinstanceseg_metadata.csv* that provides metadata information about each annotation file and can be used to link each file to its source dataset. The annotation files are structured into two main directories: train and val, each containing individual video sequence folders. Each sequence folder includes an *ann_dir* subdirectory that stores instance segmentation annotations in JSON format. The test directory is withheld for benchmarking purposes and potential competition use, but it is available upon request. A schematic representation of the dataset folder structure and a sample JSON annotation file are shown in Fig. 3.4

Annotations are formatted following the LabelMe convention. Each shape is listed under the *shapes* key, with a *label* field specifying the instrument class, *points* containing the (x, y) coordinates of the polygon defining the object's boundary, *group_id* indicating the instance ID, *image_Height* for the height of image annotated and *image_Width* for its width.

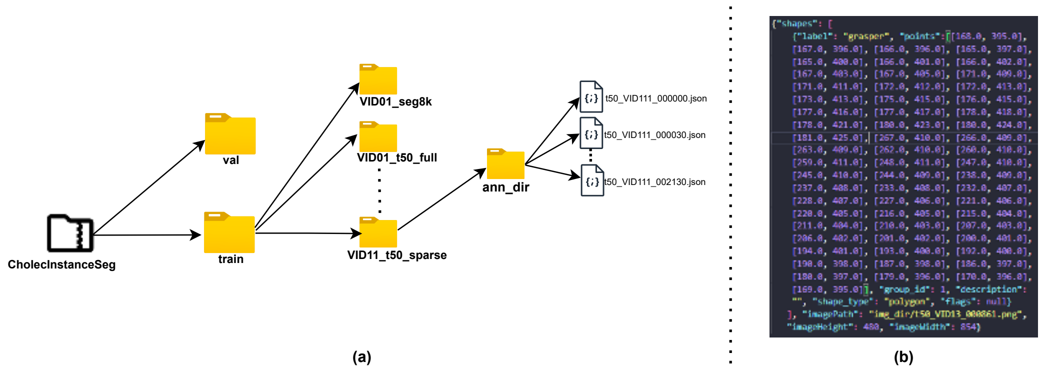


Figure 3.4: (a) Dataset directory structure for CholecInstanceSeg. Each split (e.g. train) contains subdirectories for each sequence, with further subdirectories for annotations (*ann_dir*). (b) Sample JSON annotation file showing the structure of the annotation data, including class labels ("*label*"), polygon information ("*points*") and instance IDs ("*group_id*").

3.4 Technical Validation

We provide an in-depth technical validation of the CholecInstanceSeg dataset, demonstrating its reliability and effectiveness for use in research and development of instance segmentation models. We discuss the validation through three main aspects: Data Analysis, training of baseline models, label agreement statistics, and quality control measures.

3.4.1 Data Analysis

Tool Class Distribution

Fig. 3.5(a) shows the distribution of tool classes across the entire dataset. Graspers are the most frequently used tools, followed by hooks. Tools like irrigators, clippers, scissors, and bipolars are used less frequently, with snares being the least used. This indicates a high imbalance in tool class occurrences.

Tool Instance Distribution per Frame

Fig. 3.5(b) illustrates the number of tool instances per frame. The majority of frames contain one or two tools, with a maximum of four tools in a single frame. Approximately 5,328 frames (about one-eighth of the dataset) contain no tools.

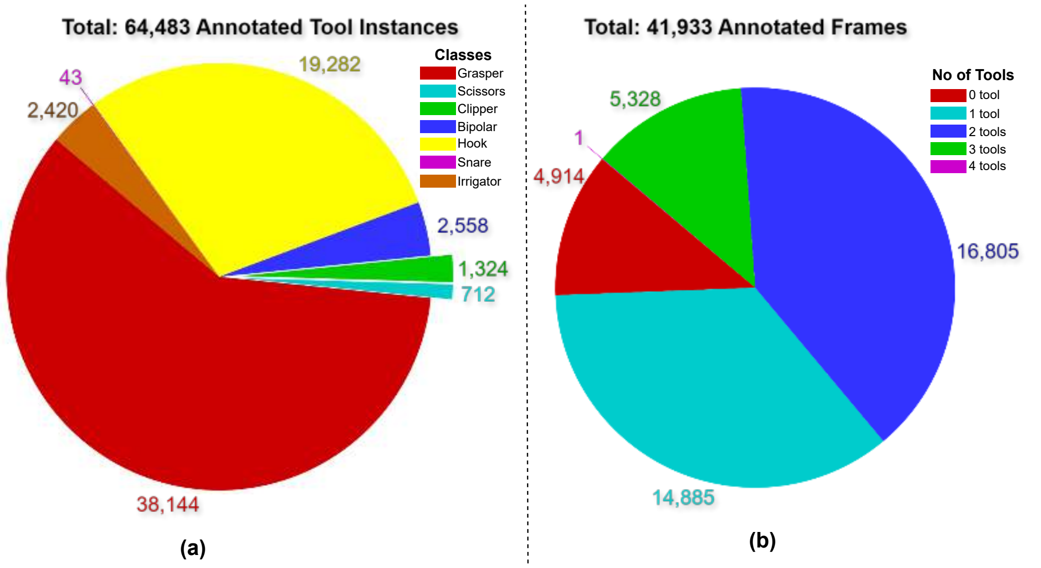


Figure 3.5: (a) Distribution of each tool class across the dataset. (b) Distribution of the number of tools per frame.

Partition and Sequence Analysis

Table 3.2 provides a summary of the number of sequences, number of frames and amount of annotated tools in each of the dataset partitions. Fig. 3.6 provides an analysis of tool instances across different sequences and their partitions.

Instance-CholecT50-full has the highest number of tool instances, whereas Instance-CholecT50-sparse has the fewest. This figure highlights the imbalance between densely and sparsely annotated sequences and the presence of a sequence with no tools in Instance-CholecSeg8k.

	Inst-CholecSeg8k	Inst-CholecT50-sparse	Inst-CholecT50-full	Inst-Cholec80-sparse	Total
No of seq	17(10 shared)	35	15(10 shared)	28	85
No of frames	8,080	2,681	28,317	2,855	41,933
No of tools	10,523	4,098	45,221	4,641	64,483

Table 3.2: Summary of Dataset Partitions in CholecInstanceSeg.

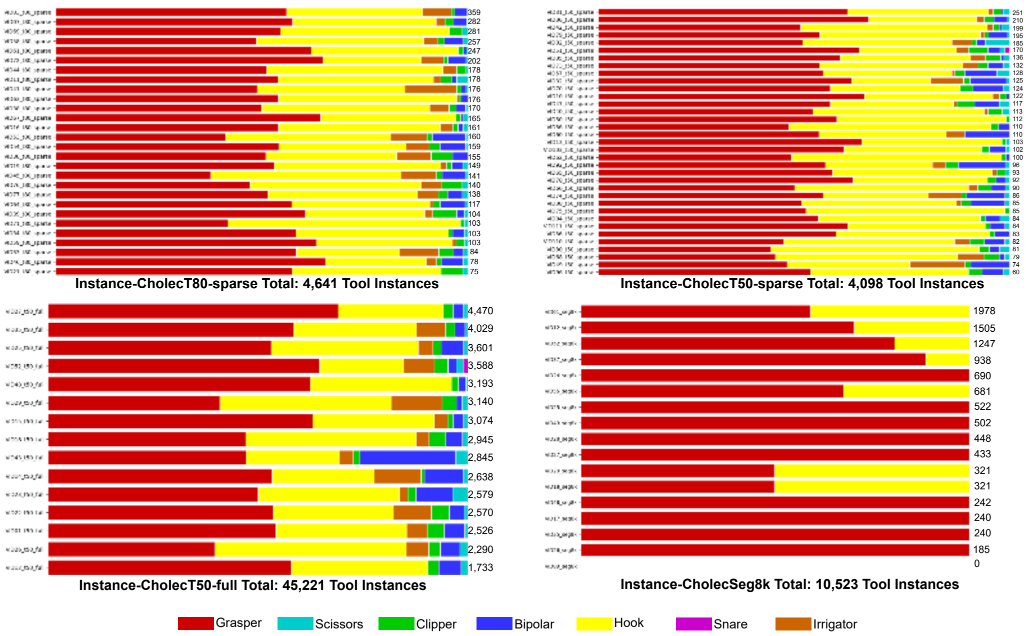


Figure 3.6: Distribution of tool instances across sequences and partitions in CholecInstanceSeg

3.4.2 Training and Evaluation of Baseline Models

Models

To validate the usability of the CholecInstanceSeg dataset, two baseline instance segmentation models, The Mask R-CNN [He 2017] and the Mask2Former [Cheng 2022] were trained. Since the objective was to establish benchmark performance metrics that future researchers can use for comparison, we decided to utilize more generalizable baseline models instead of optimized tool-specific models like [Baby

2023; Zhou 2023]. These models were trained using the training split of CholecInstanceSeg with each model's default hyperparameters.

Recommended Data Splits

To facilitate effective training, validation, and testing of neural networks for instance segmentation, we divide CholecInstanceSeg into standard training, validation, and testing subsets. This division considers the partitions and shared sequences within the dataset.

For the training dataset, we included the entire Instance-CholecT50-sparse partition, which consists of 35 sequences with 2,681 frames. This partition offers significant diversity in surgical scenes due to its varied sequence content. Additionally, 10 in Instance-CholecSeg8k and the 10 sequences in Instance-CholecT50 which come from the same video (shared) were included in the training dataset to avoid redundancy in the validation and testing subsets.

The validation dataset contains 3 out of 7 sequences which are unique to Instance-CholecSeg8k. Similarly, from Instance-CholecT50-full, 1 out of 5 sequences which are unique to Instance-CholecT50 were included in the validation dataset. Additionally, half of the sequences from Instance-Cholec80-sparse were added to the validation dataset to bolster its diversity.

The test dataset includes the remaining 4 unshared sequences from Instance-CholecSeg8k and the remaining 4 unshared sequences from Instance-CholecT50-full. The other half of the sequences from Instance-Cholec80-sparse were also included in the test dataset. This division ensures that the test set includes a wide range of scenarios (sequences), providing a comprehensive assessment of model performance.

The final split for the dataset is as follows: the training set contains 55 sequences (26,830 frames), the testing set includes 23 sequences (10,819 frames), and the validation set comprises 17 sequences (4,284 frames).

A visual representation of the recommended dataset split can be seen in Fig. 3.7. We provide scripts to convert the downloaded zip file into a train, test, and validation format based on these recommended splits.

Recommended Metric

To evaluate our models we used the standard COCO mean average precision metric (COCO mAP r@[0.50:0.95]). .

From the dataset analysis, we could tell that our dataset has a class imbalance which is captured by the mean average precision, however, it also has an imbalance among sequences, such that some sequences have a lot of tool instances and images

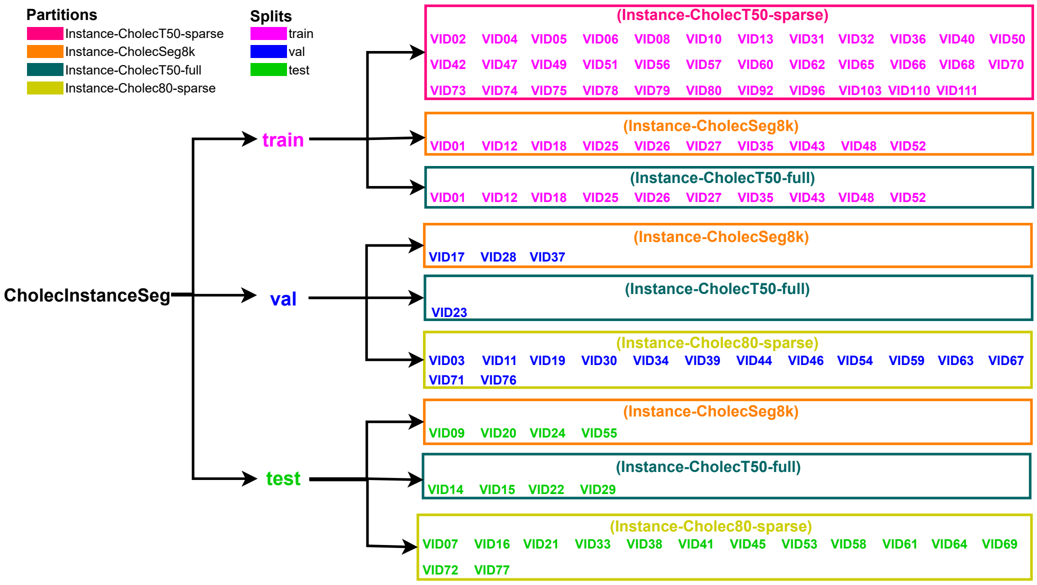


Figure 3.7: Recommended Dataset Splits for CholecInstanceSeg. The diagram shows the distribution of sequences across the training, validation, and testing datasets, highlighting the specific partitions (Instance-CholecT50-sparse, Instance-CholecSeg8k, Instance-CholecT50-full, Instance-Cholec80-sparse) and their corresponding sequences.

while others do not. Hence we wanted to also have a metric that captured how well models are performing on sequences.

COCO smAP involves a slight modification of the COCO mAP metric. The sequence average precision(sAP) for a class j over the whole dataset (sAP_j) in COCO smAP can be written as

$$sAP_j = \frac{\sum_{s=0}^{N} \int_0^1 p_{seq=s}(j,r)\,dr}{N}$$

where s represents the current sequence, N represents the number of sequences evaluated, and $p_{seq=i}(j,r)$ represents the precision at recall r for class j over a sequence s. Essentially sAP_j is calculated by taking the mean of the COCO average precision for that class for each sequence.

To measure variability in performance across sequences, we introduced the standard deviation of sequence average precision (σ_{sAP_j}).

$$\sigma_{sAP_j} = \sqrt{\frac{1}{N}\sum_{s=0}^{N}\left(\left(\int_0^1 p_{seq=s}(j,r)\right)dr - sAP_j\right)^2}$$

where s represents the current sequence, N represents the number of sequences evaluated, and $p_{seq=i}(j,r)$ represents the precision at recall r for class j over a sequence s. σ_{sAP_j} is calculated by taking the standard deviation of the COCO average precision for that class for each sequence.

Finally, the COCO smAP is calculated as a mean of the sequence average precisions across all classes

$$smAP = \frac{1}{C}\sum_{j=1}^{C} sAP_j$$

We utilize the official COCO implementation present here for calculating the COCO mAP and the metric COCO smAP metric.

Results

The performance results, presented in Table 3.3 and Table 3.4, validate the dataset's effectiveness in training robust instance segmentation models.

Notably, the hook class achieved the highest segmentation performance across both models, likely due to its visually distinct features and higher frequency in the dataset. In contrast, the scissor class exhibited the lowest performance, reflecting its smaller representation and potential challenges in detecting this class. Interestingly, the scissor class demonstrated a higher sequence mAP (smAP) but with significant variability (high standard deviation), suggesting that model performance on this class varies considerably across different sequences. Increasing the number and diversity of annotated sequences could improve dataset variability and enhance its applicability to a broader range of surgical scenarios, ensuring fewer seen instruments are better recognized by trained models.

Table 3.3: Performance of baseline instance segmentation models on CholecInstanceSeg for the mean average precision metric (mAP) providing the average precision for each class. GR - Grasper, HO - Hook, IR - irrigator, CL - clipper, BI - bipolar, SC - scissors, SN - snare.

Method	GR	HO	IR	CL	BI	SC	SN	mAP
Mask-RCNN [He 2017]	37.25	61.84	43.04	58.73	28.42	20.91	-	41.70
Mask2former [Cheng 2022]	61.10	82.16	68.38	81.30	48.02	27.89	-	61.47

3.4.3 Label Agreement

To ensure the accuracy and consistency of the annotations in CholecInstanceSeg, an analysis of label agreement statistics was conducted. Two types of label agreement were calculated. The inter-annotator agreement and manual vs semi-automatic agreement.

Table 3.4: Performance of baseline instance segmentation models on CholecInstanceSeg for the sequence mean average precision (smAP) providing the sequence average precision for each class j (sAP_j). GR - Grasper, HO - Hook, IR - Irrigator, CL - Clipper, BI - Bipolar, SC - Scissors, SN - Snare. The Standard deviation of the sequence average precision σ_{sAP_j} is provided using the ($\pm$) symbol.

Method	GR	HO	IR	CL	BI	SC	SN	smAP
Mask-RCNN	41.65 ± 12.74	64.06 ± 9.20	52.50 ± 20.09	64.85 ± 16.20	37.62 ± 20.74	40.38 ± 23.54	-	50.18
Mask2former	66.46 ± 8.52	88.90 ± 7.58	74.51 ± 20.35	84.69 ± 16.88	65.20 ± 25.36	69.76 ± 33.20	-	74.92

Inter-annotator Agreement

The inter-annotator agreement was assessed using the Panoptic Quality Metric [Kirillov 2019]. Two annotators independently annotated a subset of the dataset, and their annotations were compared. The analysis showed a high level of agreement, with a Panoptic Quality score of 91.2, indicating strong consistency between the annotators. The official implementation of the Panoptic Quality metric, available through the panopticapi repository, was utilized for this evaluation.

Manual vs Semi-automatic Agreement

To validate the consistency of our semi-automatic annotation pipeline against the manual annotation process, we measured the agreement between the two methods using the Panoptic Quality Metric. This comparison was crucial to ensure that the increased speed of the semi-automatic pipeline did not compromise annotation quality. The results demonstrated a high level of agreement, with a Panoptic Quality score of 95.7, indicating that the semi-automatic annotations were consistent with the manual annotations.

3.4.4 Quality Control

Our quality control process involved a series of techniques designed to ensure the accuracy and consistency of the CholecInstanceSeg dataset. In addition to manual quality control and a secondary review by another annotator, we leveraged specific properties of CholecInstanceSeg and tool presence labels from Cholec80. The following steps were employed for quality control:

1. checking images with more than 3 tool instances: During annotation, we observed that the most crowded scenes typically included a maximum of three separate tool instances. Therefore, we reviewed all images with more than three tool instances to identify and correct any noise. This process revealed 26 frames with errors due to noise, which were subsequently rectified.

2. utilizing tool presence labels from Cholec80: Tool presence labels from the Cholec80 parent dataset, although noisy and incomplete, were used for cross-referencing. These labels indicate which tools are present in an image but do not specify the number of tools. We cross-referenced our annotations with these tool presence labels to identify frames where an expected tool class from the tool presence labels was missing in CholecInstanceSeg labels. This method identified 267 frames with errors, achieving an approximately 48% success rate in detecting inaccuracies.

3. reducing noise for scenes with tool overlap: To reduce labelling noise for scenes that include tools occluding other tools, we checked instances that had overlaps. We set a fault tolerance of a 0.1 IoU overlap threshold between tools. All images with intersections exceeding this threshold were reviewed and corrected, resulting in the adjustment of 247 frames.

3.5 Code Availability

The relevant code for the use of this dataset is found here. This repository includes scripts that provide information about class IDs, sequences, and splits. It also contains notebooks for converting between dataset partitions and splits, utilities for converting to the COCO format [Lin 2014], visualization tools, retrieving the input images, and additional resources. The official implementation of COCO mAP and panoptic quality metric were used and they can be found on the COCO dataset GitHub page.

3.6 Detailed Annotation Guidelines for Challenging Cases

This section provides visual annotation guidance for challenging appearances encountered during the construction of CholecInstanceSeg.

Tool in fluid scenarios

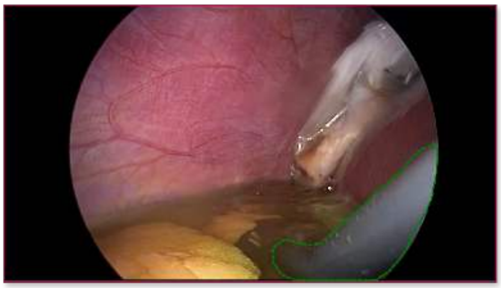
Scenario	Example
Refraction by fluid: *annotate refracted pixels*	 The green outline shows the tool's annotated refracted pixels.
Occlusion by fluid: do *not annotate*	 The blue circle highlights the tool that is fully occluded by fluid and should not be annotated.

Figure S1: Annotation guidelines for tools in fluid scenarios

Blur scenarios

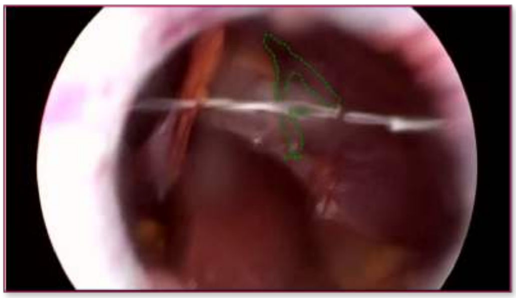
Scenario	Example
High visibility blur: *annotate based on the expected shape of the instrument*	 The green outline shows the annotation for the blurred tool which is mostly visible
Low visibility blur: *use neighbouring frames to infer the tool's shape and annotate accordingly.*	 The green outline shows the annotation for the blurred tool with low visibility
Out-of-focus blur: *annotate with the inferred shape of the tool in mind.*	 The green outline shows the annotation for a tool out focus

Figure S2: Annotation guidelines for blur scenarios.

Tissue attached to tool tip scenarios

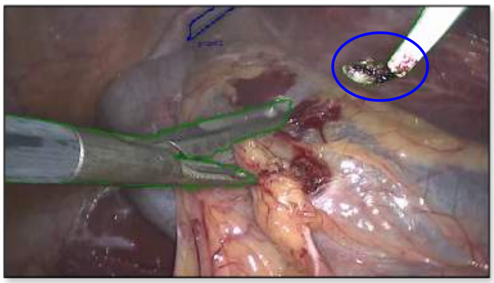
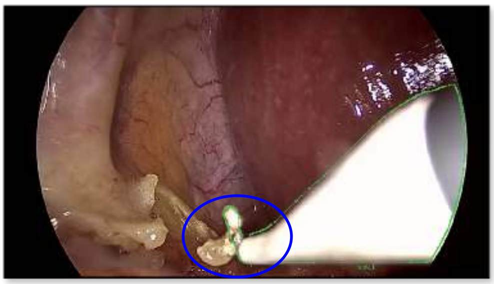
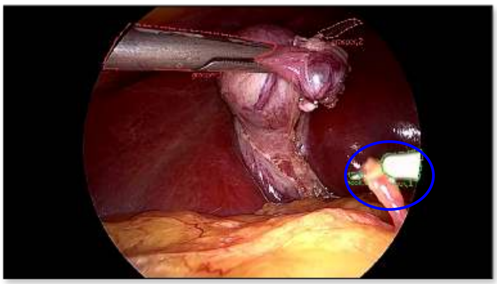
Scenario	Example
Small amount of tissue attached to tool tip: *annotate the tissue as part of the tool.*	 The blue circle highlights a tool tip annotated with a small amount of tissue.
Tissue attached to the tool tip, but the tip is clearly visible: *annotate only the tool tip; do not annotate the tissue.*	 The blue circle highlights a tool tip annotated with the tip only, as it is clearly visible.
Tissue fully occludes parts of the tool tip: annotate as as occlusion; do not annotate occluded portions	 The blue circle highlights a tool tip that is fully occluded by tissue. It is annotated as occlusion

Figure S3: Annotation guidelines for tissue attached to tool tip scenarios.

Tools at the edge of the image or far away from the camera

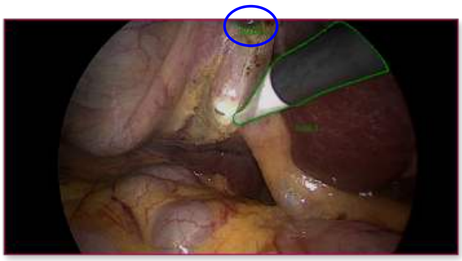
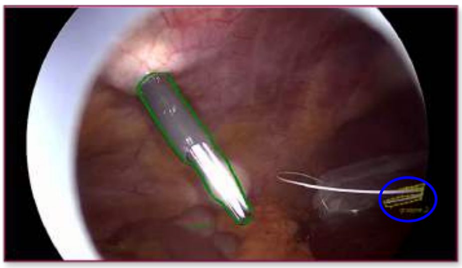
Scenario	Example
Tool at the edge of the frame: *annotators must ensure these tools are always annotated despite their partial visibility*	 The blue circle highlights an instrument partially visible at the edge of the image. Annotators should be vigilant for instruments entering or exiting the frame.
Tools distant from the camera: *tools that are situated far from the camera and might be easily overlooked; annotators must ensure these instruments are always annotated*	 The blue circle highlights an instrument distant from the camera. Annotators should take note of instruments in the background to ensure comprehensive annotation.

Figure S4: Annotation guidelines for instruments at the edge of the frame or distant from the camera.

Lens dirtiness scenarios

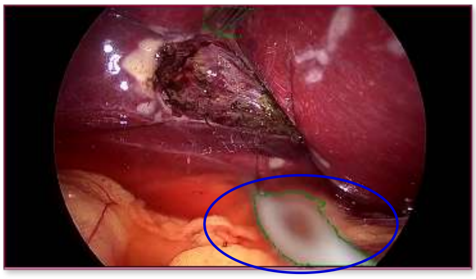
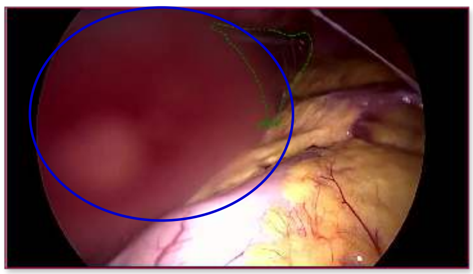
Scenario	Example
Lens dirtiness refraction: *annotate all the refracted pixels.*	 The blue circle highlights a tool with pixels affected by lens dirtiness, causing light refraction. Annotators should include all visible pixels distorted by the refraction in the annotation.
Lens dirtiness occlusion: *do not annotate occluded pixels*	 The blue circle highlights a tool partially obscured by lens dirtiness, resulting in occlusion. Annotators should exclude these occluded areas from annotation

Figure S5: Annotation guidelines for tools affected by lens dirtiness.

Low lighting and saturated lighting scenarios

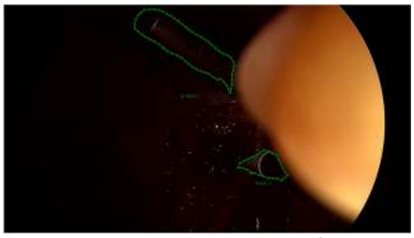
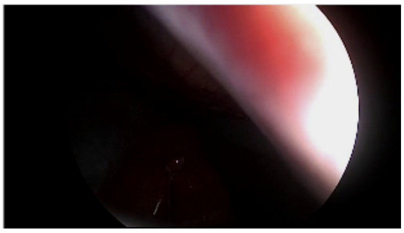
Scenario	Example
Low lighting, tool partially visible: *annotate based on the expected shape of the instrument.*	 The green outline shows annotations for a partially visible tool. Annotate the visible pixels, considering the instrument's expected shape.
Low lighting, tool not visible: *do not annotate*	 The image has low visibility such that instruments visible in previous frames are completely obscured. Do not annotate unseen instruments.
Saturated lighting, tool not visible: *do not annotate*	 The image has low visibility due to saturated lighting, making the instruments indiscernible. Do not annotate.

Figure S6: Annotation guidelines for tools in low lighting and saturated lighting.

Light reflection of tools

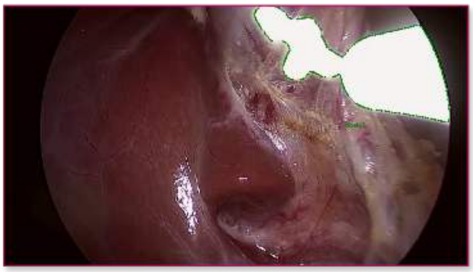
Scenario	Example
Reflection of light of a tool: *annotate based on the expected shape of the instrument.*	 The green outline shows annotations of a tool which is reflecting light.

Figure S7: Guidelines for annotating tools under light reflection conditions.

Low visibility due to smoke

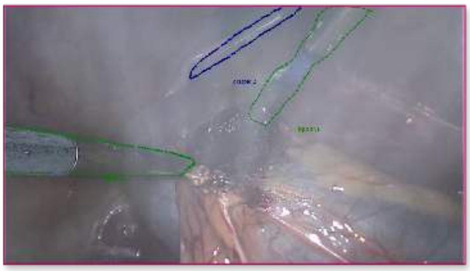
Scenario	Example
Low visibility due to smoke: *annotate based on the expected shape of the instrument.*	 Annotators are advised, to annotate based on the expected shape of the instrument when smoke leads to low visibility

Figure S8: Guidelines for annotating tools during low visibility due to smoke.

3.7 Conclusion

CholecInstanceSeg contributes instance-level instrument annotations across a large and clinically diverse collection of laparoscopic cholecystectomy videos. Its construction, quality-control process, recommended partitions, and baseline validation establish both a reusable dataset and the spatial foundation for the remaining technical chapters. The chapter demonstrates how individual instruments can be distinguished and localised, but localisation alone does not describe what each instrument is doing or which anatomy it affects.

The next chapter therefore asks whether semantic interaction labels can be attached to these spatially resolved instances. It links individual instrument masks with surgical verbs and anatomical targets, transforming surgical action-triplet understanding from frame-level recognition into an instance-level grounding and segmentation problem.

Chapter 4

Grounding Surgical Action Triplets with Instrument Instance Segmentation: A Dataset and Target-Aware Fusion Approach

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Table of contents

This chapter draws on the accepted article *Grounding Surgical Action Triplets with Instrument Instance Segmentation: A Dataset and Target-Aware Fusion Approach*, published in the *International Journal of Computer Assisted Radiology and Surgery* (2026). Full publication details are provided in the List of Publications.

4.1 Introduction

Modelling surgical instrument–tissue interactions is essential for understanding surgical actions and workflows. Action triplets ⟨`instrument, verb, target`⟩ have emerged as a concise yet powerful representation for instrument-tissue interaction, capturing interactions such as ⟨`Grasper, Grasp, Gallbladder`⟩ or ⟨`Scissors, Cut, Cystic_duct`⟩ [Nwoye 2020]. However, existing triplet recognition methods [Nwoye 2020; Nwoye 2022b; Yamlahi 2023] operate at the frame level, predicting which triplets are present in a scene without identifying the location as seen in Fig. 4.1. This lack of spatial grounding makes predictions ambiguous and reduces their clinical value, as precise action localisation is crucial for surgical skill assessment, context-aware assistance, and workflow analysis.

The preceding chapter establishes instrument instance segmentation as a spatial foundation for surgical scene understanding. This chapter extends that foundation from locating individual instruments to interpreting their interactions: which instrument is acting, what action it performs, and which anatomical target it affects. It introduces instance-level surgical action-triplet grounding by combining CholecInstanceSeg masks with CholecT50 action labels to form CholecTriplet-Seg, then develops TargetFusionNet to address the especially difficult problem of anatomical-target prediction using weak anatomical priors.

In this work, we propose grounding surgical action triplets with instrument instance segmentation, a task concisely referred to as *triplet segmentation* that unifies instrument instance segmentation with verb and target classification to produce instance-grounded ⟨`instrument, verb, target`⟩ outputs (see Fig. 4.1). Unlike frame-level recognition or coarse bounding-box detection, triplet segmentation explicitly links each predicted triplet to a corresponding pixel-level instrument instance mask. This provides fine-grained interpretable insights into surgical activities, revealing both actions and their precise spatial context.

However, existing datasets are not designed for this task: they either lack instance-level masks, do not provide complete triplet labels, or are not spatially aligned. To address this gap, we present CholecTriplet-Seg, the first large-scale dataset to pair high-quality instrument instance masks with action verb and anatomical target annotations. Building on CholecInstanceSeg [Alabi 2025] and the CholecT50 triplet labels [Nwoye 2022b], CholecTriplet-Seg provides over 30,000 annotated frames from 50 laparoscopic videos, linking each instrument instance to its associated action and anatomical target. This dataset establishes a benchmark for triplet segmentation and enables fine-grained, instance-level interaction analysis, forming a solid foundation for future extensions beyond cholecystectomy. To measure performance on this dataset, we introduce Triplet Segmentation mAP, a metric that extends existing triplet evaluation protocols to the instance level. It assesses both triplet-level and component-level accuracy, requiring models to correctly classify the triplet and accurately localise the corresponding instrument instance.

Achieving reliable anatomical target prediction remains a major bottleneck. Anatomy is inherently ambiguous and difficult to annotate at the pixel-level, even for expert surgeons. Manual anatomy segmentation masks are also prohibitively expensive to produce. Errors in target prediction usually drive overall triplet accuracy because a triplet is only correct if all three components in ⟨`instrument, verb, target`⟩ are correct. To address this challenge, we propose TargetFusionNet, a novel architecture that extends Mask2Former [Cheng 2022] with a target-aware fusion mechanism. TargetFusionNet integrates weak anatomy logits obtained from a pretrained network into the transformer decoder, improving the association between instruments and their likely targets. We demonstrate that this approach significantly improves target recognition and overall triplet accuracy compared to the existing CAM-based baseline, a standard instance segmentation baseline, and early / late fusion strategies.

Our key contributions are as follows:

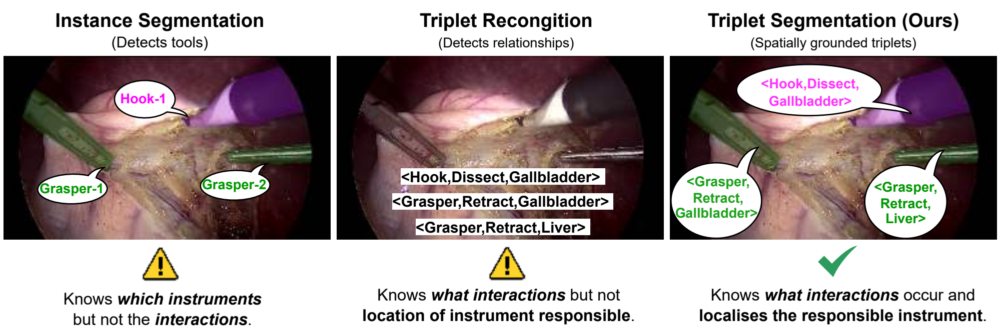


Figure 4.1: Comparison of instance segmentation, triplet recognition, and triplet segmentation. Triplet segmentation grounds ⟨`instrument, verb, target`⟩ triplets in space, unifying triplet recognition and instrument localisation for interpretable surgical scene understanding.

- Task: We define triplet segmentation, which unifies instrument instance segmentation and action–target classification for spatially grounded triplet analysis.

- Dataset: We release CholecTriplet-Seg, the first large-scale dataset linking instrument instances with verbs and targets for fine-grained triplet grounding.

- Metric: We introduce Triplet Segmentation mAP, a metric that evaluates both component and triplet-level accuracy with instance-level localisation.

- Method: We propose TargetFusionNet, a target-aware segmentation architecture that addresses target prediction bottlenecks, improving triplet grounding and establishing a new benchmark on our dataset.

4.2 Related work

Surgical triplet recognition (classification) Most prior works treat triplet recognition as a frame-level multi-label classification task, focusing on identifying which triplets are present rather than which instrument executes the action. TripNet [Nwoye 2020] and Rendezvous (RDV) [Nwoye 2022b] introduced attention-based models guided by weak spatial priors from class activation maps (CAMs), achieving strong frame-level recognition but lacking spatial grounding. Later studies explored disentangled classification [Chen 2023], self-distillation [Yamlahi 2023], and temporal reasoning [Sharma 2023a], yet none localise which instrument performs each action. Existing datasets such as CholecT50 [Nwoye 2022b] remain limited to frame-level triplet labels without explicit spatial alignment.

Surgical triplet detection (bounding box grounding) Triplet detection extends recognition by introducing bounding box spatial grounding. The CholecTriplet2022 Challenge [Nwoye 2023b] formalised this via bounding-box localisation from weak supervision. Subsequent work focused on refining frame-level recognition through disentangled classification [Chen 2023], self-distillation [Yamlahi 2023], and temporal reasoning [Sharma 2023a; Pei 2025] , but spatial grounding remains coarse since triplets are not explicitly linked to instrument instances. Bounding boxes also only offer poor localisation for elongated instruments in non axis-aligned orientation [Han 2025]. [Younis 2024] explored symbolic extensions of surgical action triplets by introducing explicit actor labels for responsibility attribution. However, this formulation does not provide pixel-level instance grounding of actions in the image.

Table 4.1: Overview of the datasets involved in constructing CholecTriplet-Seg. CholecT50 provides frame-level ⟨`instrument, verb, target`⟩ labels, CholecInstanceSeg provides dense instrument masks, and our CholecTriplet-Seg dataset unifies both by matching triplet annotations with instrument instances.

Dataset	Videos	Annotation Type	# Frames
CholecT50 [Nwoye 2022b]	50	Frame-level triplet labels	100,861
CholecInstanceSeg [Alabi 2025]	85	Instrument instance masks	41,933
CholecTriplet-Seg (Ours)	50	Triplet + instance (aligned)	30,955

Instrument instance segmentation datasets Research on surgical instrument segmentation has progressed from early binary or semantic labelling towards large-scale datasets offering fine-grained, instance-level annotations [Ahmed 2024]. Recent efforts such as PhaKIR [Rueckert 2026], GraSP [Ayobi 2024], and CholecInstanceSeg [Alabi 2025] have established comprehensive resources for training and benchmarking instrument instance segmentation models. Among these, CholecInstanceSeg provides high-quality instrument masks aligned with CholecT50 [Nwoye 2022b], forming the foundation for our CholecTriplet-Seg dataset that unifies instance segmentation with verb–target labels for grounded triplet analysis.

4.3 Material: The CholecTriplet-Seg dataset

CholecTriplet-Seg is a dataset of laparoscopic cholecystectomy videos with instrument-instance-grounded ⟨`instrument, verb, target`⟩ annotations. It is created by combining frame-level triplet labels from CholecT50 [Nwoye 2022b] with dense instrument masks from CholecInstanceSeg [Alabi 2025], synchronising corresponding frames and linking each instrument instance to its verb and target labels (see Table 4.1).

Triplet schema CholecT50 defines an annotation schema of six instruments, ten verbs, and fifteen targets, with 100 clinically valid triplet combinations established through expert consensus; we directly adopt this schema for CholecTriplet-Seg.

Alignment and quality control and ethics approval Frames from both datasets are aligned using timestamps to associate each instrument mask with a corresponding ⟨`instrument, verb, target`⟩ label. Automatic alignment was refined through manual verification to correct missing or ambiguous mappings, add absent masks and triplets, and ensure consistent assignments. Two rounds of visual inspection confirmed annotation quality, with additional statistics and discussion of dataset design trade-offs and expected sources of annotation ambiguity provided later in this chapter. The final dataset contains 30,955 annotated frames and 49,866 spatially grounded triplets from 50 cholecystectomy videos, including 15 fully annotated and 35 sparsely sampled cases (1 in 30 frames). The resulting triplet dataset exhibits a strongly long-tailed distribution, analysed in detail later in this chapter. Training, validation, and test splits follow the CholecInstanceSeg protocol. No ethics approval was required as only existing public datasets were used with additional annotations.

4.4 Methods

Our goal is to predict spatially grounded $\langle \texttt{instrument, verb, target} \rangle$ outputs by linking each instrument instance to its corresponding verb and target labels. We formalise this as the triplet segmentation task: given an input image $x \in R^{H\times W\times 3}$, the model outputs a set of instance masks $M = \{M_k\}$, each associated with a triplet label:

$$\mathcal{T} = \{\langle I_k, v_k, t_k \rangle \mid M_k \in M\},$$

where I_k, v_k, and t_k denote the instrument, verb, and target class of instance k.

While we have accurate spatial annotations for instruments, pixel-level annotations for anatomical targets are unavailable. This makes target prediction a major bottleneck in triplet segmentation. Our evaluation on CholecTriplet-Seg (see *Results*) shows that even with correct instrument masks and verb labels, poor target classification substantially limits Triplet mAP (see *Evaluation and Metrics*). Anatomical targets are particularly challenging. They often exhibit subtle appearance differences, lack distinct boundaries, and are inconsistently defined, even among expert surgeons [Owen 2022]. To address this, we propose TargetFusionNet, a model that improves target classification by incorporating weak anatomy priors derived from a pre-trained tissue segmentation network. These predictions are not treated as ground truth but as weak spatial cues that help disambiguate instrument–target interactions.

Targets represent anatomical structures (e.g., liver, gallbladder, cystic duct) that lack the clear boundaries and distinctive features of surgical instruments. Pixel-level target annotations for our triplet categories are unavailable because producing them would require extensive expert labelling and consensus-building. Without explicit target masks, target classification remains a major source of error.

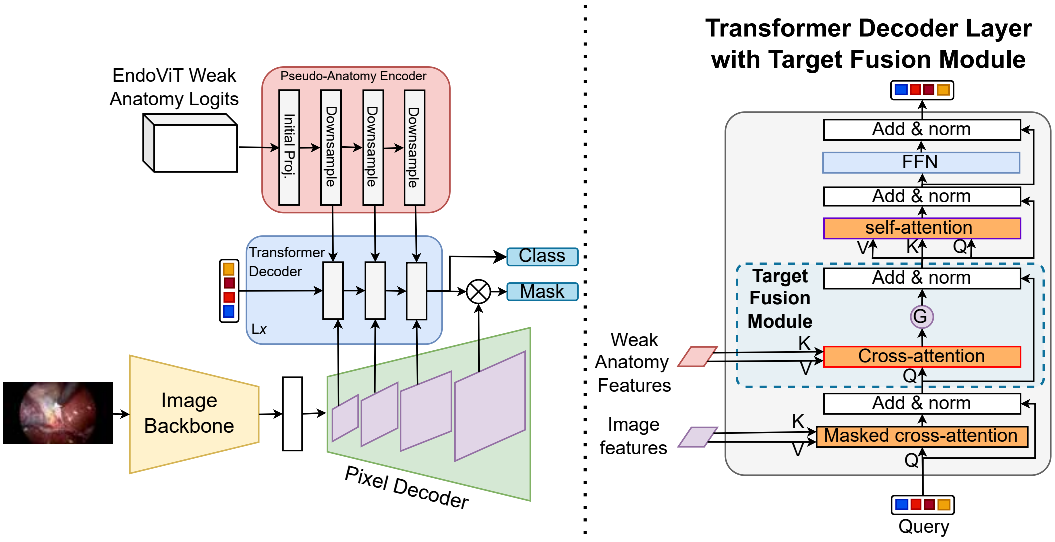


Figure 4.2: Overview of the TargetFusionNet architecture (left) and the Transformer Decoder Layer with Target Fusion Module (right). TargetFusionNet augments Mask2Former with an additional encoder that processes weak anatomy logits from EndoViT into multi-scale feature maps. These weak anatomy features serve as auxiliary memory for the transformer decoder. Each transformer decoder layer contains a Target Fusion Module, where instance queries interact with weak anatomy features via gated cross-attention. This allows the model to selectively incorporate coarse anatomical priors while refining triplet predictions.

Weak anatomy priors We incorporate coarse anatomical cues using inference logits from EndoViT [Batić 2024], trained on CholecSeg8k [Hong 2020] for tissue segmentation in cholecystectomy surgeries. EndoViT predicts six broad tissue classes that only partially overlap with the 15 anatomical target categories in our triplets, introducing class mismatches and noisy predictions. Nevertheless, these logits capture useful spatial context, such as rough anatomical boundaries (e.g., distinguishing liver regions from the gallbladder). We treat them as noisy priors rather than ground truth and integrate them into TargetFusionNet to guide target prediction. EndoViT is not used as a pretrained backbone but as a source of weak anatomical priors, providing coarse regional cues rather than enhanced visual features.

To avoid data leakage, we follow the CholecInstanceSeg split protocol [Alabi 2025], ensuring that any CholecSeg8k frames overlapping with CholecT50 appear only in the training partition. This prevents shared visual content between training and test sets, guaranteeing fair evaluation of anatomical priors.

TargetFusionNet Our TargetFusionNet, illustrated in Fig. 4.2, builds on Mask2Former [Cheng 2022], a transformer based instance segmentation model. We enhance its architecture with a target-aware fusion mechanism with three components:

- Weak anatomy encoder: The EndoViT logits are projected to the same feature dimension as the transformer decoder queries. A series of downsampling convolutions converts them into multi-scale weak anatomy feature maps that act as anatomical priors.

- Transformer decoder layers with target fusion: Mask2former transformer decoder layers are augmented with a target fusion module which attends to the weak anatomy features via a gated cross-attention as in Fig. 4.2. This allows the decoder to selectively incorporate anatomical context while refining triplet representations.

- Gated cross-attention: Each transformer query Q attends to weak anatomy features (K_t, V_t) through a learnable gating mechanism:

  $$Q' = Q + \sigma(W_g A) \odot A, \text{ where } A = \mathrm{Attn}(Q, K_t, V_t),$$

  where σ is a sigmoid function, W_g is a learnable linear layer, and A is the output of cross-attention between Q and the weak anatomy features. The gate controls how much anatomical context is injected, preventing noisy priors from dominating visual features.

Triplet output and loss Following Nwoye et al. [2022], we represent each $\langle$`instrument, verb, target`$\rangle$ combination as a single class within a 100-way classification space defined by the clinically valid triplets of CholecT50. For each predicted instance mask M_k, TargetFusionNet directly outputs a single triplet label $c_k \in [1; \ldots; 100]$, where each class c_k uniquely corresponds to a specific triplet combination. Each transformer query predicts one triplet label, while mask generation still follows the standard Mask2Former query–pixel interaction. The target-fusion gate acts only on the query embeddings before classification, enriching triplet semantics without altering the mask prediction process. The network is trained end-to-end using standard Mask2Former losses. A comparison of models with multitask loss variants, where separate losses are applied to instrument, verb, and target predictions, is included in the extended ablation analysis later in this chapter.

4.5 Experiments

We evaluate TargetFusionNet on the proposed CholecTriplet-Seg benchmark and compare it against both weakly and strongly supervised baselines to quantify the benefits of strong instance supervision and target-aware fusion.

Baselines The following models are used for comparison:

- RDV-Det [Nwoye 2023b]: the official CholecTriplet2022 baseline extending the RDV framework with class activation maps (CAMs) for weak localisation. It predicts triplets at the frame level and assigns them to instruments via CAM-derived bounding boxes, offering only coarse spatial grounding.

- RDV + Mask2Former: augments RDV-Det by replacing CAMs with instance masks predicted by a Mask2Former model trained on CholecTriplet-Seg. This isolates the benefit of strong instance segmentation while retaining RDV's triplet association mechanism.

- Mask2Former-Triplet: extends Mask2Former with a 100-class triplet head that jointly predicts instance masks and triplet labels, removing heuristic matching and providing a strong baseline for comparison with TargetFusion-Net. This model serves as an explicit architectural ablation, removing the target-aware fusion module while keeping the backbone, decoder structure, and training protocol unchanged.

Evaluation protocol Performance is assessed across triplet recognition (classification), triplet detection (bounding box grounding), and triplet segmentation settings to capture both triplet correctness and spatial grounding. Following the CholecT50 protocol [Nwoye 2023a], we report the standard recognition and detection mAP (bounding box based). To enable instance segmentation level evaluation, we introduce a new *triplet segmentation mAP* metric, which extends existing triplet evaluation to the pixel level by requiring both correct triplet classification and accurate mask localisation. This metric provides a unified framework for comparing models across frame, box, and mask-level grounding, thereby establishing the first benchmark for spatially grounded triplet segmentation. Metrics are reported for individual components (mAP_I, mAP_V, mAP_T), component pairs (mAP_{IV}, mAP_{IT}), and full triplets (mAP_{IVT}), with superscripts *rec*, *det*, or *seg* indicating the evaluation mode (e.g., mAP^{seg}_{IVT}). Further details on metric computation are provided later in this chapter and in the accompanying codebase. We also present qualitative

Table 4.2: Comprehensive results across triplet segmentation, detection (bounding box grounding), and recognition (classification). Mean Average Precision (mAP) is reported for instruments (mAP_I), verbs (mAP_V), targets (mAP_T), instrument–verb pairs (mAP_{IV}), instrument–target pairs (mAP_{IT}), and full triplets (mAP_{IVT}). **Bold** indicates the best among compared models.

Evaluation Metric	Method	mAP_I	mAP_V	mAP_T	mAP_{IV}	mAP_{IT}	mAP_{IVT}
Segmentation (mAP^{seg})	RDV-Det	0.09	0.11	0.08	0.08	0.04	0.03
	RDV + Mask2Former	48.11	32.51	16.29	14.40	11.38	8.73
	Mask2Former-Triplet	65.24	45.61	20.75	23.03	16.47	12.23
	TargetFusionNet	**67.19**	**46.27**	**21.55**	**24.93**	**17.75**	**13.47**
Detection (mAP^{det})	RDV-Det	3.28	2.86	0.29	1.43	0.77	0.55
	RDV + Mask2Former	54.28	34.48	13.54	16.56	9.30	6.72
	Mask2Former-Triplet	64.64	45.35	20.64	22.94	16.42	12.19
	TargetFusionNet	**66.52**	**46.23**	**23.70**	**24.82**	**17.74**	**13.45**
Recognition (mAP^{rec})	RDV-Det	80.55	61.44	40.83	35.93	34.47	27.17
	RDV + Mask2Former	80.55	61.44	40.83	35.93	34.47	27.17
	Mask2Former-Triplet	82.70	64.64	50.22	39.79	39.93	32.28
	TargetFusionNet	**86.61**	**71.93**	**51.19**	**44.48**	**43.24**	**34.23**

Table 4.3: Ablation of target-fusion strategies on segmentation mAP performance. **Bold** indicates the best among compared models.

Method	mAP_I^{seg}	mAP_V^{seg}	mAP_T^{seg}	mAP_{IV}^{seg}	mAP_{IT}^{seg}	mAP_{IVT}^{seg}
TargetFusionNet	67.19	46.27	**21.55**	24.93	**17.75**	**13.47**
Early-concat	**69.13**	**50.74**	21.52	**27.62**	16.83	12.21
Late-concat	59.90	40.65	20.21	23.17	16.07	10.99

results, visualising predicted masks and triplets to highlight common errors and the benefits of weak anatomical priors in TargetFusionNet.

Ablations To analyse design choices, we compare TargetFusionNet with two simplified fusion variants: 1) *early fusion*, where weak anatomical logits are concatenated with RGB inputs before the backbone; and 2) *late fusion*, where weak anatomical logits are injected after the transformer decoder via a lightweight convolutional encoder. Additional analyses later in this chapter compare the standard Mask2Former loss with multitask loss variants and evaluate how many decoder layers benefit from target fusion.

Implementation details TargetFusionNet was trained in the MMDetection [Chen 2019a] framework with a ResNet-50 [He 2016] backbone pretrained for instrument segmentation on data derived from CholecTriplet-Seg, using the AdamW optimiser [Loshchilov 2019]. Input images were resized to 1024×1024 with data augmentation comprising random horizontal flipping, cropping, and scaling. Training was performed on a single NVIDIA A100 GPU for 300k iterations, and the best checkpoint was selected based on validation mask mAP.

4.6 Results

We evaluate RDV-Det, RDV + Mask2Former, Mask2Former-Triplet, and TargetFusionNet on the CholecTriplet-Seg benchmark, reporting recognition (classification), detection (bounding box grounding), and segmentation mAP as defined in Section 4.5. Quantitative results are summarised in Table 4.2, and qualitative examples are presented in Fig. 4.3. Across all evaluation metrics, performance consistently improves with stronger spatial supervision and the introduction of target-aware fusion. Segmentation (mAP^{seg}) serves as our primary metric. RDV-Det provides negligible instance-level grounding (mAP$^{seg}_{IVT} = 0.03$), while adding strong instrument masks in RDV + Mask2Former raises accuracy to mAP$^{seg}_{IVT} = 8.73$. Mask2Former-Triplet achieves mAP$^{seg}_{IVT} = 12.23$, and TargetFusionNet attains the best triplet segmentation accuracy of mAP$^{seg}_{IVT} = 13.47$, with gains across all components, particularly in target prediction (mAP$^{seg}_{T} = 21.55$).

Similar trends are observed in detection and recognition. For detection, TargetFusionNet achieves the highest mAP$^{det}_{IVT} = 13.45$, outperforming Mask2Former-Triplet (mAP$^{det}_{IVT} = 12.19$) through more accurate target association. For recognition, which ignores spatial grounding, Mask2Former-Triplet improves over RDV baselines (mAP$^{rec}_{IVT} = 32.28$ vs. mAP$^{rec}_{IVT} = 27.17$), and TargetFusionNet further increases the triplet score to mAP$^{rec}_{IVT} = 34.23$ (+1.95).

To evaluate robustness, we use mAP$^{seg}_{IVT}$ and partition the test set into twelve subsets of 500 frames. A one-sided Wilcoxon signed-rank test shows that TargetFusionNet significantly outperforms Mask2Former-Triplet ($p = 0.026$). This subsampling strategy mitigates the small size of the original video-level test set and provides a more stable estimate of instance-level triplet performance.

Looking at the proposed ablations, as shown in Table 4.3, TargetFusionNet outperforms both early and late-fusion variants across most segmentation mAP metrics.

Qualitative evaluation To complement the quantitative analysis, Fig. 4.3 compares predictions from RDV-Det [Nwoye 2022b], RDV-Det + Mask2Former, Mask2Former-Triplet, and our TargetFusionNet across diverse scenarios. The examples highlight both strengths and remaining limitations. In case (a) and (c) where there is a single ⟨`Grasper, Retract, Gallbladder`⟩ or ⟨`Hook, Dissect,`

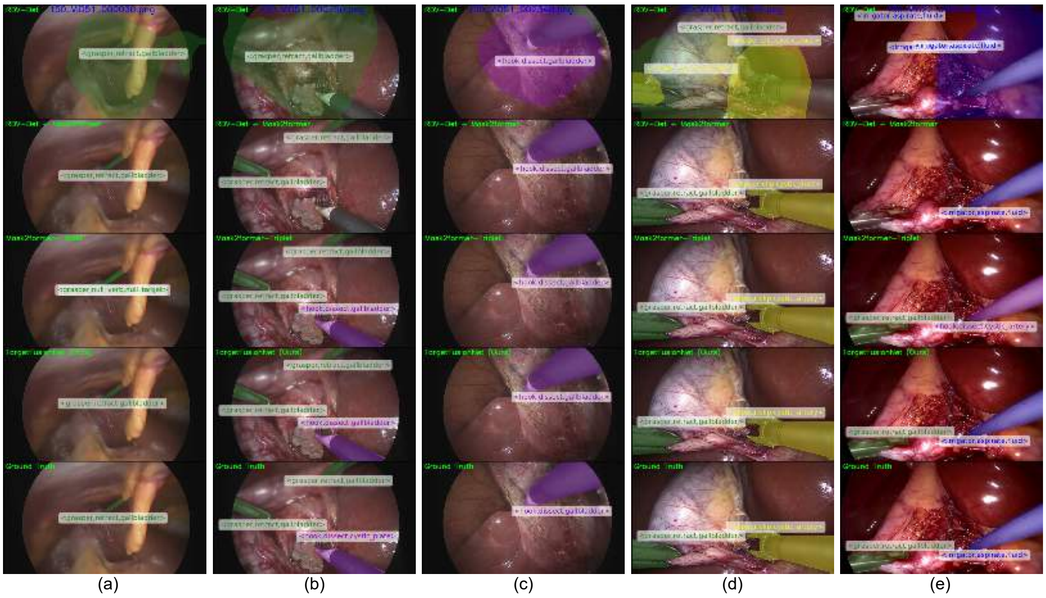


Figure 4.3: Qualitative comparisons across five frames (a–e). Models shown: RDV-Det, RDV-Det + Mask2Former, Mask2Former-Triplet, and TargetFusionNet. Triplet predictions are shown overlaid on the image. Our method achieves more accurate spatial grounding and better triplet consistency in most cases.

`Gallbladder`⟩ respectively, all models perform reliably, except RDV-Det which has weak localization ability. However, in complex scenes such as (b), (d) and (e) involving multiple active instruments, occlusions, or rare targets, TargetFusionNet achieves more consistent spatial grounding and correct instrument–target linkage. For instance in (d), it successfully distinguishes a ⟨`Clipper, Clip, Cystic Duct`⟩ from the visually similar cystic artery, and in (e), it recognises the ⟨`Irrigator, Aspirate, Fluid`⟩ where baselines misclassify the tool as a hook. These examples illustrate the advantages of target-aware fusion for robust triplet grounding. A short video comparing all evaluated methods (RDV-Det, RDV-Det + Mask2Former, Mask2Former-Triplet, and TargetFusionNet) with the ground truth is provided in the Supplementary Material.

4.7 Extended Dataset Construction and Annotation Protocol

This section provides further details on how CholecTriplet-Seg was constructed by aligning CholecT50 and CholecInstanceSeg, as described in "The CholecTriplet-Seg Dataset" section of the main paper. CholecTriplet-Seg provides spatially grounded triplet annotations ⟨instrument, verb, target⟩ for laparoscopic cholecystectomy. The dataset was created by aligning annotations from two public datasets: CholecT50 [Nwoye 2022b], which provides frame-level triplets, and CholecInstanceSeg [Alabi 2025], which contains instance segmentation masks for surgical instruments.

Frame Selection. We selected 30,955 frames corresponding to the overlapping frames between CholecT50 and CholecInstanceSeg. These frames span 50 laparoscopic cholecystectomy videos, comprising 15 full videos (with all frames overlapping) and 35 sparsely overlapping videos (around 1 frame every 30 frames).

Triplet Vocabulary. Triplets are structured as ⟨instrument, verb, target⟩, with class vocabularies derived from CholecT50: 6 instruments, 10 verbs, and 15 targets. While this results in 900 theoretically valid combinations, CholecT50 narrowed the annotation space to 100 clinically meaningful triplets based on expert feedback. Of these, 93 are observed in CholecTriplet-Seg. We adopt this clinically informed vocabulary to ensure consistency and comparability with prior work.

Annotation Format. Annotations in CholecTriplet-Seg build directly on the LabelMe-based structure used in CholecInstanceSeg [Alabi 2025]. Each annotated shape originally included a 'label' (instrument class), 'points' (polygon coordinates), and 'group_id' (instance id). To support grounded action understanding, we extended this schema by introducing two additional fields: 'verb' and 'target'. These fields assign an action and anatomical target to each instrument instance. If an instrument is represented by multiple polygons, all associated shapes share the same 'group_id', 'label', 'verb', and 'target', thereby forming a single annotated triplet.

Annotation Pipeline and Error Resolution. To construct grounded triplet annotations, we performed initial alignment by automatically matching instrument classes labels in CholecT50 and CholecInstanceSeg. The process revealed significant inconsistencies that required manual correction. We categorised these inconsistencies into two major types:

- **Matching errors** — common in frames with multiple identical instruments (typically graspers), each associated with different actions. Because CholecT50 does not ground triplets to specific instances, automatic assignment failed in these cases. Manual matching using a custom GUI tool was necessary to link each triplet to the correct mask. These issues affected **2,361 frames**.

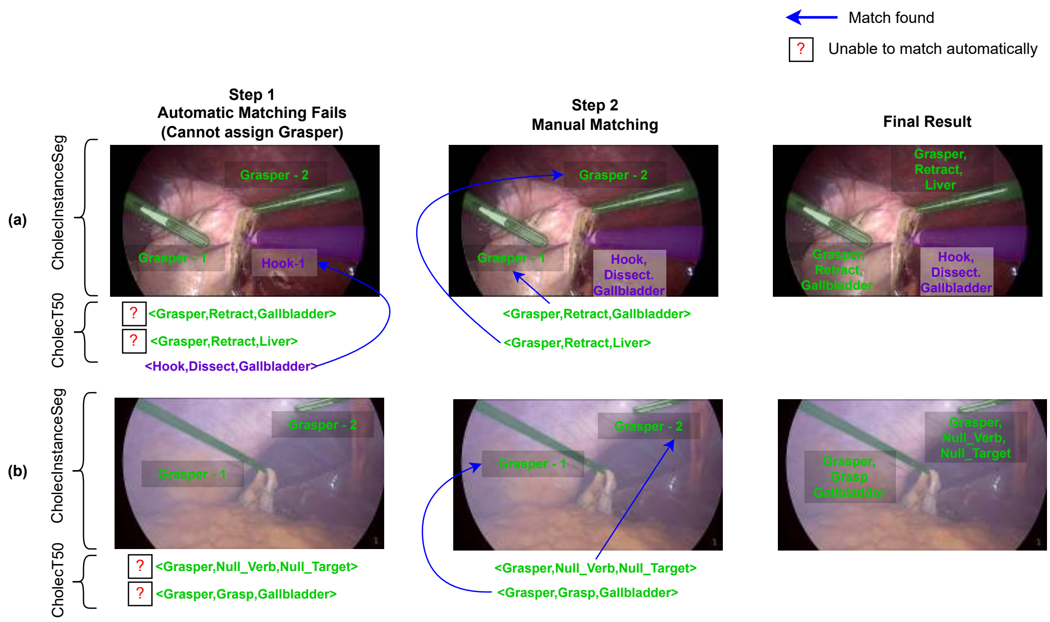


Figure 4.4: Examples of matching errors. (a) Multiple graspers with distinct actions cannot be automatically matched to triplets. (b) Similar to A, multiple graspers cannot be automatically matched. Manual matching is required.

- **Conflict errors** — frames where the segmentation and triplet annotations directly disagreed, such as missing masks, redundant triplets, or mismatched class labels. Each case required manual inspection and correction, including adding or removing triplets and masks. A total of **7,194 frames** were affected.

Visual Examples of Annotation Challenges. Figures 4.4 and 4.5 show representative examples of each error type. Matching errors typically involve multiple instruments of the same class with conflicting or ambiguous triplet assignments, making automatic matching infeasible. Conflict errors involve hard contradictions between the datasets, requiring the addition or removal of annotations to restore consistency.

Annotation Procedure. A single trained annotator verified and refined the dataset using a modified LabelMe interface that supported visualising both instance masks and triplet annotations.

Quality Control and Limitations. After merging, we conducted a full dataset pass to correct inconsistencies and enforce schema constraints. Every instrument

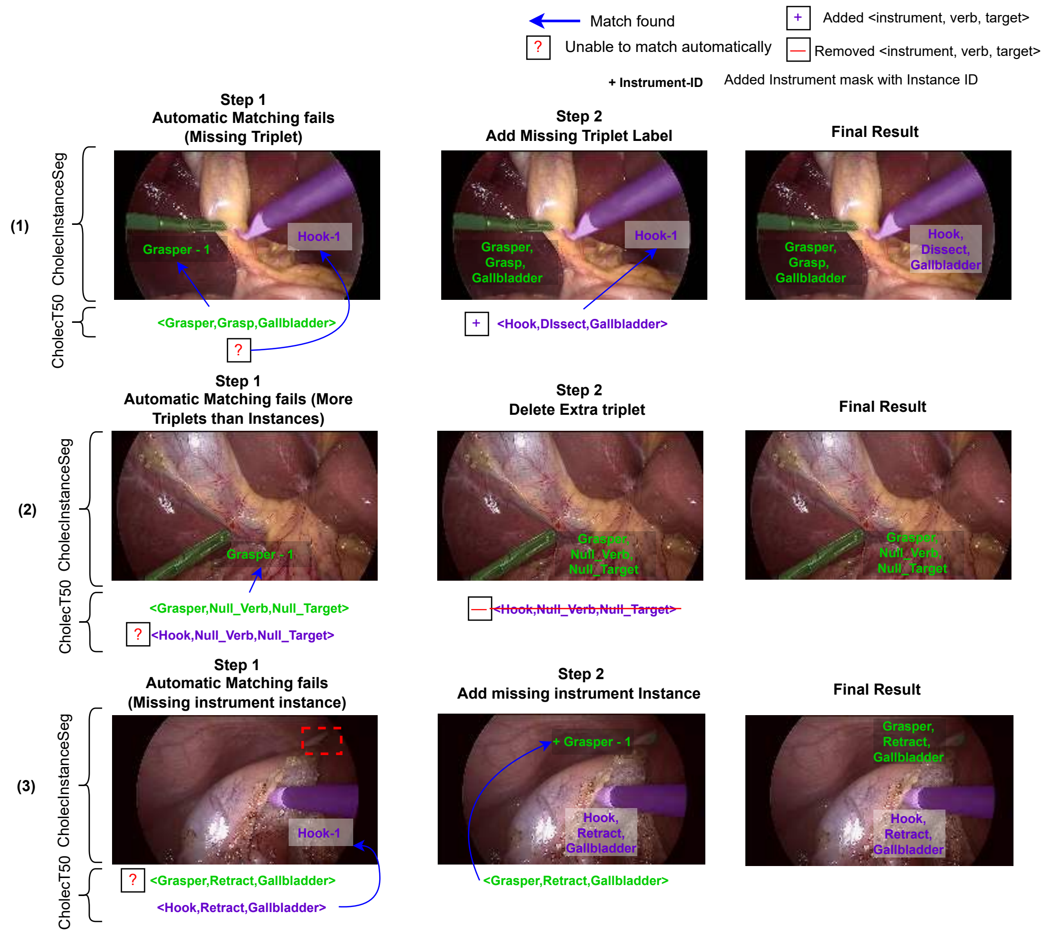
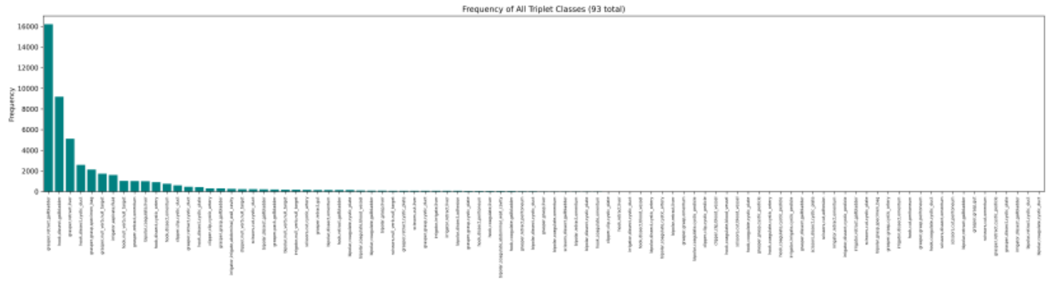


Figure 4.5: Examples of conflict errors. (1) Missing triplet that needs to be added. (2) Redundant triplet with no corresponding mask that must be removed. (3) Missing instrument instance mask needs to be added.

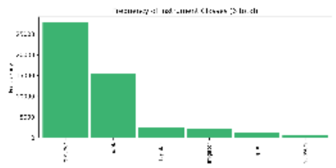
instance was checked to ensure it had been assigned a verb and target. While labels are consistent within the dataset, a few limitations remain:

- Rare triplets are underrepresented due to long-tailed frequency.
- Some actions are visually ambiguous (e.g., The actions *grasp* vs. *retract* involves holding and pulling), which caused uncertainty.
- Annotator decisions were guided by internal rules rather than clinical expert revalidation, which may affect borderline cases.

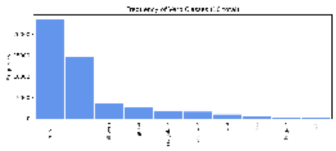
Despite these issues, the dataset reflects a grounded approach to triplet annotation and is sufficient to benchmark new models for spatially grounded action understanding.



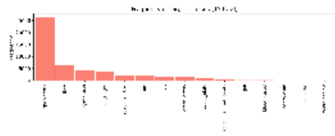
(a) Frequency of all triplet classes.



(b) Instrument class frequencies.



(c) Verb class frequencies.



(d) Target class frequencies.

Figure 4.6: Class frequency distributions in the dataset. The top row shows the frequency of all triplet classes (instrument, verb, target combinations), while the bottom row shows the distribution across individual instruments, verbs, and targets.

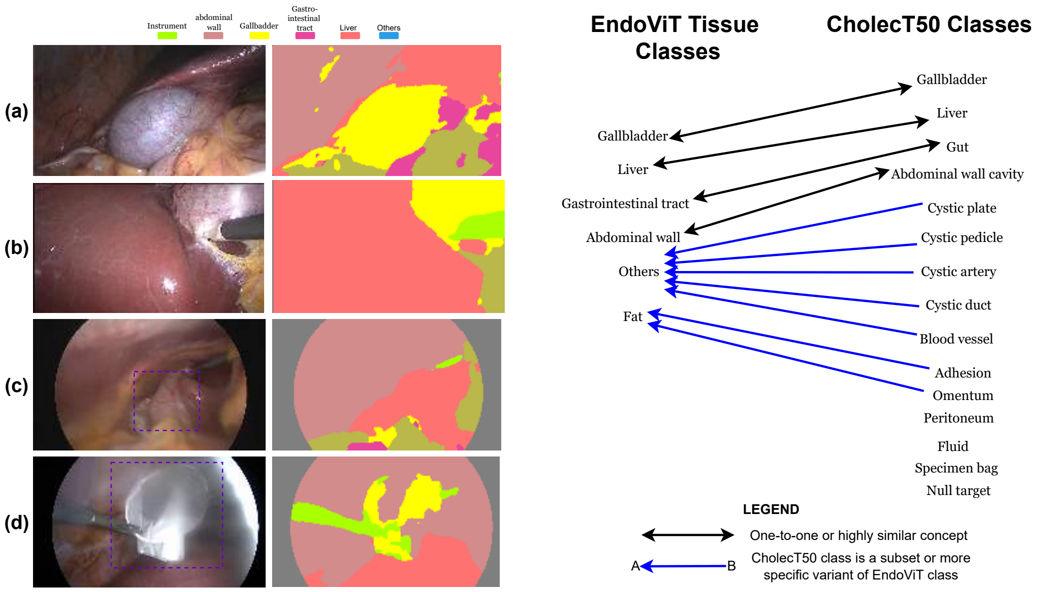
4.8 Additional Dataset Statistics

We provide additional statistics for CholecTriplet-Seg to highlight its training splits and class imbalance characteristics. We adopt the same train/val/test splits as CholecInstanceSeg. In total, CholecTriplet-Seg contains 49,866 annotated triplets. Table 4.4 gives the statistics of the dataset split

Split	# Videos	# Images	# Triplets	Avg Triplets / Frame
Train	45	22,818	37,591	1.65
Validation	1	2,496	3,341	1.34
Test	4	5,641	8,934	1.58
Total	50	30,955	49,866	1.61

Table 4.4: Dataset split statistics for triplet annotations.

Class distribution. The dataset exhibits a highly imbalanced (long-tailed) distribution, as shown in Fig. 4.6. A few frequent triplets dominate the dataset, while many others occur only sparsely. This pattern is mirrored in the individual instrument, verb, and target class distributions.



(a) EndoViT predictions: success and failure cases.

(b) Mapping between EndoViT tissue classes and CholecTriplet-Seg targets.

Figure 4.7: (a) EndoViT predictions visualised via `argmax` over softmax channels. Successful examples show well-aligned anatomical boundaries; failure cases include missing structures and misclassification of out-of-vocabulary classes. (b) Manually defined mapping between EndoViT classes and CholecTriplet-Seg targets. Solid lines indicate strong semantic alignment; dashed lines indicate subset or approximate correspondence.

4.9 EndoViT Weak Supervision Details

To complement the model description in the main text, we analyse the behaviour and alignment of the pseudo-anatomical signals provided by EndoViT. To guide grounded triplet prediction with spatial anatomical cues, we incorporate pseudo-target logits from EndoViT [Batić 2024], a segmentation model trained to identify nine laparoscopic structures (one instrument, two background classes, and six anatomical regions). Although EndoViT was not designed for our target taxonomy, its outputs provide coarse yet informative spatial priors. We opted to retain the full softmax logits during training to the model rather than filtering or masking, as missing logits could lead to more disruptive inconsistencies during training.

Qualitative assessment of EndoViT outputs. Fig. 4.7a illustrates both successful and failed examples of EndoViT's anatomical predictions. We visualise the predictions by taking the `argmax` across softmax channels to identify the dominant class per pixel. Examples (a) and (b) demonstrate strong segmentation performance, particularly in distinguishing bladder, liver, fat, and abdominal wall with accurate boundaries. In contrast, examples (c) and (d) highlight failure modes: (c) shows a missed gallbladder due to its position at the image periphery, while (d) depicts the specimen bag misclassified as gallbladder due to EndoViT lacking a dedicated specimen bag class.

Semantic alignment with CholecTriplet-Seg targets. To better understand the relevance of EndoViT predictions to our task, we manually constructed a semantic mapping between EndoViT's tissue classes and CholecTriplet-Seg targets (Fig. 4.7b). Some mappings are one-to-one (e.g., *liver* to *liver*), while others are approximate or partial overlaps (e.g., EndoViT's background covering multiple target labels). These observations support our design choice of using gated attention, allowing the model to learn when to trust or ignore weak anatomical signals during decoding.

4.10 Implementation Details

More implementation details are provided here to aid reproducibility and replication. All experiments were implemented using the MMDetection framework, which offers modular training pipelines, flexible configuration management, and support for segmentation architectures such as Mask2Former. All major dependency versions are documented in the released codebase. Training and evaluation were conducted on a single NVIDIA A100 GPU (40GB VRAM).

Model architecture. Our model extends Mask2Former with two core additions: (1) a weak target encoder that processes soft pseudo-target logits from EndoViT, and (2) a gated target fusion module that integrates these features into the transformer decoder. The weak target encoder first applies a 1×1 convolution to match the transformer's channel size, then downscales the features using a sequence of 3×3 convolutions with stride 2, batch normalisation, and ReLU activations. These multi-scale features are then injected via gated cross-attention at each decoder stage. This can be found in `downsampler_encoder.py`

We also introduced new modules to handle fusion and target-specific processing, including: (i) a classification head (`Mask2FormerTargetFusionHead.py`), (ii) a fusion-aware decoder (`targetfusionhybrid_decoder.py`), and (iii) data preprocessing logic to handle EndoViT logits.

Evaluation Library. Evaluation was performed using a customised version of the `ivtmetrics` library, originally developed for CholecT50 [Nwoye 2022b]. Due to compatibility issues, the library depends on specific versions of Python packages to run reliably. We provide a detailed README alongside the codebase, outlining the required environment and installation steps to ensure reproducibility.

Training Setup. TargetFusionNet was trained with an image resolution of 1024×1024 and a per-GPU batch size of 4. To achieve larger effective batch sizes, we employed gradient accumulation. The classification loss weight was increased from 3.0 to 4.0 to promote sharper and more confident triplet predictions. All other settings followed MMDetection defaults unless otherwise specified.

For models based on Mask2Former, we initialized from a pretrained instance segmentation checkpoint trained on CholecInstanceSeg. To ensure no data leakage and maintain consistency during fine-tuning, we adopted the same train/val/test splits as CholecInstanceSeg.

Reproducibility. All code, configuration files, and preprocessed pseudo-targets from EndoViT are publicly available at `https://github.com/labdeeman7/target_fusion_net`. Model checkpoints and evaluation scripts will be included to facilitate full reproduction of results.

4.11 Evaluation and Metrics

We evaluate triplet segmentation by measuring both triplet correctness and spatial grounding. To this end, we extend the CholecT50 protocol [Nwoye 2022b] with a new *Segmentation mAP* metric that uses instance masks (IoU ≥ 0.5) to assess spatial grounding with pixel-level precision. This provides a finer measure of instrument–tissue interaction than coarse box-level evaluation. For completeness, we also report detection and recognition metrics. Together, we end up with 3 classes of Triplet mAP metrics:

Recognition mAP mAP^{rec} evaluates which triplets are present in each frame, regardless of spatial alignment. It is computed using a **classification** style mean Average Precision (mAP), averaged across triplet classes and videos.

Detection mAP mAP^{det} assesses whether each predicted triplet is correctly linked to an instrument instance **bounding box** using bounding box IoU ≥ 0.5, following a detection-style mAP.

Segmentation mAP mAP^{seg} applies the same criteria using **instance masks** instead of bounding boxes, following segmentation mAP computation style.

We report mean Average Precision (mAP), computed per class and averaged across classes and across videos, following the CholecT50 protocol [Nwoye 2022b]. Metrics are reported for individual components (mAP_I, mAP_V, mAP_T), component pairs (mAP_{IV}, mAP_{IT}), and full triplets (mAP_{IVT}), with superscript *rec*, *det*, or *seg* indicating the evaluation mode (e.g., mAP^{seg}_{IVT}). Further implementation details are available in the accompanying codebase.

While component-level metrics (e.g., mAP_I) provide diagnostic insights, mAP_{IVT} across recognition, detection, and segmentation serves as our main benchmark for fully grounded triplet prediction. The recognition and detection metrics are implemented through the `ivtmetrics` library; a detailed description of these metrics can be found in the corresponding release paper [Nwoye 2023a]. Our repository extends this framework to include triplet-level segmentation mAP.

4.12 Ablation Results for More Model Variants

We experimented with alternative formulations to evaluate different design choices for triplet prediction. This section reports additional results for fusion strategy, multitask learning, and fusion depth variants of TargetFusionNet. These experiments assess whether changes to the fusion design, task formulation, or decoder depth can further improve triplet segmentation performance.

4.12.1 Fusion Strategy Ablations

To complement the segmentation ablation reported in the main text, we provide full results across recognition, detection, and segmentation for different fusion strategies. These variants differ in how weak anatomical priors are integrated: **Early-concat** fuses logits before the backbone, **Late-concat** injects them after the transformer decoder, and **TargetFusionNet** applies gated mid-level fusion. As shown in Table 4.5, TargetFusionNet achieves the best overall performance across all metrics, confirming that gated mid-level integration offers the most effective balance between spatial guidance and feature learning.

4.12.2 Multitask learning

One natural alternative to 100-class triplet prediction is to decompose the task into three independent classification heads: one each for instrument, verb, and target. This aligns more closely with the compositional structure of triplets and

Table 4.5: **Fusion strategy ablation across recognition, detection, and segmentation.** Mean Average Precision (mAP) reported for instruments (mAP_I), verbs (mAP_V), targets (mAP_T), instrument–verb pairs (mAP_{IV}), instrument–target pairs (mAP_{IT}), and full triplets (mAP_{IVT}). **Bold** indicates the best among fusion variants.

Evaluation Type	Method	mAP_I	mAP_V	mAP_T	mAP_{IV}	mAP_{IT}	mAP_{IVT}
Segmentation (mAP^{seg})	**TargetFusionNet**	67.19	46.27	**21.55**	24.93	**17.75**	**13.47**
	Early-concat	**69.13**	**50.74**	21.52	**27.62**	16.83	12.21
	Late-concat	59.90	40.65	20.21	23.17	16.07	10.99
Detection (mAP^{det})	TargetFusionNet	66.52	46.23	**23.70**	24.82	**17.74**	**13.45**
	Early-concat	**67.95**	**49.77**	23.25	**27.05**	16.56	12.04
	Late-concat	59.02	40.03	19.70	22.84	15.85	10.86
Recognition (mAP^{rec})	TargetFusionNet	86.61	**71.93**	**51.19**	**44.48**	43.24	34.23
	Early-concat	**88.65**	66.19	51.10	44.15	**43.41**	**34.47**
	Late-concat	77.91	64.32	47.26	40.87	36.66	30.33

Table 4.6: Segmentation performance (mAP^{seg}) for Mask2Former-Triplet and multitask learning versions. While multitask variants improve individual component scores (e.g., instrument, verb), they fail to improve overall triplet accuracy (mAP^{seg}_{IVT}), highlighting the challenge of enforcing coherent triplet structure.

Method	mAP^{seg}_I	mAP^{seg}_V	mAP^{seg}_T	mAP^{seg}_{IV}	mAP^{seg}_{IT}	mAP^{seg}_{IVT}
Mask2Former-Triplet (100 class)	65.24	45.61	20.75	23.03	16.47	**12.23**
Mask2Former-Triplet (Multitask)	**73.73**	**49.04**	20.86	**25.69**	16.65	12.12
Early-concat (Multitask)	72.53	43.24	**21.40**	21.15	**16.86**	10.30

could potentially alleviate long-tail issues, since many individual categories are more balanced than their combined triplet forms.

To explore this idea, we implemented two multitask variants for our Mask2Former-Triplet baseline. The first, Mask2Former-Triplet (Multitask), replaces the single 100-class head with three separate heads. The second variant, Early-concat (Multitask), augments the input by concatenating EndoViT's weak target logits with the RGB channels before the backbone, but still uses multitask heads.

Results are shown in Table 4.6. We observe that multitask learning improves instrument and verb segmentation scores, particularly in the absence of EndoViT signals. However, it fails to produce gains in triplet accuracy (mAP^{seg}_{IVT}). In fact, combining multitask with weak anatomical signals (Early-concat) reduces performance even further. This suggests that multitask learning, while promising, struggles to enforce coherent triplet-level structure. For this reason, we retain the 100-class formulation as the default setting for TargetFusionNet and leave further exploration for future works.

Table 4.7: Ablation of fusion depth in TargetFusionNet. We vary the number of transformer decoder layers where the target fusion module is applied. Full fusion across all 9 layers yields the highest triplet accuracy (mAP^{seg}_{IVT}).

Method	mAP^{seg}_{I}	mAP^{seg}_{V}	mAP^{seg}_{T}	mAP^{seg}_{IV}	mAP^{seg}_{IT}	mAP^{seg}_{IVT}
TargetFusionNet (9 layers)	**67.19**	**46.27**	21.55	24.93	**17.75**	**13.47**
TargetFusionNet (6 layers)	66.89	45.66	**21.92**	**26.43**	16.52	11.95
TargetFusionNet (3 layers)	64.14	44.26	21.58	23.09	16.36	11.85

Table 4.8: **Model efficiency comparison.** Measured on a single A100 (40 GB) with image size 1024x1024 input and batch size 1

Method	**Params (M)**	**FLOPs (G)**	**FPS ↑**
RDV-Det	17.1	46.3	25.83
RDV + Mask2Former	61.1	302.3	20.27
Mask2Former-Triplet	44.0	260	51.4
TargetFusionNet	50.34	628	31.8

4.12.3 Fusion depth ablations.

We also evaluated variants of TargetFusionNet with the fusion module applied only to a subset of the transformer decoder layers. Since the anatomical logits from EndoViT are weak and potentially noisy, it may be beneficial to limit their influence to early layers, allowing later stages to rely more heavily on learned visual features. This ablation helps assess whether fusion depth affects performance. Specifically, we tested fusing at 3, 6, or all 9 decoder layers. Results in Table 4.7 show that using all 9 layers provides the best overall triplet segmentation accuracy, suggesting that continuous guidance from weak targets throughout the decoder helps the model better align instrument actions with anatomical context.

4.12.4 Model Efficiency

We present computational efficiency in terms of parameters, floating-point operations (FLOPs) and inference speed for the models. All models were benchmarked on a single NVIDIA A100 (40 GB) GPU and a batch size of 1 with automatic mixed precision and all images resized to 1024x1024. Table 4.8 summarises the results. TargetFusionNet introduces only a modest computational overhead compared to Mask2Former-Triplet, while providing consistent accuracy gains across all metrics. FLOPs and parameter counts for Mask2former based models were estimated using MMDetection's *get_flops.py* script; results are approximate due to potential unsupported operations.

Table 4.9: Triplet segmentation performance grouped by triplet frequency. Triplet classes are partitioned into frequent, medium-frequency, and rare groups based on their occurrence in the training set. Performance is reported using triplet segmentation mAP (mAP_{seg}^{IVT}).

Triplet Frequency Group	No Triplet Classes	Avg. Samples / Class	mAP_{seg}^{IVT}
Frequent	10	4168.9	0.4148
Medium-frequency	21	307.9	0.2071
Rare	62	24.9	0.04

4.13 Effect of Long-Tailed Triplet Distribution on Model Performance

CholecTriplet-Seg exhibits a long-tailed distribution over clinically valid triplet combinations, with a small number of frequent triplets accounting for a large proportion of annotations and many triplets occurring only sparsely (Fig. 4.6). To analyse the effect of this imbalance on triplet segmentation performance, we group triplet classes into three frequency categories based on their occurrence in the training set.

Specifically, triplet classes are categorised as *frequent* if they occur more than **1000** times in the dataset, *medium-frequency* if they occur between **100** and **1000** times, and *rare* if they occur fewer than **100** times in the training set. These thresholds are chosen heuristically to reflect the highly skewed class distribution and to enable a coarse analysis of performance trends under long-tailed supervision, rather than to define strict or optimal class boundaries.

For each frequency group, we report the average triplet segmentation performance (mAP_{IVT}^{seg}), summarised in Table 4.9. As expected, performance decreases with decreasing triplet frequency due to limited supervision and increased semantic ambiguity.

4.14 Failure Mode Analysis

To better understand the remaining errors in triplet segmentation, we analyse common failure patterns observed across the test set. Rather than isolated prediction noise, these errors consistently arise from a small number of recurring scenarios, which we categorise below.

Missing or incorrect instrument masks. Instance-level triplet segmentation depends on accurate instrument instance prediction. We observe occasional missing, fragmented, or incorrectly assigned instrument masks, which result in incorrect triplet predictions. Joint optimisation for triplet segmentation can slightly

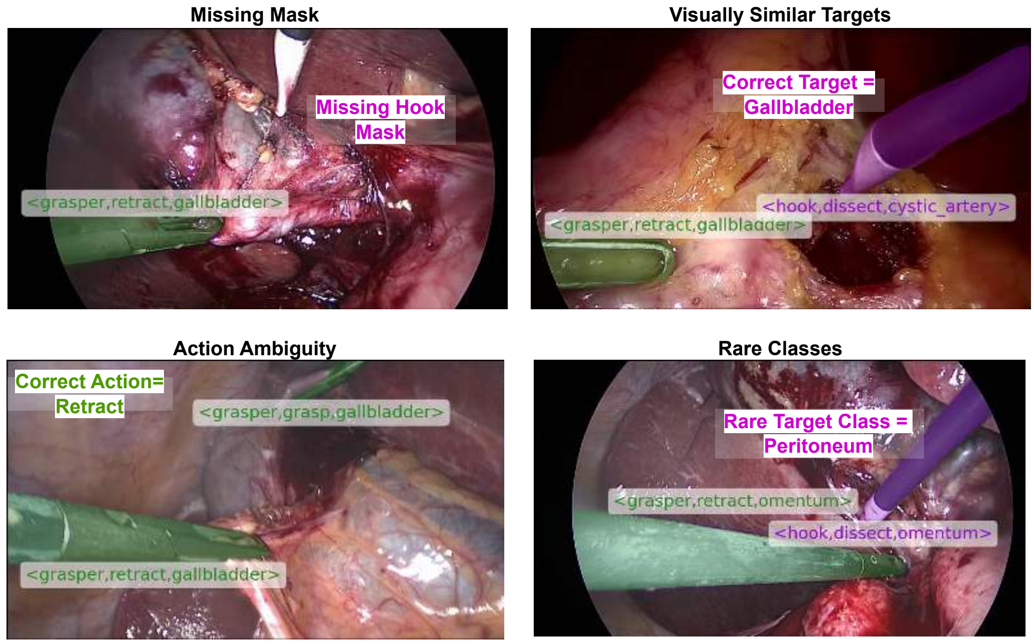


Figure 4.8: Representative failure cases in triplet segmentation. **Top left:** missing hook instrument mask leading to incorrect triplet prediction. **Top right:** confusion between boundaries of visually similar anatomical targets (gallbladder vs. cystic artery). **Bottom left:** action ambiguity between visually similar verbs (grasp vs. retract). **Bottom right:** rare target class (peritoneum) with limited training examples.

degrade raw instance mask quality compared to training instance segmentation in isolation, reflecting a trade-off between instance precision and higher-level semantic reasoning.

Rare class errors Classes with very low training frequency are more likely to be misclassified due to limited supervision. This effect is particularly pronounced for infrequent anatomical targets such as the peritoneum.

Visually similar targets Anatomically adjacent or visually similar structures (e.g., cystic artery, cystic duct, cystic pedicle, and gallbladder) are occasionally confused, especially when boundaries are ambiguous and weakly defined. While weak anatomical priors provide coarse spatial context, they do not always resolve fine-grained semantic distinctions between such targets.

Action ambiguity Some surgical actions exhibit subtle visual differences (e.g., grasp vs. retract), making them difficult to distinguish from single frames without explicit temporal context. As a result, verb misclassification can occur even when the instrument instance and anatomical target are correctly localised.

Representative examples of these failure modes are shown in Fig. 4.8, illustrating typical errors arising from missing masks, rare target classes, visually similar anatomical structures, and action ambiguity. These failure modes highlight intrinsic challenges of fine-grained laparoscopic scene understanding and motivate future work on improved instance segmentation, anatomical modelling, better metrics, and temporal reasoning.

4.15 Discussion and conclusion

In this work, we introduced the task of triplet segmentation, released the CholecTriplet-Seg benchmark, and proposed TargetFusionNet, which combines strong instance supervision with weak anatomical priors for fine-grained and interpretable surgical scene understanding. Our experiments demonstrate that models trained with strong instrument supervision achieve the best overall balance across instrument, verb, and target predictions, significantly outperforming weakly supervised approaches such as RDV-Det. Instance-level instrument linking improves performance by reducing action ambiguity in multi-instrument scenes and enforcing spatial consistency between predicted actions and their visual evidence. Despite these advances, two key challenges remain. First, while triplet-optimised models achieve the highest triplet mAP^{seg}_{IVT}, their instrument masks (mAP^{seg}_{I}) are slightly less precise than those from single-task segmentation models [Alabi 2025]. This trade-off highlights the need for hybrid or multi-stage architectures that preserve mask quality while enabling effective triplet reasoning. Second, target segmentation still remains a bottleneck, as low target mAP directly constrains overall triplet accuracy. Target prediction is intrinsically harder than instrument/verb prediction due to high visual similarity and ambiguous boundaries between anatomical structures, which contributes to the lower target mAP. A more detailed breakdown of failure patterns with representative examples is provided in the preceding analysis. Improving anatomical modelling beyond weak priors could further enhance spatial grounding. Future extensions could incorporate temporal reasoning to exploit action continuity across frames and integrate relation-aware modules to capture richer inter-instrument and tissue relationships.

Within the thesis, these findings connect its spatial and semantic strands. Strong supervision resolves the ambiguity of associating an action with a visible instrument instance, while the results identify anatomical-target prediction and rare interactions as the principal remaining difficulties. TargetFusionNet addresses localisation and semantic interaction prediction within a unified discriminative architecture. Chapter 5 investigates a complementary factorisation: specialist instrument grounding first establishes the actor and its spatial extent, after which a multimodal model predicts the associated verb and anatomical target.

Chapter 5

Two-Stage Multimodal Interaction Prediction

Table of contents

This chapter forms the basis of the manuscript in preparation *Multimodal Large Language Models for Interaction Prediction in Two-Stage Surgical Triplet Segmentation.* The List of Publications records its status and authorship.

5.1 Introduction

The preceding chapters develop a progression from individual surgical-vision tasks to spatially grounded instrument–tissue interactions. Chapter 2 identifies the need to connect complementary perceptual and semantic information; Chapter 3 provides the instrument-instance annotations required to distinguish individual tools; and Chapter 4 combines those spatial annotations with surgical action triplets to establish CholecTriplet-Seg and TargetFusionNet. Together, these contributions show that instrument instances can be localised reliably, but that predicting the verb and, especially, the anatomical target associated with each instance remains substantially more difficult.

This imbalance motivates a two-stage formulation. Instrument instance segmentation is a dense prediction problem for which specialist architectures are well suited. Once the instruments have been localised, interaction prediction is instead a structured semantic task: the model must interpret the action, anatomy, and surrounding scene for each known instance. Requiring one architecture to learn both capabilities jointly may not be necessary. Separating them also provides a controlled setting in which interaction prediction can first be studied with ground-truth instrument instances and then evaluated as part of a complete pipeline with predicted instances.

Multimodal large language models (MLLMs) offer a natural candidate for the second stage because they combine visual input with textual instructions and can generate structured outputs. This chapter investigates whether an MLLM can serve as a viable interaction predictor when instrument instances are supplied, rather than asking it to perform dense localisation. The resulting framework uses a fixed Mask2Former for instrument grounding and Qwen3-VL-8B-Instruct [Bai 2025] for ⟨`verb, target`⟩ prediction. The complete triplet is formed by combining the fixed instrument identity from Stage 1 with the interaction predicted by Stage 2.

Importantly, Stage 2 is not formulated as conventional frame-level surgical triplet recognition. Frame-level recognition determines whether a triplet occurs somewhere in an image, but need not associate it with a particular physical instrument when several instances are visible. Here, each supplied instance is spatially grounded by a mask and assigned a unique identifier. Stage 2 predicts a ⟨`verb, target`⟩ interaction for each identifier, preserving the association between the semantic prediction and its instrument mask. The complete system therefore produces spatially grounded triplet instances rather than an unordered set of frame-level triplet labels.

The study first establishes Direct SFT as the principal baseline and then evaluates controlled changes to the second-stage formulation. Sequential Dialogue Prediction decomposes interaction prediction across staged dialogue responses; temporal variants supply short windows of neighbouring frames; an anatomical-context variant supplies coarse EndoViT-derived evidence around an estimated instrument tip; ontology-guided prompting supplies valid ⟨`verb, target`⟩ combinations; and distilled-reasoning supervision uses rationales generated by a larger teacher model. These variants are evaluated as individual adaptations, not as a factorial or cumulative ablation study.

The chapter addresses four research questions:

1. Can an MLLM serve as a viable second-stage interaction predictor when instrument instances are provided?

2. How do sequential factorisation, short temporal context, anatomical priors, ontology guidance, and distilled-reasoning supervision affect interaction prediction?

3. Can the complete two-stage system remain competitive when the instrument instances are predicted?

4. Which limitations prevent the observed differences from being interpreted as definitive improvements?

The principal contribution is the two-stage formulation and its empirical evaluation. The results show that Direct SFT is a strong and difficult-to-improve baseline; that Sequential Dialogue Prediction and three-frame context are promising for complete-triplet prediction under ground-truth grounding; and that additional context or longer generated supervision does not automatically improve performance. The chapter further demonstrates that stronger component metrics do not necessarily yield a commensurate increase in complete-triplet AP because the instrument, verb, and target must all be correct for the same instance.

5.2 Related Work

5.2.1 Grounded Surgical Triplet Prediction

Surgical action triplets represent an interaction as the $\langle$`instrument, verb, target`$\rangle$ tuple. TripNet introduced triplet recognition as a frame-level task, while Rendezvous subsequently modelled relationships between instruments, verbs, and targets through attention mechanisms [Nwoye 2020; Nwoye 2022b]. Later work investigated temporal modelling, self-distillation, and multi-task optimisation, including MT4MTL-KD and TERL [Gui 2023; Gui 2024]. These methods improved frame-level recognition but did not directly associate every interaction with a precise instrument instance.

Chapter 4 addresses this limitation by introducing CholecTriplet-Seg and triplet segmentation [Alabi 2026]. Its Mask2Former-Triplet baseline and TargetFusionNet jointly learn instrument localisation and semantic triplet labels, with TargetFusionNet incorporating weak anatomical priors to support target prediction. The present work retains the same grounded task and evaluation protocol but decomposes localisation and interaction prediction into specialist stages.

5.2.2 Multimodal Models in Surgical Vision

MLLMs have increasingly been studied for surgical visual question answering, dialogue, and scene interpretation. SurgicalGPT combines visual representations with an autoregressive language model for surgical visual question answering, while Surgical-LLaVA adapts instruction-tuned vision–language models to surgical images and videos [Seenivasan 2023a; Jin 2024]. Related triplet-recognition work has also used MLLM-generated semantic descriptions as auxiliary information within discriminative models [Zhang 2025]. In contrast, this chapter uses an MLLM itself as the structured interaction-prediction stage after specialist instrument grounding.

5.3 Two-Stage Multimodal Interaction Prediction

5.3.1 Problem Formulation

Let $I \in \mathbb{R}^{H\times W\times 3}$ denote a laparoscopic image. Stage 1 produces a set of grounded instrument instances

$$\mathcal{S} = \{s_i\}_{i=1}^{N}, \qquad s_i = (m_i, c_i, \mathrm{id}_i), \tag{5.1}$$

where s_i denotes the ith physical instrument instance in the image, m_i is its binary mask, c_i is its fixed instrument category, and id_i is the within-image textual identifier used to associate the model output with that instance. Stage 2 predicts

$$f_\theta(I, \mathcal{S}) \rightarrow \mathcal{Y}, \qquad \mathcal{Y} = \{(v_i, t_i)\}_{i=1}^{N}, \tag{5.2}$$

where v_i and t_i denote the verb and target. The final triplet for instance i is $\tau_i = (c_i, v_i, t_i)$. Stage 2 receives the instrument identity as a fixed input and cannot revise it.

The index i preserves the correspondence between each predicted interaction and the supplied grounded instance. Thus, although Stage 2 performs semantic interaction prediction, its output remains instance-specific: for example, it predicts the interaction associated with `grasper_2`, not merely that a grasper interaction occurs somewhere in the frame.

Two grounding settings are used. Ground-truth grounding supplies the annotated masks, categories, and instance count, isolating Stage 2 from localisation errors. Predicted grounding supplies the outputs of Stage 1 and therefore evaluates the complete system. False-positive Stage 1 instances are passed to Stage 2, whereas missed instruments are absent and cannot be recovered.

5.3.2 Stage 1: Fixed Instrument Grounding

Stage 1 uses the Mask2Former architecture, backbone, augmentation, optimisation, checkpoint-selection criterion, and inference configuration described for instrument instance segmentation in Chapter 3. For the present study, it is retrained on the CholecTriplet-Seg training split, including the corrected instrument masks available for this task-specific subset. The detector is fixed throughout all Chapter 5 experiments. Predicted masks are retained using the established Mask2Former confidence filtering and are passed to Stage 2 without refinement. Ground-truth and predicted masks use the same overlay-generation procedure; only their source differs.

5.3.3 Stage 2: Structured Interaction Prediction

For single-frame experiments, the Stage 2 input is one RGB image onto which every grounded instrument mask is rendered translucently. Each instance additionally receives a bounding box and an identifier comprising its instrument class and an image-level index, such as `hook_1` or `grasper_2`. Instrument classes have fixed colours; multiple instances of the same class share a colour and are distinguished by their identifiers. Indices follow the order in which instances are received from

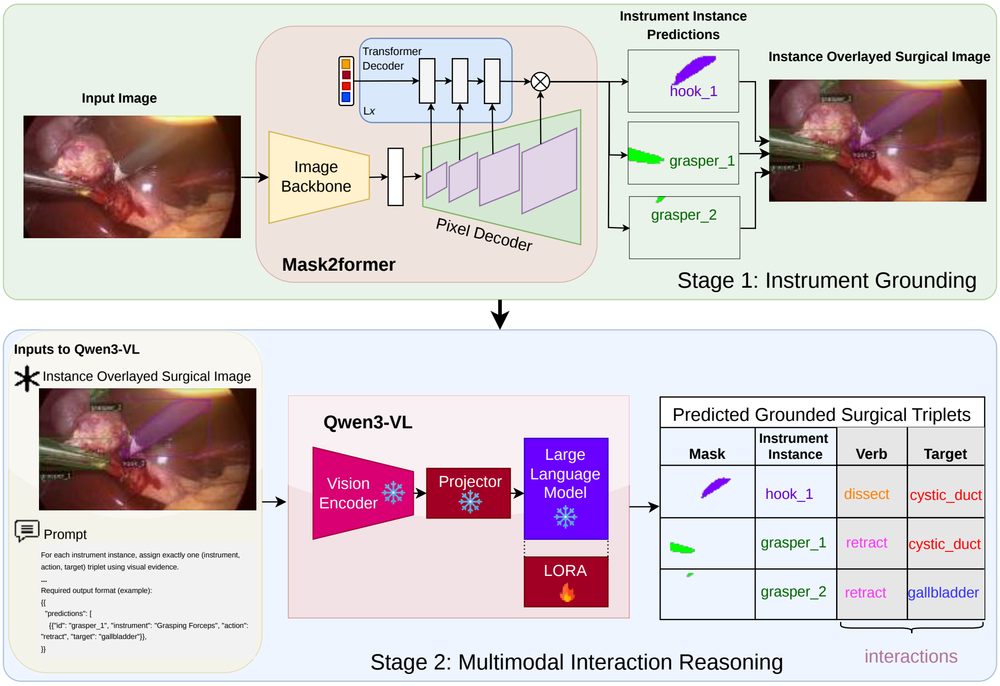


Figure 5.1: The two-stage framework. Stage 1 uses Mask2Former to localise and classify instrument instances. Stage 2 receives a single image containing translucent instance overlays, bounding boxes, and textual identifiers, and uses Qwen3-VL-8B-Instruct to predict one ⟨`verb, target`⟩ interaction per grounded instance. The complete triplets combine the fixed Stage 1 instrument classes with the Stage 2 predictions.

the annotation or detector and have no semantic meaning.

The textual instruction lists the allowable verb and target labels and specifies a structured response containing an identifier, verb, and target for every supplied instance. The label vocabulary includes the `null_verb` and `null_target` categories introduced in Chapter 4. All instances in an image are processed jointly in one autoregressive response. Generation is unconstrained: Direct SFT does not enforce valid triplets during decoding or apply ontology-based filtering. Malformed outputs, invalid labels, duplicate identifiers, and omitted instances are treated as missing predictions and receive no credit.

5.3.4 Direct Supervised Fine-Tuning

Direct SFT trains Qwen3-VL-8B-Instruct to generate the structured ⟨`verb, target`⟩ response directly from the grounded visual input and instruction. The base model is loaded with 4-bit NF4 quantisation and bfloat16 computation. QLoRA adapters are applied to the attention and MLP projection layers, with rank $r = 64$, scaling factor $\alpha = 128$, and dropout 0.05; the vision encoder, multimodal projector, and base language model remain frozen. Supervised loss is computed only over assistant-response tokens.

Figure 5.2 illustrates representative ambiguities observed with this direct formulation. These cases motivated the controlled adaptations considered next: finer anatomical evidence, short-range temporal context, knowledge of valid label relationships, and richer supervision.

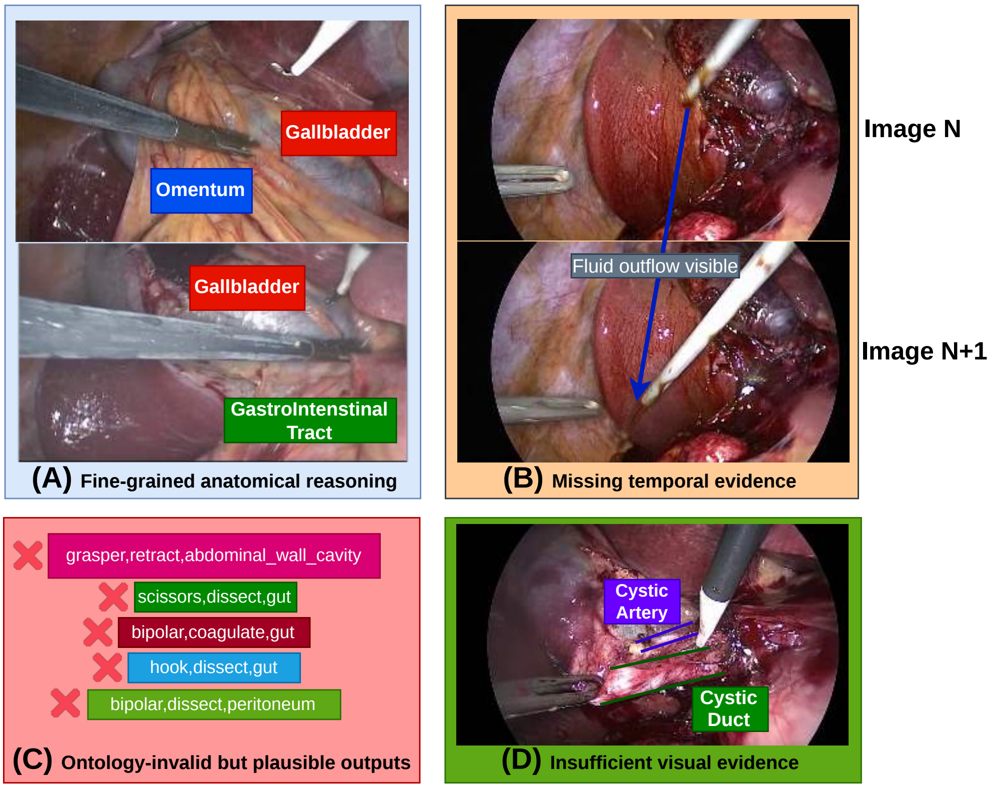


Figure 5.2: Representative challenges motivating the Stage 2 adaptations: fine-grained anatomical ambiguity, temporal evidence absent from an isolated frame, plausible outputs outside the benchmark ontology, and cases in which the available image evidence is insufficient to distinguish neighbouring structures.

5.3.5 Interaction-Prediction Adaptations

These methods change the Stage 2 input or supervision while retaining the principal Direct SFT optimisation configuration.

Sequential Dialogue Prediction

Sequential Dialogue Prediction factorises the structured response across three assistant turns, following the motivation of divide-and-discern dialogue tuning [Zhang 2026]. The first turn predicts one verb for each grounded instance. The second predicts the targets while retaining the verb response as dialogue context. A final turn then produces the complete ⟨`instrument, verb, target`⟩ response from the accumulated dialogue. During training, the preceding responses are teacher-forced and supervised loss is applied to all three assistant responses. During inference, each later turn receives the model outputs generated in the preceding turns, so early errors can propagate to the final prediction. The model is trained independently from the same public base model as Direct SFT, with otherwise identical QLoRA and optimisation settings. The complete dialogue templates and response schemas are provided in Section 5.9.

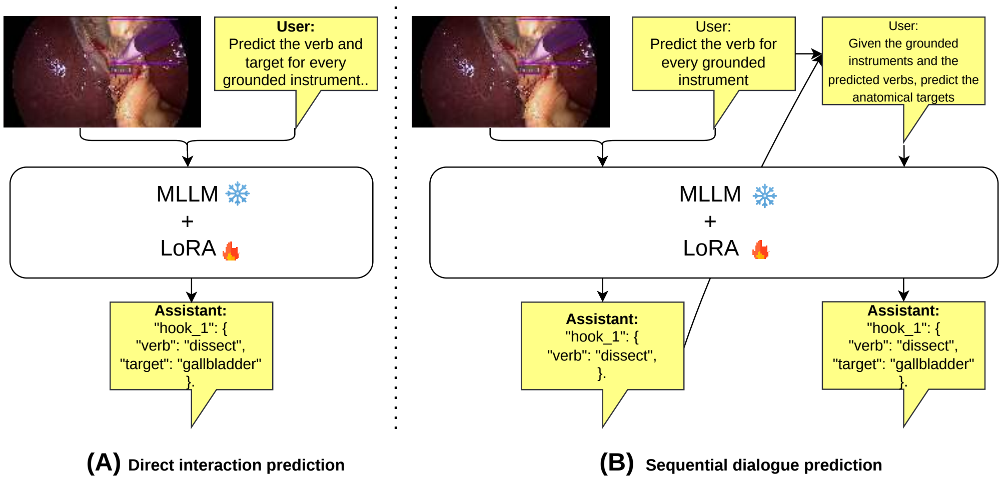


Figure 5.3: Direct interaction prediction and Sequential Dialogue Prediction. Direct SFT predicts verbs and targets jointly. The sequential formulation first predicts verbs and then predicts targets conditioned on the preceding dialogue response; a final dialogue turn serialises the complete triplet response.

Temporal Visual Context

The temporal variants retain centre-frame interaction prediction but replace the single image with an ordered native multimodal input. The three-frame model uses $(t-1, t, t+1)$ and the five-frame model uses $(t-2, t-1, t, t+1, t+2)$, where neighbouring inputs are immediately adjacent source-video frames. Every frame contains overlays derived from its own instrument masks; no temporal tracking or mask propagation is used. The prompt explicitly identifies the current centre frame as the prediction target and states that the neighbouring frames provide context only. Consequently, only the centre frame supplies triplet supervision. Windows remain within the same video and split, and at video boundaries only available neighbouring frames are used.

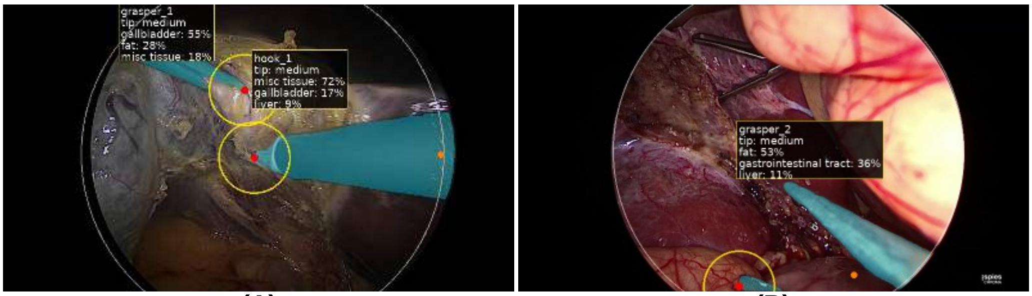
The total visual-token budget is held equal to the single-frame setting and is therefore shared across the temporal frames. The comparison between three and five frames is consequently exploratory rather than a controlled temporal-window ablation: the five-frame input contains more visual content within the same budget. Apart from the temporal input, training and validation follow the Direct SFT configuration. An earlier cached-feature implementation was technically inappropriate and is not included in the study.

Anatomical Context Priors

This adaptation supplies coarse anatomy around each instrument tip as text in addition to the standard visual input. It uses the same EndoViT checkpoint and label space described in Chapter 4. A heuristic first estimates the laparoscopic field boundary and identifies instrument-mask pixels close to either that boundary or an image edge. Their mean position defines the estimated instrument base, and the mask pixel furthest from this base defines the distal tip. A circular neighbourhood centred on the endpoint has radius one-eighth of the smaller image dimension. Instrument pixels are excluded from this neighbourhood. Each remaining pixel is assigned its highest-probability EndoViT class; classes are ranked by their proportion of the valid region, and the three most prevalent classes and proportions are appended to the prompt. Anatomical context is supplied during both training and inference. The model is trained independently from the same base model with the Direct SFT configuration. The complete heuristic and prior format are documented in Section 5.9.

Ontology-Guided Prompting

Ontology-Guided Prompting augments the instruction with the valid ⟨`verb, target`⟩ combinations for each supplied instrument class. These combinations are taken without modification from the 100-triplet ontology introduced in Chapter 4. Ontology information is included during both fine-tuning and evaluation, but decoding remains autoregressive and unconstrained. The experiment therefore evaluates prompt-level access to the ontology rather than constrained prediction.



(A) (B)

Figure 5.4: Representative examples of the heuristic instrument-tip estimation used to extract local anatomical context. The estimated instrument tip (red dot) defines the centre of a circular region (yellow) from which the local anatomical distribution is obtained using EndoViT predictions. The accompanying text reports the three most prevalent predicted anatomical classes and their pixel proportions within the selected region. (A) The heuristic estimates the tips of both visible instruments, producing anatomically relevant local context. (B) A limitation of the heuristic: the grasper contains two plausible distal extremities, making the true contact point ambiguous. The selected tip may consequently yield less representative anatomical context. Qualitative inspection nevertheless indicated that the heuristic generally produced reasonable tip estimates across the examined examples.

Distilled-Reasoning Supervision

Qwen3.7-Max, a closed-weight commercial teacher model, was prompted with the grounded image and ground-truth labels to generate a structured rationale for each instance. Each response describes contact evidence, action evidence, a primary target hypothesis, a contrastive target hypothesis, uncertainty, and the final triplet. Rationales were generated for all splits, but only training-split rationales were used as supervision. Approximately ten rationales were inspected informally; they appeared plausible, but neither systematic quality control nor clinical expert validation was performed.

The Distilled Rationale Only model is trained exclusively to generate the complete rationale followed by the final structured prediction. The multitask model uses equal numbers of Direct SFT and distilled-rationale examples. The same trained multitask model is evaluated in two modes: *Multitask—Direct SFT* requests the concise response, whereas *Multitask—Distilled Rationale* requests the rationale before the final prediction. Only the final structured triplet is scored. Rationale and answer tokens have equal loss weight, so the longer rationale accounts for most supervised response tokens. All student models otherwise follow the Direct SFT configuration.

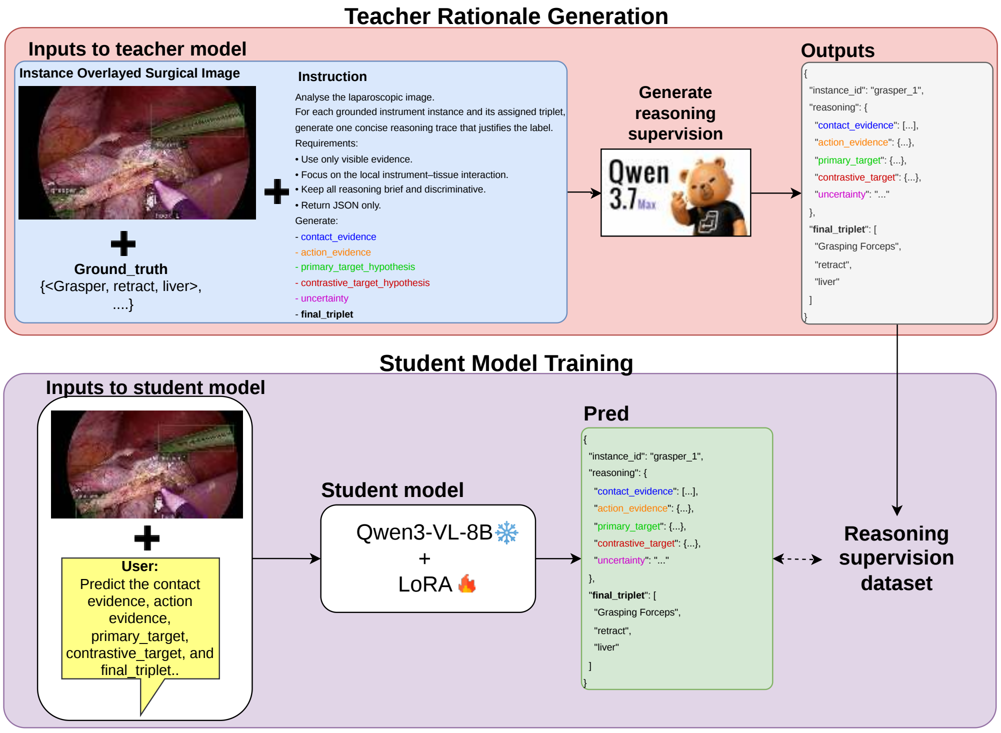


Figure 5.5: Distilled-reasoning supervision. Qwen3.7-Max generates structured rationales from grounded images and ground-truth triplets. A Qwen3-VL-8B-Instruct student is then trained to reproduce the rationale and final structured prediction.

5.3.6 Implementation and Training Configuration

All Stage 2 experiments use the 8B Instruct variant of Qwen3-VL and the same principal parameter-efficient training configuration unless an adaptation-specific change is listed below. Each adaptation is initialised independently from the public base model rather than from a converged Direct SFT adapter. Stage 1 training is inherited from Chapter 3 and is not repeated here.

No manually enforced text sequence-length limit is reported because the recovered launch configuration does not establish that such a limit affected the final experiments. The same fixed total visual-token budget is used throughout. A single-frame experiment allocates this budget to one image, whereas the three- and five-frame configurations share it across the images in their temporal window. Consequently, increasing the number of frames does not increase the overall visual allocation.

Table 5.1: Common Stage 2 QLoRA and optimisation configuration.

Setting	Configuration
Base model	Qwen3-VL-8B-Instruct
Parameter-efficient adaptation	QLoRA
Base-model quantisation	4-bit NF4 with double quantisation
Computation precision	bfloat16
Trainable modules	LoRA adapters in attention and MLP projection layers
Attention projections	Query, key, value, and output projections
MLP projections	Gate, up, and down projections
LoRA rank	64
LoRA scaling factor	128
LoRA dropout	0.05
Vision encoder	Frozen
Supervised objective	Cross-entropy over assistant-response tokens; instruction tokens masked
Optimiser	Fused AdamW
Initial learning rate	2×10^{-4}
Learning-rate schedule	3% warm-up followed by linear decay to zero
Weight decay	0.01
Maximum gradient norm	1.0
Per-device training batch size	1
Gradient-accumulation steps	32
Effective batch size	32
Maximum training duration	5 epochs
Random seed	42
Validation cadence	Every 5% of planned optimisation steps
Early stopping	Patience of 5 validation evaluations
Checkpoint retention	At most 10 checkpoints
Model selection	Lowest loss on the official validation split
Test-set use	Final evaluation only

Adaptation-specific changes

All configurations use the same official training, validation, and test partitions. Frames without visible grounded instruments are excluded from Stage 2 because there is no instance for which an interaction can be predicted. Checkpoint selection is based exclusively on the official validation split; the test split is not used for early stopping or model selection.

Table 5.2: Changes relative to the common Direct SFT configuration. Unlisted training and optimisation settings remain unchanged.

Configuration	Change relative to Direct SFT
Sequential Dialogue Prediction	Three supervised assistant responses predict the action, target, and final triplet in sequence; preceding reference responses are supplied through teacher forcing during training.
Three-frame temporal context	Ordered inputs $(t-1, t, t+1)$ share the common total visual-token budget; only frame t is supervised.
Five-frame temporal context	Ordered inputs $(t-2, t-1, t, t+1, t+2)$ share the same total visual-token budget; only frame t is supervised.
Anatomical Context Prior	The EndoViT-derived local anatomy block is appended to the Direct SFT prompt during training and evaluation.
Ontology-Guided Prompting	Valid action–target combinations are appended to the prompt during training and evaluation; decoding remains unconstrained.
Distilled Rationale Only	The assistant target contains the complete teacher-generated structured rationale followed by the final triplet.
Multitask	Equal numbers of Direct SFT and distilled-rationale examples are combined in one training set; the resulting model is evaluated using either response instruction.

5.4 Experimental Protocol

5.4.1 Dataset and Samples

All experiments use the official CholecTriplet-Seg training, validation, and test partitions introduced in Chapter 4: 22,818 training frames from 45 videos, 2,496 validation frames from one video, and 5,641 test frames from four videos. The chapter reuses the established annotations without further relabelling or cleaning. Frames containing no visible instrument instance are excluded because Stage 2 predicts interactions only for grounded instances. Each single-frame sample contains all grounded instances and associated triplets in one annotated frame. All adaptations use the same data partitions as Direct SFT.

5.4.2 Evaluation

Performance is measured using the mask-based IVTMetrics segmentation protocol defined in Chapter 4. The reported measures are instrument AP (AP_I), verb

AP (AP_V), target AP (AP_T), instrument–verb AP (AP_{IV}), instrument–target AP (AP_{IT}), and complete-triplet AP (AP_{IVT}). The same matching, overlap, assignment, and class-averaging rules are used here.

Ground-truth grounding fixes the instrument masks, classes, and instance count; AP_I is therefore 100 by construction, while the remaining metrics are computed from Stage 2 outputs. Predicted grounding uses the fixed Stage 1 detector and includes both grounding and interaction errors. Only Direct SFT is evaluated with predicted instances; all adaptations are evaluated under ground-truth grounding. Small differences between configurations are therefore interpreted as observations rather than definitive improvements.

5.5 Results

5.5.1 Interaction Prediction with Ground-Truth Grounding

Table 5.3 reports the principal comparison under ground-truth grounding. Direct SFT reaches $AP_{IVT} = 14.56$, establishing that the MLLM can perform structured interaction prediction when instrument instances are supplied. Three-frame temporal context gives the highest observed AP_{IVT} at 15.89, an absolute increase of 1.33 points, and also gives the highest AP_V and AP_{IV}. Sequential Dialogue Prediction reaches $AP_{IVT} = 15.31$, 0.75 points above Direct SFT, despite slightly lower component metrics. These differences are promising observations, but the available evidence does not establish definitive superiority.

Table 5.3: Principal Stage 2 results under ground-truth instrument grounding. $AP_I = 100$ by construction and is omitted. Bold values indicate the highest observed result in each column.

Method	AP_V	AP_T	AP_{IV}	AP_{IT}	AP_{IVT}
Direct SFT	55.99	**26.35**	31.03	21.87	14.56
Sequential Dialogue Prediction	55.92	26.21	30.48	21.61	15.31
Anatomical Context Priors	53.35	26.27	31.48	**23.15**	14.96
Temporal Visual Context, 3 frames	**56.31**	26.32	**32.37**	21.66	**15.89**

The anatomical-prior model produces a mixed result. It attains the highest AP_{IT} in the principal table and $AP_{IVT} = 14.96$, but AP_T remains effectively unchanged from Direct SFT and AP_V declines. The experiment therefore does not show that this prompt-level anatomical representation improves target classification. It instead suggests that coarse context and heuristic tip extraction can alter instance–target associations without providing a consistent overall benefit.

5.5.2 Exploratory and Distilled-Supervision Results

Table 5.4 collects adaptations that yielded negative or mixed results. Five-frame context falls to $\mathrm{AP}_{IVT} = 11.61$. Because the same total visual budget is shared across more frames, this result shows that adding frames under this formulation does not automatically help; it does not isolate temporal-window length as a causal factor. Ontology-Guided Prompting reaches $\mathrm{AP}_{IVT} = 14.44$, providing no evidence that listing valid combinations in the prompt improves the principal outcome.

Table 5.4: Exploratory adaptations under ground-truth grounding. The Direct SFT row is repeated as a reference. The two multitask rows evaluate the same trained model with different inference instructions.

Method	AP_V	AP_T	AP_{IV}	AP_{IT}	AP_{IVT}
Direct SFT	55.99	26.35	31.03	21.87	14.56
Temporal Visual Context, 5 frames	53.52	22.10	27.72	17.39	11.61
Ontology-Guided Prompting	55.81	24.48	33.07	20.68	14.44
Distilled Rationale Only	53.22	20.88	27.30	13.59	8.69
Multitask—Direct SFT	52.86	25.68	31.70	21.60	14.57
Multitask—Distilled Rationale	54.90	25.18	33.20	22.49	14.32

Distilled Rationale Only supervision performs poorly, particularly on target-related measures. The multitask model recovers most Direct SFT performance, but neither evaluation mode improves AP_{IVT} meaningfully: Direct SFT evaluation gives 14.57 and distilled-rationale evaluation gives 14.32. Thus, the experiments do not establish that the generated rationales are beneficial. One plausible explanation is that equal token weighting causes the long rationale to dominate the objective while the short final answer determines evaluation; this remains a hypothesis because answer-token reweighting was not tested.

5.5.3 Complete Two-Stage System

Table 5.5 evaluates Direct SFT with Stage 1 predictions and compares it with the Chapter 4 systems under the same dataset and IVTMetrics mask-segmentation protocol. The published comparison values are reproduced from that chapter. The two-stage system obtains $\mathrm{AP}_I = 80.64$, which is fixed by Stage 1 and is not attributable to the MLLM. It also produces substantially higher AP_V, AP_T, AP_{IV}, and AP_{IT} than TargetFusionNet. Complete-triplet performance is essentially comparable: $\mathrm{AP}_{IVT} = 13.52$ for the two-stage system and 13.47 for TargetFusionNet. The 0.05-point difference is too small to support an improvement claim.

The reduction from $\mathrm{AP}_{IVT} = 14.56$ under ground-truth grounding to 13.52 in

Table 5.5: Mask-based triplet segmentation using predicted instrument instances. Comparison results are reproduced from Chapter 4. Bold values indicate the highest observed score in each column.

Method	AP_I	AP_V	AP_T	AP_{IV}	AP_{IT}	AP_{IVT}
RDV-Det [Nwoye 2023b]	0.09	0.11	0.08	0.08	0.04	0.03
Mask2Former-Triplet [Alabi 2026]	65.24	45.61	20.75	23.03	16.47	12.23
TargetFusionNet [Alabi 2026]	67.19	46.27	21.55	24.93	17.75	13.47
Two-stage Direct SFT	**80.64**	**51.89**	**25.27**	**29.01**	**18.73**	**13.52**

the complete system reflects compounded errors. False-negative Stage 1 instances never reach Stage 2, false positives generate additional interaction predictions, and incorrect instrument identities cannot be corrected by the MLLM. Moreover, improvements in individual components need not translate proportionally to AP_{IVT} because all three labels and the spatial association must be correct simultaneously.

5.5.4 Error Analysis

Figures 5.6–5.8 present row-normalised action and target confusion matrices for the official test-set predictions of Direct SFT, three-frame temporal context, and the multitask model evaluated using the Direct SFT response instruction. Each cell reports the percentage of examples belonging to the ground-truth class in that row that were assigned to the predicted class in that column. The matrices are descriptive diagnostics from the same runs reported in the main chapter and complement, rather than replace, the official mask-based AP evaluation.

The action matrices expose several recurrent ambiguities. In Direct SFT, irrigation is predominantly predicted as aspiration: 90.4% of ground-truth irrigation examples are assigned to aspiration, leaving no correct irrigation predictions. Three-frame temporal context reduces this one-directional collapse and restores a 55.8% diagonal response for irrigation. However, 25.0% of irrigation examples remain assigned to aspiration, while 36.4% of aspiration examples are assigned to irrigation. Temporal information therefore helps distinguish these classes but does not resolve their ambiguity. Coagulation is also sometimes predicted as dissection, while grasp, pack, and the no-action class show substantial cross-class errors that vary between configurations.

Target prediction remains more diffuse than action prediction. Frequent structures such as the gallbladder and liver form prominent prediction destinations, whereas several less common or visually similar targets have weak diagonals. Confusion between the cystic artery and cystic duct is present across the configurations, and gastrointestinal tract, peritoneum, cystic plate, and no-target examples are frequently assigned to other abdominal structures. These observations are consistent with visual similarity and class imbalance as plausible contributors, but the matrices alone cannot establish their causes. In particular, they do not quantify annotation ambiguity or demonstrate that any apparent label inconsistency is systematic.

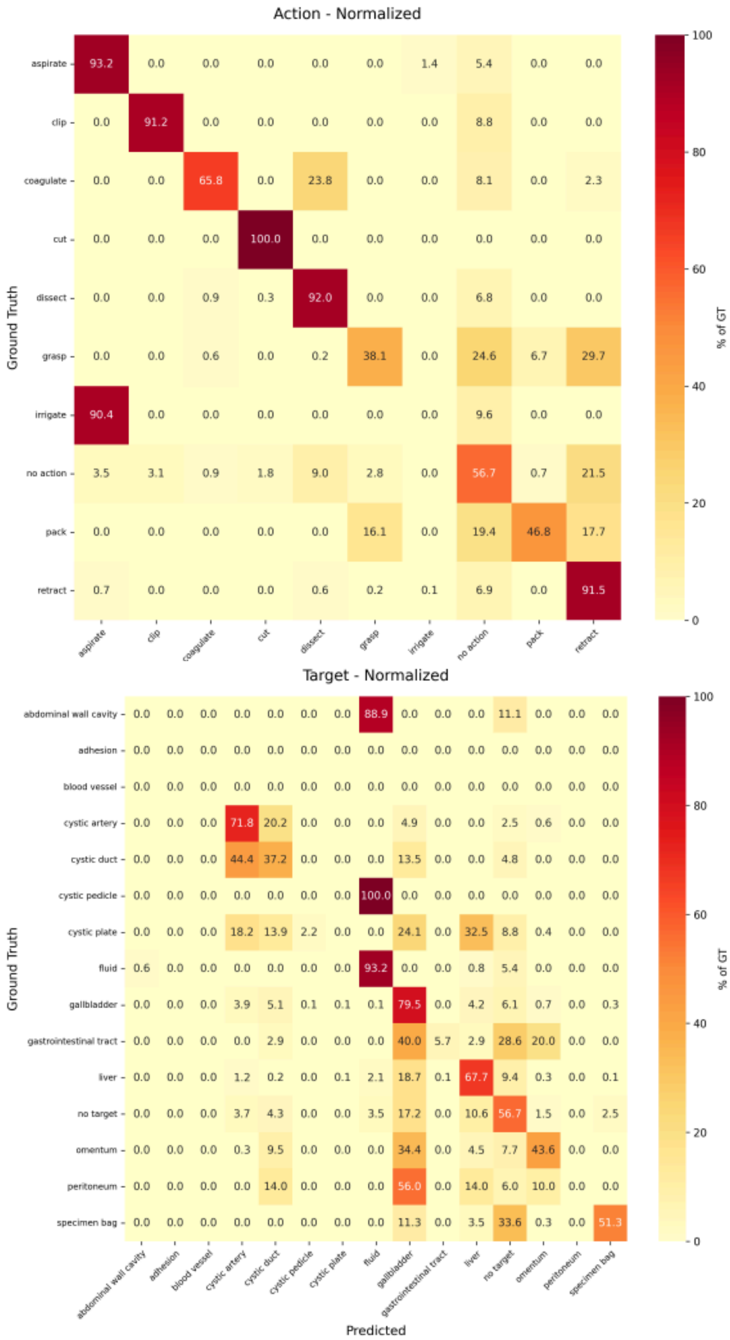


Figure 5.6: Row-normalised confusion matrices for Direct SFT under ground-truth instrument grounding. The upper matrix reports actions and the lower matrix reports targets. Rows denote ground-truth classes and columns denote predicted classes.

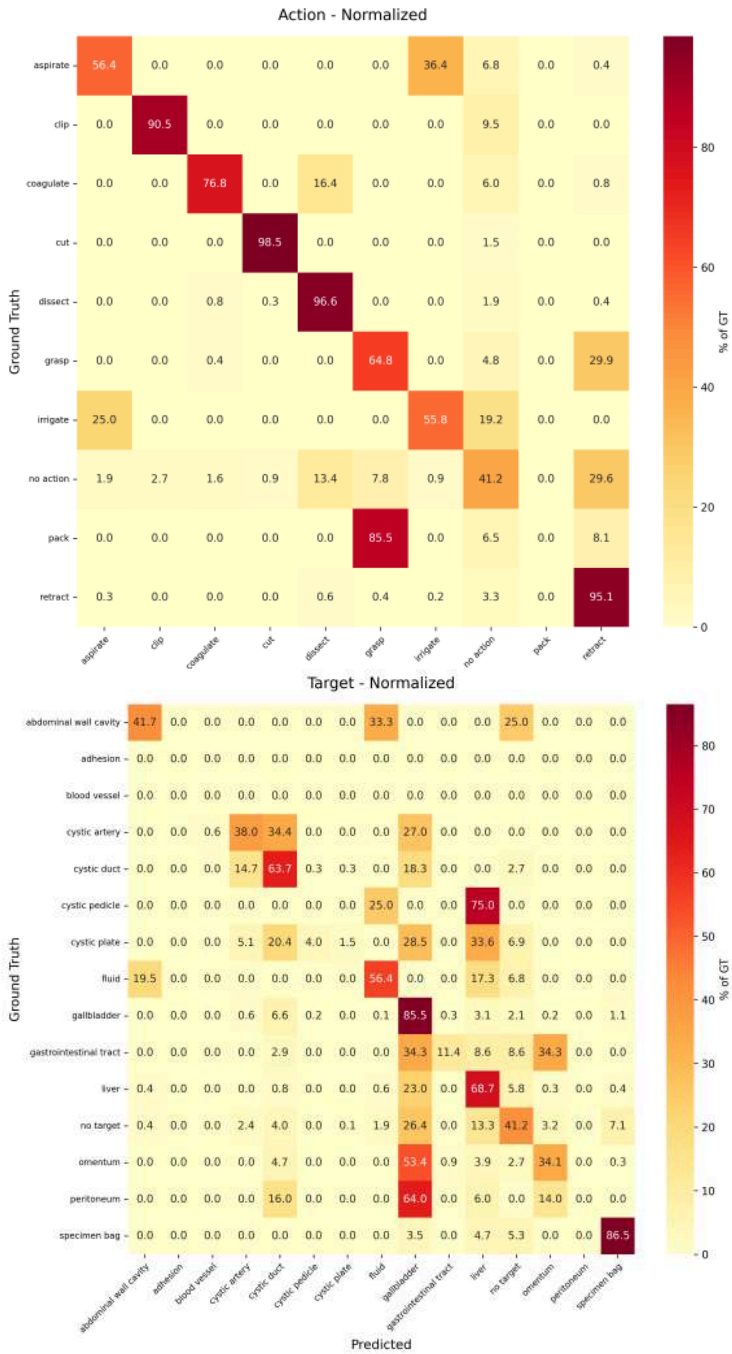


Figure 5.7: Row-normalised confusion matrices for three-frame temporal context under ground-truth instrument grounding. The upper matrix reports actions and the lower matrix reports targets. Rows denote ground-truth classes and columns denote predicted classes.

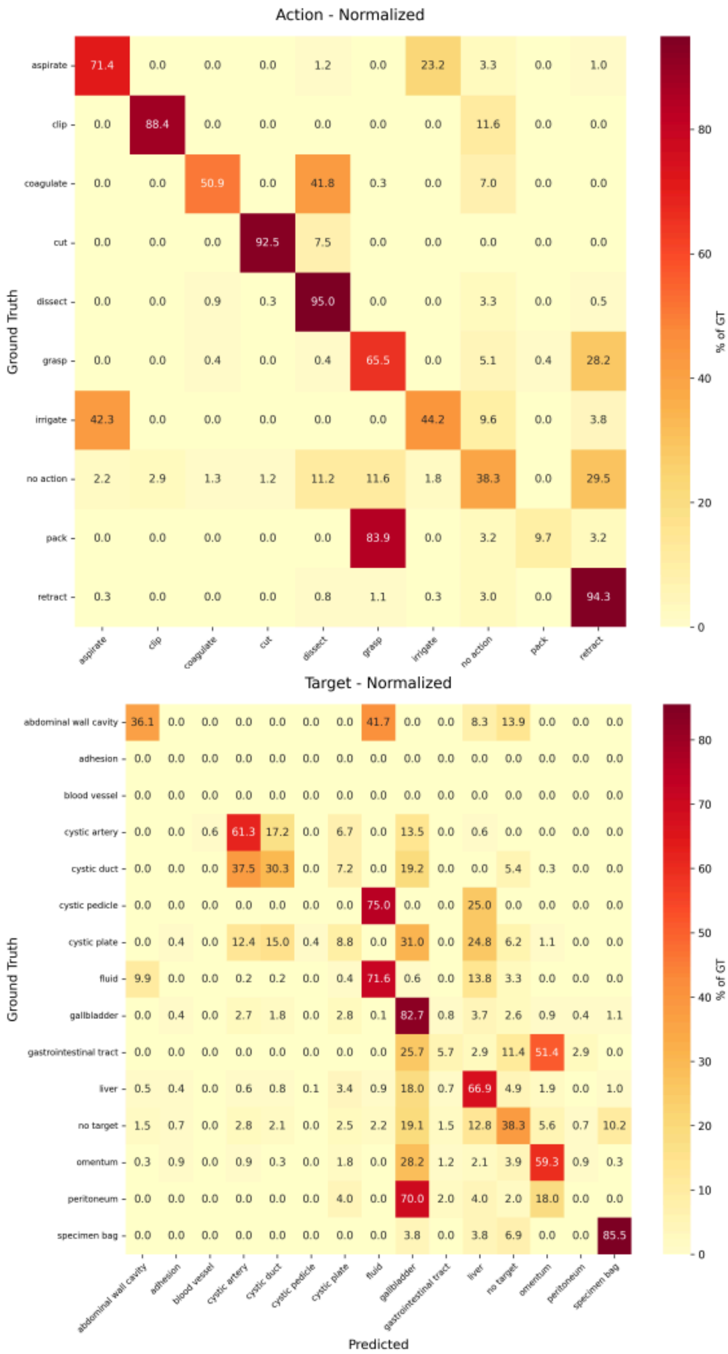


Figure 5.8: Row-normalised confusion matrices for the multitask model evaluated with the Direct SFT response instruction under ground-truth instrument grounding. The upper matrix reports actions and the lower matrix reports targets. Rows denote ground-truth classes and columns denote predicted classes.

5.6 Discussion

5.6.1 Viability of the Two-Stage Formulation

The principal finding is that an MLLM can act as a viable second-stage interaction predictor when instrument instances are supplied. Direct SFT requires no explicit rationale, anatomical prompt, or ontology constraint, yet provides a strong reference point under ground-truth grounding and remains competitive in the complete predicted-grounding pipeline. The decomposition focuses the MLLM on semantic interaction prediction while leaving dense localisation to a specialist model. This does not prove that two-stage modelling is universally preferable to end-to-end learning, but it establishes the formulation as credible and exposes the contribution and failure modes of each stage more clearly.

The complete-system results also clarify the relationship to TargetFusionNet. Component-level scores are substantially stronger for the two-stage Direct SFT pipeline, but AP_{IVT} is essentially unchanged. Complete-triplet evaluation is conjunctive: better verb or target recognition contributes only when localisation, instrument identity, and the remaining semantic component are also correct for the same instance. Consequently, component gains should not be presented as equivalent to complete interaction gains.

5.6.2 Task Factorisation and Temporal Evidence

Sequential Dialogue Prediction and three-frame context produce the strongest observed AP_{IVT} results among the principal Stage 2 experiments. Their effects are nevertheless metric-specific. Sequential prediction increases AP_{IVT} while slightly decreasing every reported component or pair metric relative to Direct SFT. It may encourage more coherent complete predictions, but the present experiment does not quantify error propagation across dialogue turns, and the difference between teacher-forced training responses and generated inference responses remains important.

Three-frame input gives a larger AP_{IVT} increase and modest improvements in AP_V and AP_{IV}. This is consistent with the premise that neighbouring frames can reveal motion or interaction progression absent from a still image. However, the experiment uses ground-truth grounding and only one seed. The five-frame result further shows that additional visual context is not automatically useful when a fixed token budget is divided among more frames. It should not be interpreted as a controlled demonstration that three frames are intrinsically optimal.

5.6.3 Why Additional Priors and Longer Supervision Did Not Consistently Help

The anatomical and ontology experiments provide useful negative evidence about prompt-level integration. EndoViT context is conceptually connected to the weak anatomical priors investigated in Chapter 4, but the present conversion of dense predictions into three textual class proportions does not improve AP_T. This result does not show that anatomy is unhelpful; it shows that this heuristic tip region and prompt representation do not reliably improve the final task.

Similarly, access to valid ⟨`verb, target`⟩ combinations does not improve AP_{IVT}. Because generation remains unconstrained, the model may not consistently use the supplied ontology. The experiment therefore evaluates contextual guidance, not the potential value of constrained decoding.

Distilled-reasoning supervision also fails to outperform Direct SFT. The rationales were generated at scale but were not clinically validated, and only a small informal sample was inspected. Equal weighting also directs most response-token loss towards reproducing the rationale rather than the evaluated triplet. These design properties provide plausible explanations for the negative result, but their individual effects were not tested and should not be treated as established causes.

Across these experiments, anatomical targets remain the principal semantic challenge. Dense target annotation could provide more direct spatial supervision than the weak local priors evaluated here, but delineating interacting anatomy requires substantial expert effort and is often ambiguous at instrument boundaries. Moreover, the annotation burden grows as additional procedures introduce new structures and interaction ontologies. Dense target labels are therefore valuable where they can be obtained, but are unlikely to provide the only scalable route to grounded interaction prediction.

5.7 Limitations

The study has four principal limitations. First, only Direct SFT is evaluated with predicted grounding. The sequential, temporal, anatomical, ontology, and distilled-supervision variants are assessed only with ground-truth instances, so their behaviour in the complete system is unknown.

Second, adaptations are evaluated independently. No controlled combinations test whether sequential factorisation and temporal context are complementary, and the reported differences have not been subjected to formal uncertainty analysis.

Third, malformed generations and verb-to-target error propagation were not quantified systematically. Invalid outputs receive no credit, but their frequency may differ between formulations and remains an unmeasured source of variability. Finally, all experiments use CholecTriplet-Seg and laparoscopic cholecystectomy. Generalisation to other procedures, acquisition conditions, and interaction ontologies has not been demonstrated.

5.8 Future Research

The immediate priority is to evaluate the strongest Stage 2 variants with predicted Stage 1 grounding, thereby testing whether their observed advantages persist in the complete pipeline. Temporal predicted grounding may require instance association or mask propagation because independently detected instances are not guaranteed to correspond across frames. More explicit temporal integration may also use motion and procedural continuity more effectively than distributing a fixed visual budget across independent images. Subsequent comparative studies should include repeated runs and controlled combinations of the strongest adaptations.

Sequential prediction should be tested with imperfect verb conditioning during training so that robustness to first-turn errors can be measured directly. Controlled combinations of the strongest adaptations would determine whether task factorisation and temporal evidence are complementary. Curriculum-style training could also initialise specialised variants from a converged Direct SFT adapter rather than training each independently from the base model.

For distilled supervision, future work should prioritise rationale quality control and objectives that increase the relative weight of final answer tokens. Comparison-based or contrastive auxiliary tasks may offer a more directly verifiable form of supervision than long free-text rationales. Finally, evaluation on additional procedures and datasets is required to establish whether two-stage interaction prediction transfers beyond cholecystectomy. ProstaTD, a fully supervised surgical triplet-detection dataset for robot-assisted prostatectomy, provides a relevant external benchmark for investigating this cross-procedure generalisation [Chen 2026].

5.9 Additional Methodological and Experimental Details

5.9.1 Prompt and Response Specifications

All Stage 2 configurations combine one or more grounded visual inputs with textual instructions and autoregressively generate JSON-formatted predictions. The templates below reproduce the scientifically relevant prompt content while replacing sample-specific image names and instance lists with braces. Whitespace and Python-specific string escaping are omitted. The requested schemas are formatting instructions rather than decoding constraints: generation is not programmatically restricted to valid JSON or valid triplets.

Shared label vocabulary and serialisation

The prompts use the six instrument labels, ten action strings, and fifteen target strings inherited from CholecTriplet-Seg:

```
Instruments = ["Grasping Forceps", "Bipolar Forceps", "Monopolar Hook",
               "Laparoscopic Scissors", "Clip Applier", "Suction-Irrigator"]
Actions = ["grasp", "retract", "dissect", "coagulate", "clip", "cut",
           "aspirate", "irrigate", "pack", "no action"]
Targets = ["gallbladder", "cystic plate", "cystic duct", "cystic artery",
           "cystic pedicle", "blood vessel", "fluid",
           "abdominal wall cavity", "liver", "adhesion", "omentum",
           "peritoneum", "gastrointestinal tract", "specimen bag",
           "no target"]
```

The literal strings `no action` and `no target` are the serialised forms of the `null_verb` and `null_target` categories used elsewhere in the thesis. Similarly, labels are rendered as natural-language strings with spaces, such as `cystic duct`, rather than internal identifiers such as `cystic_duct`. Although the generated object repeats the instrument class, Stage 2 does not revise the supplied grounding identity; this field serialises the complete triplet associated with the instance identifier.

Direct supervised fine-tuning

The Direct SFT system instruction defines the response object and the handling of inactive instruments:

```
Return ONLY valid JSON. Output an object with a single key "predictions",
containing a list of items. Each item must have keys: id, instrument,
action, target. Use only the allowed labels. If idle: action="no action",
target="no target". No text outside JSON.
```

The user instruction supplies the grounded image, the shared label vocabulary, and the identifiers that must be labelled:

```
You are analysing a laparoscopic cholecystectomy frame.
The image contains multiple surgical instrument instances predicted by a
neural network. Each instance is marked in the image with a name and ID
(e.g. grasper_1, hook_2).
```

```
For EACH instance ID, assign exactly one (instrument, action, target)
triplet using visual evidence.

Allowed labels:
{shared instrument, action, and target vocabularies}

Required output format (example):
{
  "predictions": [
    {"id": "grasper_1", "instrument": "Grasping Forceps",
     "action": "retract", "target": "gallbladder"},
    {"id": "hook_2", "instrument": "Monopolar Hook",
     "action": "dissect", "target": "cystic plate"}
  ]
}

Instrument Instances to label (from the overlay): {instrument_list}
Now produce the JSON for this image.
```

Every training response follows the same schema, with one object per grounded instance:

```
{
  "predictions": [
    {
      "id": "grasper_1",
      "instrument": "Grasping Forceps",
      "action": "retract",
      "target": "gallbladder"
    }
  ]
}
```

Sequential Dialogue Prediction

Sequential Dialogue Prediction uses the same grounded image throughout a three-stage dialogue. An earlier instrument-identification turn was not used in the reported experiment. The first retained turn requests the action for each instance:

```
System: You are a surgical scene understanding assistant.
        Return valid JSON only.

User: For each labelled instrument instance in this image, identify the
      action being performed. Return JSON only.

Assistant response:
{"predictions": [{"id": "grasper_1", "action": "retract"}]}
```

The target turn is conditioned on the preceding dialogue response:

```
User: For each labelled instrument instance, identify the target being
      acted upon. Return JSON only.

Assistant response:
{"predictions": [{"id": "grasper_1", "target": "gallbladder"}]}
```

The final turn requests the complete serialised prediction:

```
User: For each labelled instrument instance, predict the final
      instrument-action-target triplet. Return JSON only.

Assistant response:
{
  "predictions": [
    {"id": "grasper_1", "instrument": "Grasping Forceps",
     "action": "retract", "target": "gallbladder"}
  ]
}
```

During training, the complete three-stage dialogue is supplied with the reference assistant response at each stage. During inference, the generated action and target responses remain in the conversation before the final request. Only the final triplet response is evaluated.

Ontology-Guided Prompting

Ontology-Guided Prompting retains the Direct SFT user instruction and response schema but extends the system instruction with the valid action–target combinations associated with each instrument class. The complete ontology supplied to the model is:

```
Valid Combinations (Instrument: {Action: [Targets]}):

Grasping Forceps:
  dissect: [cystic plate, gallbladder, omentum]
  grasp: [cystic artery, cystic duct, cystic pedicle, cystic plate,
          gallbladder, gastrointestinal tract, liver, omentum,
          peritoneum, specimen bag]
  pack: [gallbladder]
  retract: [cystic duct, cystic pedicle, cystic plate, gallbladder,
            gastrointestinal tract, liver, omentum, peritoneum]
  no action: [no target]

Bipolar Forceps:
  coagulate: [abdominal wall cavity, blood vessel, cystic artery,
              cystic duct, cystic pedicle, cystic plate, gallbladder,
              liver, omentum, peritoneum]
  dissect: [adhesion, cystic artery, cystic duct, cystic plate,
            gallbladder, omentum]
```

```
  grasp: [cystic plate, liver, specimen bag]
  retract: [cystic duct, cystic pedicle, gallbladder, liver, omentum]
  no action: [no target]

Monopolar Hook:
  coagulate: [blood vessel, cystic artery, cystic duct, cystic pedicle,
              cystic plate, gallbladder, liver, omentum]
  cut: [blood vessel, peritoneum]
  dissect: [blood vessel, cystic artery, cystic duct, cystic plate,
            gallbladder, omentum, peritoneum]
  retract: [gallbladder, liver]
  no action: [no target]

Laparoscopic Scissors:
  coagulate: [omentum]
  cut: [adhesion, blood vessel, cystic artery, cystic duct, cystic plate,
        liver, omentum, peritoneum]
  dissect: [cystic plate, gallbladder, omentum]
  no action: [no target]

Clip Applier:
  clip: [blood vessel, cystic artery, cystic duct, cystic pedicle,
         cystic plate]
  no action: [no target]

Suction-Irrigator:
  aspirate: [fluid]
  dissect: [cystic duct, cystic pedicle, cystic plate, gallbladder,
            omentum]
  irrigate: [abdominal wall cavity, cystic pedicle, liver]
  retract: [gallbladder, liver, omentum]
  no action: [no target]

Each predicted (instrument, action, target) MUST exactly match one of the
allowed triplets listed above. Do NOT output any combination outside this
list.
```

This instruction exposes ontology information but does not enforce it through constrained decoding or post-processing. Invalid generations can therefore still occur and receive no credit under the common evaluation procedure.

Distilled Reasoning Supervision

The final rationale-generation template was designed to produce concise evidence for a supplied training triplet. Qwen3.7-Max received the grounded image, its instance annotations, and the reference triplets. Its system instruction was:

```
You are an expert surgical reasoning assistant specialised in laparoscopic
cholecystectomy analysis. Generate concise reasoning for provided surgical
action triplets. Focus on local instrument-target interaction and anatomical
discrimination.
```

```
Use the dataset target ontology:
[gallbladder, cystic plate, cystic duct, cystic artery, cystic pedicle,
 blood vessel, fluid, abdominal wall cavity, liver, adhesion, omentum,
 peritoneum, gastrointestinal tract, specimen bag, no target]

Rules:
- Use only visible evidence.
- Do not invent anatomy not visible.
- Ground the reasoning in visible instrument-target interaction.
- If no dataset target is visibly engaged, use "no target" and therefore
  "no action". This includes idle/floating instruments.
- Mention non-target objects such as needles or gauze when relevant.
- Keep reasoning compact and discriminative.
- Return only valid JSON.
```

The accompanying user instruction imposed explicit brevity constraints and supplied the reference triplets to be justified:

```
Analyse the provided laparoscopic cholecystectomy frame.

Image name: {image_name}
The image contains these labelled instrument instances and their provided
triplets:
{instances_json}

For EACH instrument instance, generate ONE concise structured reasoning
trace. Use the provided triplet only as the label being justified.

Brevity rules:
- contact_evidence: exactly 2 observations, each <= 15 words
- action support: 1 sentence, <= 15 words
- primary target rationale: 1 sentence, <= 15 words
- contrastive rationale: 1 sentence, <= 12 words
- contrastive rejection: 1 sentence, <= 15 words
- uncertainty: 1 sentence, <= 10 words

Use only visible evidence; do not invent anatomy or rely on unsupported
procedural assumptions. Do not add confidence scores or text outside JSON.
```

The generated supervision follows this schema:

```
{
  "image": "{image_name}",
  "instances": [
    {
      "instance_id": "grasper_1",
      "instrument": "Grasping Forceps",
      "reasoning": {
        "contact_evidence": ["...", "..."],
        "action_evidence": {
          "primary_action": "retract",
```

```
        "support": "..."
      },
      "primary_target_hypothesis": {
        "target": "gallbladder",
        "why_plausible": "..."
      },
      "contrastive_target_hypothesis": {
        "target": "liver",
        "why_plausible": "...",
        "why_rejected": "..."
      },
      "uncertainty": "..."
    },
    "final_triplet": {
      "instrument": "Grasping Forceps",
      "action": "retract",
      "target": "gallbladder"
    }
  }
 ]
}
```


For student supervision, the generated rationale and final triplet are used as the assistant response. The student receives the grounded image and instance identifiers but not the reference triplet supplied to the teacher. In Distilled Rationale Only evaluation, the student is instructed to generate the structured rationale followed by `final_triplet`; only the latter is extracted for metric computation. The term *distilled rationale* describes the source and form of supervision and does not imply that the teacher's explanations were clinically validated.

Multitask prompts

The multitask training set contains equal numbers of two supervision types. Direct SFT examples use the Direct SFT prompt and concise `predictions` response schema above. Distilled-rationale examples use the grounded student instruction and the structured rationale response ending in `final_triplet`. No additional multitask-specific prompt is introduced.

At evaluation, *Multitask—Direct SFT* uses the Direct SFT prompt and requests the concise `predictions` object. *Multitask—Distilled Rationale* requests the structured rationale and final triplet. These are two inference modes of the same multitask-trained model, not separately trained models. In both cases, evaluation uses only the final ⟨`instrument, verb, target`⟩ prediction.

5.9.2 Instrument-Tip Anatomical Context Heuristic

The Anatomical Context Prior experiment converts dense EndoViT predictions into a compact textual summary local to each grounded instrument. It uses the EndoViT checkpoint and anatomical label space described in Chapter 4; the procedure below concerns only localisation of the context region and construction of the text supplied to Stage 2.

Laparoscopic field and instrument-base estimation

For an RGB image, pixels whose maximum channel intensity exceeds 12 are treated as belonging to the visible laparoscopic field. If these pixels occupy less than 10% of the image, the field estimate is rejected. Otherwise, the field centre is defined by the median horizontal and vertical coordinates of the valid pixels. The field radius is the 98th percentile of their Euclidean distances from this centre. Estimates containing fewer than 100 valid pixels or having radius below 20 pixels are rejected.

All polygons associated with a grounded instrument instance are rasterised into a single binary mask. Mask pixels are designated as possible shaft-entry pixels when they lie within 20 pixels of an image boundary or within 20 pixels of the estimated laparoscopic-field boundary. The mean coordinate of these candidate pixels defines the estimated instrument base,

$$\hat{\mathbf{b}}_i = \frac{1}{|\mathcal{B}_i|} \sum_{\mathbf{x} \in \mathcal{B}_i} \mathbf{x}, \tag{5.3}$$

where $\mathcal{B}_i$ denotes the selected boundary pixels for instance i. The estimated distal tip is the mask pixel furthest from this base:

$$\hat{\mathbf{p}}_i = \underset{\mathbf{x} \in m_i}{\operatorname{argmax}} \left\| \mathbf{x} - \hat{\mathbf{b}}_i \right\|_2^2. \tag{5.4}$$

No separate shaft-orientation model is used. If no boundary candidate is available, tip estimation fails and no anatomical text prior is supplied for that instance. This deliberately conservative behaviour avoids substituting an arbitrary mask centroid for an uncertain entry point.

Local anatomical distribution

A disk centred on $\hat{\mathbf{p}}_i$ defines the local context region. Its radius is

$$r = \operatorname{round}\left(\frac{1}{8} \min(H, W)\right), \tag{5.5}$$

where H and W are the image dimensions. Approximately four candidate radii were inspected qualitatively before selecting this value; this was a design choice rather than a quantitative radius ablation. Pixels belonging to the instrument mask are removed from the disk so that the distribution represents surrounding tissue rather than the instrument itself.

EndoViT logits are converted to per-pixel class probabilities, after which every pixel is assigned its argmax class. Three configured non-context channels (indices 0, 7, and 8) are excluded. For each remaining anatomical class c, the reported value is the proportion of valid region pixels assigned to that class,

$$q_{i,c} = \frac{\sum_{\mathbf{x}\in\mathcal{R}_i} \mathbb{1}[\hat{c}(\mathbf{x}) = c]}{|\mathcal{R}_i|}, \tag{5.6}$$

where $\mathcal{R}_i$ is the valid disk region after removal of instrument pixels and excluded channels. Thus, the values are argmax-label proportions, not averages of the raw class probabilities. The three classes with the largest proportions are inserted into the prompt. If tip estimation, EndoViT loading, or valid-region construction fails, the prior is omitted rather than replaced with a fabricated value.

Textual prior supplied to Stage 2

The anatomy block is appended to the standard Direct SFT user instruction during both training and evaluation. A representative instance is formatted as follows:

```
Weak local anatomy prior from a separate segmentation model, computed near
the estimated instrument tip:

Instance grasper_1:
1. liver: 0.53
2. abdominal wall: 0.47
3. gastrointestinal tract: 0.00

The tip location is estimated automatically and may be imperfect. The
anatomy prior is weak and may be wrong. Pay close attention to the image
first, and use this prior only as contextual supporting evidence. The final
target must still be chosen from the surgical triplet target ontology.
```

The EndoViT names describe contextual anatomy and are not required to match the triplet target strings exactly; for example, `abdominal wall` is contextual evidence whereas `abdominal wall cavity` is a triplet target label. The MLLM remains responsible for selecting its final target from the benchmark vocabulary. The prior does not modify the visual input, constrain decoding, or replace the image evidence. Figure 5.4 illustrates both a successful estimate and a failure mode caused by ambiguous distal mask geometry.

5.9.3 Distilled Rationale Example and Quality-Control Scope

The following example is reproduced from the final structured-rationale dataset and illustrates the supervision format used by Distilled Rationale Only and the distilled-rationale component of multitask training. The teacher received the grounded image and the reference triplet `Monopolar Hook-dissect-omentum`; consequently, agreement between `final_triplet` and the reference label is a property of the supervision-generation task and is not an independent prediction result.


```
{
  "reasoning_predictions": [
    {
      "id": "hook_1",
      "reasoning": {
        "contact_evidence": [
          "Hook tip contacts dark, vascularized tissue fold adjacent to liver edge.",
          "Instrument applies traction to separate this tissue from underlying structures."
        ],
        "action_evidence": {
          "primary_action": "dissect",
          "support": "Mechanical separation of tissue planes via hook traction and
    potential cautery."
        },
        "primary_target_hypothesis": {
          "target": "omentum",
          "why_plausible": "The engaged tissue is a fatty, vascular fold consistent with
    omental attachment being cleared."
        },
        "contrastive_target_hypothesis": {
          "target": "gallbladder",
          "why_plausible": "Located in the gallbladder fossa region near the organ.",
          "why_rejected": "The instrument engages soft, mobile tissue distinct from the
    firm gallbladder wall."
        },
        "uncertainty": "Visual similarity between omentum and peritoneal adhesions limits
    definitive distinction."
      },
      "final_triplet": {
        "instrument": "Monopolar Hook",
        "action": "dissect",
        "target": "omentum"
      }
    }
  ]
}
```


This example was selected to demonstrate the complete structured response rather than to establish the factual correctness of its explanation. Approximately 30,000 training images received generated rationales, of which approximately ten were inspected informally by the author. The inspected examples appeared plausible, but the review was neither systematic nor performed by a clinician. The rationale corpus therefore cannot be regarded as clinically validated. In addition, rationale and final-answer tokens receive equal token-level loss weight, so the substantially longer rationale contributes most of the supervised objective. These limitations motivate interpreting the experiment as *Distilled Reasoning Supervision*, not as validation of the generated reasoning process.

5.10 Conclusion

This chapter formulates surgical triplet segmentation as specialist instrument grounding followed by MLLM-based interaction prediction. Direct SFT establishes a viable Stage 2 baseline. Under ground-truth grounding, Sequential Dialogue Prediction and three-frame temporal context produce promising observed gains in AP_{IVT}, while prompt-level anatomical priors, ontology guidance, five-frame context, and distilled-reasoning supervision show that additional context or longer supervision does not inherently improve interaction prediction. The mixed component-level effects require the relative differences between adaptations to be interpreted cautiously.

When the full system uses predicted masks, grounding errors reduce performance. Nevertheless, the two-stage Direct SFT pipeline produces substantially stronger component scores and complete-triplet AP comparable to TargetFusionNet. The results therefore establish two-stage multimodal interaction prediction as a viable culmination of the thesis's technical progression: Chapter 3 establishes the spatial substrate, Chapter 4 connects that substrate to grounded interactions, and this chapter demonstrates that semantic interaction prediction can be studied as a specialised second stage. The resulting framework provides a coherent basis for continued research without making the contribution demonstrated here dependent on additional experiments.

The following chapter brings these contributions together. It considers what the review, datasets, grounded benchmark, target-aware architecture, and two-stage formulation collectively establish about surgical scene understanding, before discussing the limitations of strict triplet evaluation and the most important directions for extending the work beyond laparoscopic cholecystectomy.

Chapter 6

Discussion and Conclusion

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6.1 Overview

This thesis has investigated how surgical scene understanding can progress from recognising activities at frame level towards spatially grounded interpretation of instrument–tissue interactions. The central premise is that a useful description of a surgical scene should identify not only which instruments, actions, and anatomical targets are present, but also which visible instrument instance is responsible for each interaction. Establishing this association requires appropriate annotations, evaluation protocols, and models: semantic interaction labels alone do not identify the actor, while instrument masks alone do not describe its surgical role.

The work develops this argument progressively. Chapter 2 reviews multitask learning and surgical scene understanding and identifies the absence of strong instance-level spatial grounding as an important limitation of increasingly fine-grained activity-recognition tasks. Chapter 3 addresses the corresponding data gap through CholecInstanceSeg, establishing large-scale instrument instance supervision in laparoscopic cholecystectomy. Chapter 4 then connects instrument masks with verb and anatomical target labels through CholecTriplet-Seg and evaluates TargetFusionNet as a target-aware model for the resulting triplet-segmentation task.

Finally, Chapter 5 investigates a complementary decomposition in which a specialist model grounds instrument instances and a multimodal large language model (MLLM) predicts their interactions.

This chapter synthesises the principal contributions and answers the thesis research questions. It then considers the broader implications of spatial grounding, the difficulty of anatomical target prediction, and the interpretation of comparatively low complete-triplet scores. The chapter closes by discussing limitations, responsible use of surgical video data, and future work led by cross-procedure evaluation of the grounded interaction formulation.

6.2 Development of Spatially Grounded Surgical Interaction Understanding

6.2.1 Lessons from the Review of Surgical Scene Understanding

The review in Chapter 2 provides the intellectual starting point for this thesis. Surgical scene understanding had developed through tasks including instrument detection and segmentation, surgical phase and workflow recognition, action recognition, and anatomical understanding. These predictions describe related aspects of the same operative scene, and multitask learning offered a useful framework for considering their relationships. Nevertheless, multitask learning was not itself the central technical objective of the subsequent thesis. Where it was investigated, it was useful in some settings but did not consistently determine complete-interaction performance. Its principal role in the thesis is therefore conceptual: it helped expose the need to connect multiple forms of surgical information rather than treating each prediction in isolation.

Surgical action triplets represented an important step towards this richer description. By representing an interaction as the $\langle$`instrument, verb, target`$\rangle$ tuple, the triplet formulation expressed more surgical meaning than instrument presence or workflow phase alone. However, the dominant formulation remained frame-level: a method could recognise that an interaction occurred somewhere in the image without identifying which visible instrument instance performed it. This distinction becomes important whenever several instruments are present, and particularly when two instruments share the same class. In such a scene, a frame-level label cannot unambiguously associate different actions or targets with the corresponding instances.

The review also showed that large-scale instrument instance segmentation was comparatively uncommon in routine human laparoscopic video. Work towards spatially aware action recognition was emerging, but the field did not yet have an established mask-based, instance-level triplet-segmentation dataset and benchmark. These limitations were connected. Strongly grounding surgical triplets required instance annotations, while instance masks became substantially more informative when they were linked to the surgical role of each instrument.

The main conclusion drawn from the review was therefore not simply that more tasks should be predicted together. Rather, the next step was to connect semantic interaction labels to the individual visual instances responsible for them. This observation motivated the dataset and modelling contributions that followed.

6.2.2 CholecInstanceSeg: Establishing the Spatial Foundation

CholecInstanceSeg addresses the scarcity of large-scale instrument instance annotations in laparoscopic surgery. It contains 41,933 annotated frames from 85 laparoscopic cholecystectomy procedures and 64,483 instrument instances. In contrast to instrument-presence or semantic-segmentation labels, its instance identifiers distinguish individual instruments belonging to the same semantic class. This distinction supplies the representation required to ask which particular instrument is involved in an interaction.

Constructing the dataset required more than aggregating existing files. Annotations originating from different public datasets had to be converted into a common representation, reconciled, refined, and subjected to technical validation. The validation experiments in Chapter 3 demonstrate that contemporary instance-segmentation models can learn from the resulting annotations, while the accompanying analyses document the dataset composition and its limitations. CholecInstanceSeg therefore contributes both a reusable public resource and evidence that instrument instances can be modelled at meaningful scale in routine laparoscopic scenes.

Within the thesis, its significance extends beyond instrument segmentation. CholecInstanceSeg establishes individual instrument instances as the spatial units around which later interaction labels can be organised. It changes the available question from “which instrument classes are visible?” to “where is each instrument, and what is each one doing?” The subsequent construction of CholecTriplet-Seg depends upon precisely this change in representation.

6.2.3 CholecTriplet-Seg: Connecting Space and Semantics

CholecTriplet-Seg connects instrument-mask supervision with the ⟨`instrument, verb, target`⟩ annotations of CholecT50. The resulting dataset contains 30,955 frames and 49,866 grounded triplet instances from 50 laparoscopic cholecystectomy videos. Each interaction is associated with an instrument mask, enabling triplet prediction to be evaluated as an instance-level spatial task rather than only as frame-level recognition.

This reformulation is conceptually important. A grounded prediction must identify the instrument instance, localise it, and associate it with the correct verb and anatomical target. The task consequently distinguishes between interactions performed by different instruments in the same frame and links semantic predictions to visible evidence. CholecTriplet-Seg thus brings together the two gaps identified by the review: the absence of large-scale instrument instance supervision and the weak spatial grounding of surgical action triplets.

The benchmark reports separate performance for instruments, verbs, targets, their component pairs, and complete triplets. This decomposition is necessary because complete-triplet performance alone does not reveal whether an error arose from localisation, instrument identity, action recognition, or target prediction. At the same time, the complete metric remains important because a meaningful interaction description requires these elements to be coherent for the same instance.

The contribution of CholecTriplet-Seg is therefore not limited to increasing annotation volume. It defines a more demanding scientific problem and provides the data and evaluation protocol required to study it. By doing so, it advances the triplet task from asking what occurs in a frame towards identifying which instrument performs each action and upon which anatomical target.

6.2.4 TargetFusionNet: Target-Aware Grounded Prediction

TargetFusionNet addresses the principal semantic difficulty exposed by grounded triplet prediction: anatomical target identification. Instruments often have distinctive shapes and direct mask supervision, whereas anatomical targets may have poorly defined boundaries, similar appearances, partial visibility, and substantial deformation. The local appearance around an instrument may also be insufficient to distinguish adjacent structures such as the cystic duct, cystic artery, cystic pedicle, and gallbladder.

The model extends an instance-segmentation architecture with a gated mechanism that fuses weak anatomical information with instrument queries. The Chapter 4 experiments show that strong instance supervision substantially improves grounded triplet performance over weak localisation, and that TargetFusionNet improves the complete triplet result relative to the evaluated strongly supervised baselines. The ablations further show that the manner and depth of fusion matter:

anatomical information is not useful merely because it is available, but must be integrated into the learned representation in a form that supports the task.

The long-tail and failure analyses qualify this result. Performance declines for less frequent triplets, and confusions remain among anatomically adjacent or visually similar targets. Target prediction consequently continues to constrain complete-triplet performance. TargetFusionNet should therefore be understood both as a positive modelling result and as evidence that anatomical target association remains an unresolved challenge. It demonstrates the value of explicit target-aware modelling without implying that weak anatomical information removes the need for better target supervision, contact localisation, or contextual modelling.

6.2.5 Two-Stage Multimodal Interaction Prediction

Chapter 5 explores a complementary formulation with specialised spatial and semantic components. Mask2Former first predicts instrument instances, after which Qwen3-VL-8B-Instruct predicts the corresponding verb and target for each grounded instance. This decomposition asks whether a general multimodal model can build upon the spatial foundation established by the preceding chapters without also learning dense localisation from the same objective.

Direct supervised fine-tuning establishes a viable Stage 2 baseline. Under ground-truth grounding, Sequential Dialogue Prediction and three-frame temporal context produce the strongest observed complete-triplet results among the principal adaptations. The complete Direct SFT pipeline also produces substantially stronger component scores and essentially comparable complete-triplet AP relative to TargetFusionNet when predicted instrument instances are used. These findings show that MLLM-based interaction prediction can be connected successfully to a specialist grounding model.

The remaining adaptations also provide useful guidance. Five-frame context, textual anatomical priors, ontology guidance, and distilled-rationale supervision do not consistently improve the principal outcome. More context or longer supervision is therefore not inherently helpful; its value depends on representation, objective, and integration. The two-stage work establishes a promising continuation of the thesis rather than a final resolution of interaction prediction. Its most important conclusion is that specialist grounding and multimodal semantic prediction form a workable system whose temporal modelling and cross-procedure generalisability can now be investigated further.

6.3 Answers to the Research Questions

6.3.1 Progress from Perceptual Tasks to Spatially Grounded Interaction Understanding

Surgical scene understanding progressed from individual perceptual and workflow tasks towards increasingly fine-grained descriptions of surgical activity. Multitask learning provided one useful framework for relating these predictions, while surgical action triplets offered a compact representation of instrument–tissue interactions. However, the triplet task remained predominantly frame-level and could not reliably associate an interaction with a particular instrument in a multi-instrument scene. At the same time, suitable large-scale instance annotations were scarce.

The principal gap was therefore not simply the need to predict more related outputs. It was the need to spatially ground semantic interaction predictions to the visual instrument instances responsible for them. This conclusion links the review directly to the construction of CholecInstanceSeg and CholecTriplet-Seg.

The answer to the first research question is therefore that progress towards fine-grained interaction recognition exposed spatial grounding, rather than the prediction of additional frame-level labels alone, as the missing capability required to associate surgical action triplets with individual instrument instances.

6.3.2 Construction of Large-Scale Instrument Instance Supervision

CholecInstanceSeg demonstrates that existing public laparoscopic datasets can be extended into a large-scale instrument instance-segmentation resource. Its 41,933 frames and 64,483 instances provide sufficient scale to train and compare modern instance-segmentation methods, while distinguishing individual instruments within multi-instrument scenes. Technical validation confirms that the resulting annotations support learnable and measurable instance segmentation.

The answer to the second research question is therefore affirmative. The resulting dataset also fulfils a broader role: it provides the spatial substrate needed to connect interaction labels to individual instruments in the later chapters.

6.3.3 Grounding Surgical Action Triplets

CholecTriplet-Seg demonstrates that surgical action triplets can be grounded to instrument instance masks. By linking each visible instrument to its verb and target, it enables spatially interpretable predictions and mask-based evaluation of complete interactions. The associated benchmark separates component, pair, and full-triplet performance so that different sources of error remain observable.

The third research question can therefore also be answered affirmatively. The contribution changes triplet recognition from a frame-level presence problem into an instance-level association problem and establishes the dataset and metric framework needed to study it.

6.3.4 Weak Anatomical Priors and Target Prediction

The TargetFusionNet experiments provide evidence that weak anatomical priors can improve grounded triplet prediction when fused into an instance-segmentation architecture. The result is nevertheless conditional on how the prior is represented and integrated. The prompt-level anatomical experiment in Chapter 5, which supplies EndoViT-derived context from a heuristic instrument-tip region, does not reliably improve target classification.

These findings are complementary rather than contradictory. They indicate that anatomical information can be useful, but that its presence alone is insufficient. Effective use requires an appropriate spatial correspondence, representation, and learned integration mechanism. The continuing difficulty of target prediction also suggests that richer contact-region or target supervision may be more valuable than increasingly elaborate prompt descriptions of coarse anatomy.

The fourth research question is therefore answered conditionally: weak anatomical priors can improve target and complete-triplet prediction when they are incorporated through the learned fusion mechanism evaluated in TargetFusionNet, but the prompt-level experiment shows that this benefit does not generalise automatically to every representation of anatomical context.

6.3.5 Multimodal Second-Stage Interaction Prediction

Chapter 5 demonstrates that a fine-tuned MLLM can act as a viable second-stage interaction predictor when instrument instances are supplied. Direct SFT establishes the principal baseline, while sequential factorisation and short temporal context provide promising directions for further development. When predicted masks are used, the complete pipeline remains competitive with TargetFusionNet despite the additional errors introduced by Stage 1.

The fifth research question is therefore answered affirmatively at the level of formulation: MLLM-based interaction prediction is viable as part of a two-stage grounded system. Continued work is required to assess the strongest adaptations under predicted grounding and determine how well the formulation transfers beyond cholecystectomy. These next steps extend, rather than invalidate, the evidence established in the chapter.

6.4 Principal Scientific and Technical Insights

6.4.1 Spatial Grounding as an Interaction Representation

Spatial grounding is more than an additional output attached to action recognition. It changes the unit of prediction from the frame to the instrument instance. This makes it possible to distinguish multiple actors, associate semantic labels with visible evidence, and penalise an otherwise correct interaction assigned to the wrong instrument. The task is consequently more demanding than frame-level recognition, but also more interpretable and more precise.

The thesis demonstrates two ways of modelling this relationship. TargetFusionNet learns masks and triplet labels within a unified architecture, whereas the two-stage framework treats grounded instances as an interface to an MLLM. The unified approach can share representations across localisation and interaction prediction. The decomposed approach allows each stage to specialise and makes their contributions easier to inspect, but it also makes Stage 2 dependent on Stage 1: missed instruments are absent from the interaction input, false positives create unnecessary predictions, and instrument classes cannot currently be revised by the MLLM. The results do not establish that one design is universally preferable. They show instead that accurate spatial grounding and coherent semantic interaction prediction are both necessary, regardless of whether they are learned jointly or separately.

6.4.2 Dataset Construction Enables New Scientific Questions

The dataset contributions are not merely preparatory steps for model development. They determine which scientific questions can be asked and evaluated. Before instrument instance masks are available, a model cannot be assessed on whether it distinguishes two graspers in the same frame. Before masks are aligned with verbs and targets, a triplet model cannot be assessed on whether it assigns the correct interaction to the correct grasper.

CholecInstanceSeg enables the first question by defining individual instruments spatially. CholecTriplet-Seg enables the second by attaching semantic interactions to those instances. Together, they convert a previously weakly grounded recognition problem into a strongly supervised instance-association problem. This progression illustrates a broader lesson for surgical data science: advances in annotation design can change the scientific scope of a task, not only its dataset size.

6.4.3 Anatomical Targets Remain the Principal Semantic Challenge

Across Chapters 4 and 5, target prediction is more difficult than instrument grounding and frequently constrains the complete interaction result. Several properties of the problem contribute to this pattern. Anatomical structures deform, overlap, and change appearance during surgery; their boundaries may be obscured or intrinsically weak; and adjacent structures may look similar in a restricted laparoscopic field of view. The relevant target may also be defined by the instrument's contact or surgical intent rather than by the dominant tissue visible in the image.

The distribution of targets and complete triplets compounds these visual difficulties. A small number of interactions occur frequently, while many valid combinations are rare. Models consequently receive limited supervision for some of the distinctions on which complete-triplet correctness depends. TargetFusionNet shows that learned anatomical fusion can help, but neither it nor the prompt-level prior resolves the underlying ambiguity. Dense target annotation could provide richer supervision, but it demands substantial expert time, remains ambiguous around contact boundaries, and becomes harder to scale as new procedures expand the anatomical and interaction vocabulary. Better contact localisation, selectively richer target supervision, and representations that integrate spatial and temporal evidence therefore remain important directions.

6.4.4 Why Complete-Triplet Performance Is Comparatively Low

The complete-triplet AP values reported in Chapters 4 and 5 are substantially lower than the instrument and several component-level scores. These values reflect genuine model limitations, but they must also be interpreted in relation to the strict task definition. A grounded triplet is correct only when the model localises the appropriate instance and assigns its instrument, verb, and target labels coherently. An error in any one component makes the complete prediction incorrect.

This conjunctive requirement causes errors to compound. A target improvement contributes to AP_{IVT} only when localisation, instrument identity, and verb are also correct for the same instance. This explains why the two-stage model can produce substantially stronger component results than TargetFusionNet without producing a correspondingly large complete-triplet increase. Component metrics and AP_{IVT} answer different questions: the former diagnose individual capabilities, while the latter measures exact grounded interaction correctness.

The task also contains a strongly long-tailed set of clinically valid triplets. Rare combinations receive fewer examples, yet class-averaged AP ensures that their performance remains visible rather than allowing common interactions to dominate the score. Anatomical ambiguity, temporally ambiguous verbs, missed or imperfect masks, and uncertain transitional frames add further failure routes. Spatial evaluation is appropriately harder than frame-level recognition because a correct semantic label attached to the wrong instrument is not a correct grounded interaction.

These factors do not make low AP uninformative. On the contrary, they expose the difficulty that the new task was designed to measure. They do, however, mean that AP_{IVT} should not be interpreted in the same way as accuracy for a single, balanced classification problem. The component, frequency-stratified, and confusion analyses are necessary for understanding what lies beneath the aggregate score.

6.4.5 Not All Errors Are Equal

Exact complete-triplet evaluation treats all non-reference predictions as incorrect, but their severity and meaning can differ considerably. Consider a reference interaction `<clipper, clip, cystic duct>`. A prediction of `<clipper, clip, cystic artery>` preserves the instrument and action and confuses two anatomically adjacent structures. A prediction of `<grasper, retract, liver>` describes an entirely different interaction. Both fail exact AP_{IVT}, although the former indicates substantially more relevant understanding of the scene.

Similar distinctions occur between aspiration and irrigation, grasp and retract, or cystic duct and cystic pedicle. A single-component near miss differs from a prediction in which localisation and every semantic component are incorrect. An ontology-valid but visually ambiguous alternative also differs from a malformed or clinically incompatible output. Exact AP remains necessary because a deployed structured prediction ultimately requires a definite and correctly grounded answer, but it does not describe these differences in error structure.

6.4.6 The Case for Top-k and Richer Evaluation

Future evaluation should retain strict AP while supplementing it with ranked measures. Top-k triplet or component evaluation would test whether the reference interaction remains among a model's highest-ranked candidates. In the preceding example, a model that ranks cystic duct second behind cystic artery has narrowed the prediction to a plausible anatomical neighbourhood, whereas a model whose leading candidates are unrelated has failed more fundamentally. Both may have the same top-1 outcome, but they motivate different modelling responses.

Top-k evaluation should not be presented as a replacement for the established benchmark or merely as an easier score. Its value is diagnostic: it distinguishes narrowly misranked predictions from complete misunderstandings. It could be combined with component-aware results, frequency-stratified reporting, and ontology-aware groupings of related actions or targets. Such measures would provide a richer account of model behaviour while preserving AP_{IVT} as the criterion for exact grounded correctness.

The present thesis does not report top-k results because the evaluated pipelines produce or retain a single final structured prediction rather than a consistently calibrated ranking of triplets. Implementing this evaluation therefore requires models or inference procedures that expose comparable candidate scores. It is a concrete recommendation for continued work, particularly when evaluating transfer to a new procedure, where the nature and severity of errors may differ from those observed in cholecystectomy.

6.5 Limitations

6.5.1 Dataset Scope and Generalisability

The technical contributions focus on laparoscopic cholecystectomy and build upon established public datasets from this procedure. This focus enables coherent development from instrument instances to grounded triplets, but limits conclusions about other operations, institutions, imaging systems, and instrument sets. CholecInstanceSeg and CholecTriplet-Seg also inherit long-tailed class distributions. Rare instruments, targets, and triplets remain difficult to model and may be underrepresented in aggregate examples despite their influence on class-averaged metrics.

6.5.2 Annotation and Task Definition

Instance-mask construction and alignment with triplet labels require substantial annotation and verification effort. Instrument boundaries can be obscured, and target or action labels may be ambiguous in isolated frames. The datasets were carefully constructed and inspected, but should be understood as research annotations rather than perfect or exhaustive clinical ground truth. The current evaluation generally uses one reference interaction per grounded instance; it does not represent the degree to which an alternative target or action might also be visually plausible.

The task definition grounds an interaction through the instrument mask because dense target masks are unavailable. This provides a clear and useful actor-centric evaluation, but does not directly localise the instrument–tissue contact or the complete anatomical extent of the target. Future annotations that represent contact regions or uncertainty could enable more detailed evaluation.

6.5.3 Modelling Limitations

No model in the thesis fully resolves anatomical target prediction or the long tail of grounded triplets. TargetFusionNet depends on weak anatomical predictions, while the two-stage system propagates missed or incorrect instrument instances into Stage 2. Temporal information is investigated only in the final technical chapter and is not yet connected to temporally associated predicted instances across frames.

Chapter 5 establishes MLLM viability but represents an active continuation of the research. Most adaptations are evaluated under ground-truth grounding, so the strongest sequential and temporal formulations require broader evaluation within the complete predicted-grounding system. This qualification limits comparative claims but does not alter the demonstrated feasibility of the two-stage formulation.

6.6 Ethical and Responsible Dataset Considerations

The dataset contributions are derived from existing surgical video resources and must be used within their original governance, licensing, and ethical conditions. Surgical videos can contain sensitive clinical information even when conventional patient identifiers are absent. Responsible release therefore requires attention to de-identification, access conditions, documentation, and restrictions inherited from the source datasets.

Representativeness is another important consideration. A benchmark concentrated on one procedure and a limited set of institutions, equipment, surgeons, and workflows cannot represent the full variation encountered in surgery. Long-tailed labels may also produce uneven performance across instruments and interactions. Reporting only an aggregate score can conceal these differences; class-level, frequency-stratified, and failure analyses are therefore important parts of responsible evaluation.

The methods developed in this thesis are retrospective research systems rather than clinically validated decision-support tools. Spatial masks and structured triplets make outputs more inspectable, but do not guarantee their correctness. The present contribution concerns the representation, annotation, and computational modelling of surgical interactions. Any later clinical use would require separate validation under the intended conditions and appropriate human oversight.

6.7 Future Work

6.7.1 Cross-Procedure Evaluation with ProstaTD

The most immediate future direction is to test whether the grounded interaction formulation transfers beyond laparoscopic cholecystectomy. ProstaTD is an external, multi-source dataset for fully supervised surgical triplet detection in robot-assisted prostatectomy [Chen 2026], released through its official repository. It provides an opportunity to examine the procedure dependence of the datasets, ontology, grounding strategy, and interaction-prediction models developed in this thesis.

A possible extension would adapt its spatial annotations towards mask-based instrument grounding, using SAM2-assisted mask generation followed by systematic manual quality assurance where required. A two-stage model could then be evaluated with procedure-appropriate instrument, action, and target labels. This experiment would test whether specialist grounding followed by multimodal interaction prediction remains viable under different anatomy, instruments, viewpoints, and workflows. It would also reveal which aspects of the formulation transfer directly and which require procedure-specific adaptation.

6.7.2 Annotation-Efficient Expansion

Expanding grounded interaction datasets manually is expensive. Foundation segmentation models offer a possible means of accelerating mask construction, but generated annotations must remain subject to targeted human correction and quality control. Future pipelines should prioritise uncertain masks, rare classes, ambiguous contacts, and inconsistent instance–interaction alignments for review. This would extend the semi-automated principles used in the thesis while preserving the reliability required for grounded evaluation.

6.7.3 Improved Anatomical and Temporal Modelling

Future target modelling should investigate learned instrument–tissue contact localisation and richer use of dense anatomical predictions. The Chapter 5 textual prior compresses the local anatomy into three class proportions and depends on a heuristic tip estimate; alternative representations could retain spatial structure and learn which anatomical evidence is relevant to each instrument and action.

Temporal modelling should similarly move beyond independently rendered frames sharing a fixed visual-token budget. Consistent association of instrument instances across time would allow the model to connect motion and interaction progression to the same actor. This is particularly relevant to actions such as aspiration and irrigation, or grasping and retraction, that may be difficult to distinguish from a still image. Evaluation under predicted grounding will require tracking, mask propagation, or another means of maintaining instance correspondence across neighbouring frames.

6.7.4 Continued Development of Multimodal Interaction Prediction

The sequential and three-frame formulations in Chapter 5 provide the most immediate candidates for further study. The first priority is to evaluate them within the complete predicted-grounding pipeline, both separately and in controlled combinations; repeated runs should then support more reliable comparison of the resulting systems. Sequential prediction could be made more robust to an incorrect first-turn verb, while curriculum-style adaptation could initialise specialised variants from a converged Direct SFT model rather than independently from the base MLLM.

Structured-output reliability also warrants attention. Constrained generation or lightweight ontology-aware validation may reduce malformed or incompatible predictions, provided that such processing is evaluated transparently rather than silently correcting outputs. If distilled-rationale supervision is revisited, answer-token weighting and systematic rationale quality control would be necessary to avoid allowing long, unverified explanations to dominate the objective.

6.7.5 Richer Evaluation of Grounded Interactions

Cross-procedure work should retain exact mask-based AP to preserve continuity with CholecTriplet-Seg, while adding top-k component and triplet measures where calibrated candidate rankings are available. Frequency-stratified evaluation and ontology-aware error groupings would clarify whether transfer failures arise from rare labels, new anatomy, unfamiliar instruments, or fundamentally unrelated predictions. This richer evaluation would recognise that not all errors are equal without weakening the requirement for exact grounded correctness.

6.8 Conclusion

This thesis advances surgical scene understanding from recognising which interactions occur in a frame towards identifying which instrument instance performs each action and upon which anatomical target. The review of multitask learning and surgical scene understanding identified the absence of strong spatial grounding as a central limitation of increasingly fine-grained activity recognition. CholecInstanceSeg then established the large-scale instrument-instance supervision needed to distinguish individual actors in laparoscopic scenes.

Building upon this spatial foundation, CholecTriplet-Seg connected instrument masks with verbs and anatomical targets and established triplet segmentation as a strongly supervised, instance-level task. TargetFusionNet demonstrated that target-aware fusion can improve grounded triplet prediction, while its results and failure analyses showed that anatomical target association and long-tailed complete interactions remain difficult. The final technical chapter demonstrated that grounding and semantic interaction prediction can also be separated successfully: a specialist segmentation model can supply instrument instances to an MLLM that predicts their interactions.

Together, these contributions provide datasets, benchmarks, and modelling approaches for spatially grounded surgical interaction understanding. They also show why complete-triplet performance remains challenging: localisation, instrument identity, action, target, and their association must all be correct simultaneously, and exact evaluation treats near semantic or anatomical confusions in the same way as unrelated errors. Future top-k and component-aware measures can provide a richer diagnosis, but should complement rather than replace strict grounded evaluation.

The immediate next step is to determine whether this formulation generalises beyond cholecystectomy. Cross-procedure evaluation using an external dataset such as ProstaTD would test the transfer of instrument grounding and multimodal interaction prediction to different anatomy, instruments, and workflows. By establishing the required spatial and semantic foundations, this thesis makes that broader investigation possible and provides a clear direction for the continued development of grounded surgical scene understanding.

6.9 Impact Statement

The principal impact of this thesis is the establishment of spatially grounded surgical interaction understanding as a concrete and measurable research problem. Before this work, surgical action triplets were predominantly evaluated as frame-level labels, making it possible to recognise that an interaction occurred without identifying the instrument instance responsible for it. By introducing instrument-instance resources, grounded triplet annotations, and corresponding evaluation tasks, this thesis provides the surgical data science community with a basis for developing and comparing models that connect semantic predictions to visible actors in the operative scene.

CholecInstanceSeg contributes a large-scale, open research resource for laparoscopic instrument instance segmentation. Its distinction between separate instruments of the same class supports research beyond conventional instrument-presence and semantic-segmentation settings. CholecTriplet-Seg builds upon this foundation by associating instrument masks with verb and anatomical target labels. Together, these datasets enable questions about localisation, action recognition, target association, and complete grounded interactions to be studied within a connected experimental framework. Their accompanying benchmarks and baseline models provide reference points against which subsequent methods can be assessed.

The methodological contributions also have broader research value. TargetFusionNet demonstrates how weak anatomical information can be incorporated into grounded interaction prediction, while the two-stage formulation shows that specialist visual grounding can be combined with a multimodal large language model for structured interaction prediction. The results expose persistent difficulties in anatomical target identification, long-tailed interactions, and exact triplet evaluation. Identifying these limitations is itself useful: it directs future work towards improved anatomical and temporal modelling, cross-procedure validation, and evaluation measures that distinguish narrowly plausible errors from unrelated predictions while retaining strict grounded AP.

The work has also supported research exchange and community building. Its themes have contributed to publications spanning surgical instrument segmentation, surgical action understanding, and multimodal modelling, alongside engagement through the Large Models Meet Surgical Data Science workshop and the development of a surgical data science collaboration between King's College London and Obafemi Awolowo University Teaching Hospitals Complex. These activities extend the value of the thesis beyond its individual experiments by encouraging discussion, shared resources, and research capacity in surgical artificial intelligence.

The systems developed here are retrospective research methods and have not been clinically validated. Their immediate impact is therefore scientific rather than clinical: they provide data, task definitions, evidence, and models that make fine-grained surgical interaction understanding more reproducible and inspectable. In the longer term, reliably grounding actions to instrument instances could support more informative surgical video analysis, skill and workflow research, and context-aware computer-assisted intervention. Realising such applications will require validation across procedures, institutions, equipment, and clinical settings, together with appropriate governance and human oversight.

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